

# On the Cartan Graphs of Nichols Algebras over Coquasi-Hopf Algebras

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## Abstract

Over an algebraically closed field of characteristic zero, let  $H$  be a coquasi-Hopf algebra with bijective antipode, and let  $M$  be a tuple of finite-dimensional simple Yetter–Drinfeld modules over  $H$ . We prove that, if  $M$  admits all reflections, then its associated semi-Cartan graph is a Cartan graph. We characterize the finiteness of this Cartan graph by tensor decomposability of  $\mathcal{B}(M)$  and obtain a finite-dimensionality criterion of Nichols algebras. We also show that braided monoidal equivalences preserve reflections and the associated Cartan graphs. As applications, we prove that every Cartan graph associated with a diagonal type tuple over a finite abelian group equipped with an abelian 3-cocycle is covered by one arising from a diagonal type tuple over some finite abelian group  $G$  with trivial associator, and that their real-root sets agree at corresponding objects. We also construct a Cartan graph of the former kind that cannot be obtained from any diagonal tuple in  ${}^G\mathcal{YD}$ .

## 1 Introduction

Throughout the paper,  $\mathbb{k}$  is an algebraically closed field of characteristic zero,  $H$  is a coquasi-Hopf algebra with bijective antipode, and

$$\mathcal{C} = {}^H_H\mathcal{YD}$$

denotes the category of Yetter–Drinfeld modules over  $H$ . We write  $\mathbb{I} = \{1, \dots, \theta\}$ , where  $\theta \geq 2$ , and let  $M = (M_1, \dots, M_\theta)$  be a tuple of finite-dimensional simple objects of  $\mathcal{C}$ . We use the notation

$$\mathcal{B}(M) = \mathcal{B}(M_1 \oplus \dots \oplus M_\theta).$$

Nichols algebras occur in the bosonization description of graded pointed Hopf algebras and play a central role in the lifting method; see [2, 3, 5]. One of the structural tools in their study is the reflection theory of tuples of simple Yetter–Drinfeld modules. When the iterated braided adjoint actions terminate, their maximal nonzero degrees determine a generalized Cartan matrix; if all successive reflections exist, they give rise to a Weyl groupoid. This construction was developed in the Hopf algebra setting in [1, 13, 14, 15]. For Nichols algebras of diagonal type, the associated Weyl groupoid and arithmetic root system led to the classification of finite arithmetic root systems [11, 12]. In this setting, reduced expressions and convex orders also enter the construction of PBW bases and presentations by generators and relations [4, 7].

Nichols algebras over coquasi-Hopf algebras are braided Hopf algebras in braided monoidal categories with possibly nontrivial associativity constraints. Consequently, the duality, bosonization, coinvariant, and braided adjoint constructions used in reflection theory must be interpreted inside the corresponding braided monoidal category. Such Nichols algebras arise in

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the study of pointed coquasi-Hopf algebras and pointed tensor categories; see, for example, [17, 16, 18].

The reflection construction used in this paper was established in [19, 20]. Fix  $i \in \mathbb{I}$  and suppose that, for every  $j \neq i$ , there is an integer  $m_{ij}^M \geq 0$  such that

$$(\text{ad } M_i)^{m_{ij}^M}(M_j) \neq 0, \quad (\text{ad } M_i)^{m_{ij}^M+1}(M_j) = 0,$$

and that the first of these objects is finite-dimensional. The  $i$ -th reflected tuple is given by

$$R_i(M)_i = M_i^*, \quad R_i(M)_j = (\text{ad } M_i)^{m_{ij}^M}(M_j) \quad (j \neq i).$$

If  $M$  admits all reflections, the tuples obtained from  $M$  determine a semi-Cartan graph

$$\mathcal{G}(M) = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, (A^{[P]})_{[P] \in \mathcal{X}}), \quad \mathcal{X} = \{[P] \mid P \in \mathcal{F}_\theta(M)\}.$$

The fact that this is a semi-Cartan graph follows from [20]. The first objective of the present paper is to prove the remaining Cartan-graph axioms.

## 1.1 Reflection theory and Cartan graphs

For this purpose, we study one-sided coideal subalgebras and their behavior under reflections. Put

$$K_i^M = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)}, \quad L_i^M = {}^{\text{co } \mathcal{B}(M_i)}\mathcal{B}(M).$$

The braided monoidal equivalence associated with the canonical Hopf pairing between  $\mathcal{B}(M_i^*)$  and  $\mathcal{B}(M_i)$  gives the isomorphism

$$\mathcal{B}(R_i(M)) \simeq \Omega_i(K_i^M) \# \mathcal{B}(M_i^*)$$

from [20]. In Section 3, the comparison of one-sided coideal subalgebras yields an algebra isomorphism in  $\mathcal{C}$ ,

$$T_i^M : L_i^{R_i(M)} \longrightarrow K_i^M,$$

and bijections between the relevant graded one-sided coideal subalgebras.

Let  $\kappa = (i_1, \dots, i_l)$  be an  $[M]$ -reduced sequence and set

$$\beta_k = \text{id}_{[M]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}), \quad 1 \leq k \leq l.$$

Using the maps  $T_i^M$ , we construct finite-dimensional simple objects  $M_{\beta_k}$  of degree  $\beta_k$  and a graded right coideal subalgebra  $E^M(\kappa) \subseteq \mathcal{B}(M)$  such that multiplication induces an isomorphism of graded objects

$$\mathcal{B}(M_{\beta_l}) \otimes (\cdots (\mathcal{B}(M_{\beta_2}) \otimes \mathcal{B}(M_{\beta_1}))) \xrightarrow{\sim} E^M(\kappa).$$

These objects and factorizations are used to prove (CG3'). The comparison of the factorizations associated with the two alternating reduced sequences in each rank-two subsystem gives (CG4'). We obtain the following result in Theorem 4.6.

**Theorem A.** *Let  $M$  be a tuple of finite-dimensional simple Yetter–Drinfeld modules over a coquasi-Hopf algebra with bijective antipode. If  $M$  admits all reflections, then  $\mathcal{G}(M)$  is a Cartan graph.*

## 1.2 Tensor decompositions and finite-dimensionality

After establishing Theorem A, we compare the finiteness of  $\mathcal{G}(M)$  with a factorization property of  $\mathcal{B}(M)$ . An  $\mathbb{N}_0^\theta$ -graded object  $V$  in  $\mathcal{C}$  is called tensor decomposable if there are finite-dimensional simple objects  $Q_1, \dots, Q_n$  of pairwise distinct degrees  $\beta_1, \dots, \beta_n \in \mathbb{N}_0^\theta \setminus \{0\}$  such that

$$V \simeq \mathcal{B}(Q_1) \otimes \cdots \otimes \mathcal{B}(Q_n)$$

as an  $\mathbb{N}_0^\theta$ -graded object. This definition is applied to  $\mathcal{B}(M)$  with the grading  $\deg M_i = \alpha_i$ .

We prove that tensor decomposability already implies the existence of all reflections. Tensor decomposability is also preserved by each reflection, and the degrees of the factors are transformed by the corresponding simple reflection. These statements give Theorem 5.8.

**Theorem B.** *Let  $M = (M_1, \dots, M_\theta)$  be a tuple of finite-dimensional simple Yetter–Drinfeld modules. The following conditions are equivalent.*

- (1)  *$M$  admits all reflections and  $\mathcal{G}(M)$  is a finite Cartan graph.*
- (2)  *$M$  admits all reflections and  $\mathcal{B}(P)$  is tensor decomposable for some  $P \in \mathcal{F}_\theta(M)$ .*
- (3)  *$\mathcal{B}(M)$  is tensor decomposable.*

Suppose that these conditions hold. Let  $P \in \mathcal{F}_\theta(M)$ , and choose a reduced expression  $\kappa = (i_1, \dots, i_l)$  of a longest morphism with target  $[P]$ . If

$$\beta_k = \text{id}_{[P]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}),$$

then

$$\Delta_+^{[P], \text{re}} = \{\beta_1, \dots, \beta_l\},$$

and multiplication induces an isomorphism of graded objects

$$\mathcal{B}(P_{\beta_l}) \otimes (\cdots (\mathcal{B}(P_{\beta_2}) \otimes \mathcal{B}(P_{\beta_1}))) \xrightarrow{\sim} \mathcal{B}(P).$$

Thus the factors are indexed by the positive real roots at  $[P]$ . The Cartan graph determines their number and degrees, but finite-dimensionality also requires the Nichols algebra of each simple factor to be finite-dimensional. Combining this factorization with the behavior of the factors under morphisms of the Weyl groupoid gives Theorem 5.12.

**Theorem C.** *Let  $M = (M_1, \dots, M_\theta)$  be a tuple of finite-dimensional simple Yetter–Drinfeld modules. Then  $\mathcal{B}(M)$  is finite-dimensional if and only if  $M$  admits all reflections,  $\mathcal{G}(M)$  is finite, and  $\mathcal{B}(P_i)$  is finite-dimensional for every  $P \in \mathcal{F}_\theta(M)$  and every  $i \in \mathbb{I}$ .*

In the forward implication, finite-dimensionality gives the existence of every successive reflection. The factorizations associated with reduced sequences bound their lengths and imply that the real-root sets are finite. Conversely, a reduced expression of a longest morphism gives the displayed factorization, and every factor is isomorphic to a component  $P_i$  for a suitable  $P \in \mathcal{F}_\theta(M)$ .

## 1.3 Braided monoidal equivalences

Let  $H'$  be another coquasi-Hopf algebra with bijective antipode and let

$$(F, \xi) : {}^H_H \mathcal{YD} \longrightarrow {}^{H'}_{H'} \mathcal{YD}$$

be a  $\mathbb{k}$ -linear braided monoidal equivalence. The functor  $F$  sends Nichols algebras to Nichols algebras and preserves the images of the iterated braided adjoint maps. Hence it preserves the integers  $m_{ij}^M$ , whenever they are defined, and satisfies

$$F(R_i(M)) \simeq R_i(F(M)).$$

Applying this compatibility to successive reflections proves Theorem 6.2, together with the subsequent corollary.

**Theorem D.** *The tuple  $M$  admits all reflections if and only if  $F(M)$  does. If these conditions hold, the map*

$$[P] \longmapsto [F(P)]$$

*induces an isomorphism  $\mathcal{G}(M) \simeq \mathcal{G}(F(M))$  of Cartan graphs. It also induces an isomorphism of their Weyl groupoids, and the real-root sets at corresponding objects are equal.*

In particular, for tuples admitting all reflections, twists of coquasi-Hopf algebras preserve the associated Cartan graphs.

## 1.4 Applications

We first apply these results to Nichols algebras of diagonal type. Let  $G$  be a finite abelian group and let  $\Phi$  be a normalized abelian 3-cocycle on  $G$ . For a diagonal tuple  $M = (M_1, \dots, M_\theta)$  in  ${}^G\mathcal{YD}^\Phi$ , we compute the iterated braided adjoint actions and the braiding matrices of the reflected tuples. For a tuple of Cartan type, all reflections exist, every tuple in  $\mathcal{F}_\theta(M)$  remains of Cartan type, and the associated Cartan graph is standard. Its finiteness is equivalent to the finite type of the common Cartan matrix, and Theorem C gives the corresponding finite-dimensionality criterion.

We then compare the Cartan graphs of diagonal tuples in twisted and ordinary Yetter–Drinfeld categories. First, we construct a rank-two diagonal tuple over a finite abelian group equipped with an abelian 3-cocycle whose Cartan graph is not isomorphic to the Cartan graph of any diagonal tuple in  ${}^K\mathcal{YD}$  for any finite abelian group  $K$ . Consequently, the isomorphism classes obtained in ordinary Yetter–Drinfeld categories form a proper subset of those obtained in the presence of abelian 3-cocycles.

Second, let  $M$  be an arbitrary diagonal tuple in  ${}^G\mathcal{YD}^\Phi$ . By pulling  $M$  back to a suitable finite abelian group  $G$  and applying a 2-cochain twist, we obtain a diagonal tuple

$$M' \in {}^G\mathcal{YD}$$

with trivial associator. The tuple  $M$  admits all reflections if and only if  $M'$  does. Under these equivalent conditions, there is a covering

$$\mathcal{G}(M') \longrightarrow \mathcal{G}(M),$$

and the real-root sets at objects related by the covering map are equal. Thus  $\mathcal{G}(M)$  is a quotient of a Cartan graph arising from a diagonal tuple in an ordinary Yetter–Drinfeld category, but it need not be isomorphic to that graph.

This covering transfers the following results from ordinary Nichols algebras of diagonal type. If  $M$  admits all reflections, its real-root system is finite if and only if its generalized Dynkin diagram occurs in Heckenberger’s classification [12]. Moreover,

$$\mathrm{GKdim} \mathcal{B}(M) < \infty \iff M \text{ admits all reflections and } \mathcal{G}(M) \text{ is finite.}$$

The proof uses the equality of the real-root sets under the covering, the equality of the multi-graded Hilbert series of the two Nichols algebras, and the corresponding result for ordinary diagonal type [6].

## 1.5 Organization of the paper

Section 2 recalls semi-Cartan graphs, Cartan graphs, Weyl groupoids, root systems, coquasi-Hopf algebras, Yetter–Drinfeld modules, bosonization, Nichols algebras, and the reflection construction from [19, 20]. Section 3 studies one-sided coideal subalgebras and the maps  $T_i^M$  associated with reflections. Section 4 constructs the simple objects and graded right coideal subalgebras associated with reduced sequences and proves that  $\mathcal{G}(M)$  is a Cartan graph. Section 5 introduces tensor decomposable Nichols algebras, proves the factorization indexed by positive real roots, and establishes the finite-dimensionality criterion. Section 6 proves the invariance of reflections, Cartan graphs, Weyl groupoids, and real-root sets under braided monoidal equivalences. Section 7 treats diagonal tuples, compares the twisted and ordinary diagonal cases by a covering of Cartan graphs, derives criteria for finiteness of the real-root system and for finite Gelfand–Kirillov dimension.

## 2 Preliminaries

In this section, we briefly recall the fundamental definitions and fix the notation for coquasi-Hopf algebras and Yetter–Drinfeld modules. For a more detailed exposition and proofs of standard properties, we refer the reader to [19].

### 2.1 Cartan graphs and root systems

We start from the definition of semi-Cartan graphs, which plays an important role in this paper, we follow the notation in [15]. One may refer to that book for detailed proofs and various examples.

**Definition 2.1.** Let  $\mathbb{I}$  be a non-empty finite set,  $\mathcal{X}$  a non-empty set,  $r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}$ ,  $A : \mathbb{I} \times \mathbb{I} \times \mathcal{X} \rightarrow \mathbb{Z}$  maps. For all  $i, j \in \mathbb{I}$  and  $X \in \mathcal{X}$  we write  $r_i(X) = r(i, X)$ ,  $a_{ij}^X = A(i, j, X)$  and  $A^X = (a_{ij}^X)_{i, j \in \mathbb{I}} \in \mathbb{Z}^{\mathbb{I} \times \mathbb{I}}$ . The quadruple  $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, (r_i)_{i \in \mathbb{I}}, (A^X)_{X \in \mathcal{X}})$  is called a semi-Cartan graph if for all  $X \in \mathcal{X}$ , the matrix  $A^X$  is a generalized Cartan matrix, and the following axioms hold.

(CG1) For all  $i \in \mathbb{I}$ , the map  $r_i$  satisfies  $r_i^2 = \text{id}_{\mathcal{X}}$ .

(CG2) For all  $i \in \mathbb{I}$ ,  $X \in \mathcal{X}$ ,  $A^X$  and  $A^{r_i(X)}$  have the same  $i$ -th row.

Kac–Moody Lie algebras, Kac–Moody Lie superalgebras and Nichols algebras over coquasi-Hopf algebras provide an abundant class of examples of semi-Cartan graphs [20].

Given semi-Cartan graphs, we can construct Weyl groupoids, and define the sets of real roots. Given a semi-Cartan graph, its Weyl groupoid is defined as follows.

**Definition 2.2.** We denote by  $\mathcal{D}(\mathcal{X}, \text{End}(\mathbb{Z}^{\mathbb{I}}))$  to be the category with objects  $\text{Ob } \mathcal{D}(\mathcal{X}, \text{End}(\mathbb{Z}^{\mathbb{I}})) = \mathcal{X}$ , and morphisms

$$\text{Hom}(X, Y) = \{(Y, f, X) \mid f \in \text{End}(\mathbb{Z}^{\mathbb{I}})\},$$

where the composition of morphisms is defined by

$$(Z, g, Y) \circ (Y, f, X) = (Z, gf, X), \text{ for all } X, Y, Z \in \mathcal{X}, f, g \in \text{End}(\mathbb{Z}^{\mathbb{I}}).$$

Let  $\alpha_i, 1 \leq i \leq \theta$  be the standard basis of  $\mathbb{Z}^{\mathbb{I}}$ , and

$$s_i^X \in \text{Aut}(\mathbb{Z}^{\mathbb{I}}), s_i^X(\alpha_j) = \alpha_j - a_{ij}^X \alpha_i, \text{ for all } j.$$

We call the smallest subcategory of  $\mathcal{D}(\mathcal{X}, \text{End}(\mathbb{Z}^{\mathbb{I}}))$  which contains all morphisms  $(r_i(X), s_i^X, X)$  with  $i \in \mathbb{I}, X \in \mathcal{X}$  the Weyl groupoid of  $\mathcal{G}$ , denoted by  $\mathcal{W}(\mathcal{G})$ .

Now let  $X, Y \in \mathcal{G}$ ,  $\omega \in \text{Hom}(X, Y)$ . The following definition is parallel to that in Lie algebra. We call

$$l(\omega) = \min\{k \mid \omega = s_{i_1} \cdots s_{i_k}, i_1, \dots, i_k \in \mathbb{I}\}$$

the length of  $\omega$  and the decomposition  $(i_1, \dots, i_k)$  is called the reduced decomposition of  $\omega$ . Here we omit the subscript for simplicity

For each  $X \in \mathcal{X}$ , the set of real roots of  $\mathcal{G}$  at  $X$  is defined as follows:

$$\Delta^{X, \text{re}} = \{\omega(\alpha_i) \in \mathbb{Z}^{\mathbb{I}} \mid \omega \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), X), i \in \mathbb{I}\}.$$

We say  $\mathcal{G}$  is a finite Cartan graph if  $\Delta^{X, \text{re}}$  is finite for each  $X \in \mathcal{X}$ .

The elements

$$\Delta_+^{X, \text{re}} = \Delta^{X, \text{re}} \cap \mathbb{N}_0^{\mathbb{I}}, \Delta_-^{X, \text{re}} = \Delta^{X, \text{re}} \cap -\mathbb{N}_0^{\mathbb{I}}$$

are called positive and negative real roots, respectively. For any  $X \in \mathcal{X}$  and  $i, j \in \mathbb{I}$ , let

$$m_{ij}^X = |\Delta^{X, \text{re}} \cap (\mathbb{N}_0 \alpha_i + \mathbb{N}_0 \alpha_j)|.$$

With the above notations, we can give the definition of a Cartan graph.

**Definition 2.3.** We say that a semi-Cartan graph  $\mathcal{G}$  is a **Cartan graph** if the following hold.

(CG3) For any  $X \in \mathcal{X}$ , the set  $\Delta^{X, \text{re}} = \Delta_+^{X, \text{re}} \cup \Delta_-^{X, \text{re}}$ .

(CG4) For any  $i \neq j \in \mathbb{I}$  and  $X \in \mathcal{X}$ , if the cardinality  $m_{ij}^X := |R^X \cap (\mathbb{N}_0 \alpha_i + \mathbb{N}_0 \alpha_j)|$  is finite, then  $(r_i r_j)^{m_{ij}^X}(X) = X$ .

There are many examples showing that a semi-Cartan graph may not satisfy (CG3) or (CG4), see [15, Example 9.1.26, Example 9.2.3].

We are going to introduce root systems over semi-Cartan graphs.

**Definition 2.4.** Let  $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$  be a semi-Cartan graph. For all  $X \in \mathcal{X}$ , let  $(R^X)_{X \in \mathcal{X}}$  be a subset of  $\mathbb{Z}^{\mathbb{I}}$  with the following properties.

- (1)  $0 \notin R^X$  and  $\alpha_i \in R^X$  for all  $X \in \mathcal{X}$  and  $i \in \mathbb{I}$ .
- (2)  $R^X \subseteq \mathbb{N}_0^{\mathbb{I}} \cup -\mathbb{N}_0^{\mathbb{I}}$  for all  $X \in \mathcal{X}$ .
- (3) For any  $X \in \mathcal{X}$  and  $i \in \mathbb{I}$ ,  $s_i^X(R^X) = R^{r_i(X)}$ .
- (4) If  $i, j \in \mathbb{I}$  and  $X \in \mathcal{X}$  such that  $i \neq j$  and  $m_{ij}^X$  in Definition 2.3 is finite, then  $(r_i r_j)^{m_{ij}^X}(X) = X$ .

Then we say that the pair  $(\mathcal{G}, (R^X)_{X \in \mathcal{X}})$  is a **root system** over  $\mathcal{G}$ . A root system over  $\mathcal{G}$  is said to be reduced if for all  $X \in \mathcal{X}$  and  $\alpha \in R^X$  the roots  $\alpha$  and  $-\alpha$  are the only rational multiples of  $\alpha$  in  $R^X$ . A root system is said to be finite if for all  $X \in \mathcal{X}$ ,  $R^X$  is finite.

If  $\mathcal{G}$  is a Cartan graph, the pair  $(\mathcal{G}, (\Delta^{X, \text{re}})_{X \in \mathcal{X}})$  is automatically a reduced root system over  $\mathcal{G}$ . Furthermore, if  $(R^X)_{X \in \mathcal{X}}$  is finite, then  $R^X = \Delta^{X, \text{re}}$  for each  $X \in \mathcal{X}$ .

**Remark 2.5.** It was proved that there is an equivalence between a semi-Cartan graph with a reduced root system  $(\mathcal{G}, (R^X)_{X \in \mathcal{X}})$  and a crystallographic Tits arrangement. A crystallographic Tits arrangement consists of a pair  $(\mathcal{A}, T)$ , where  $\mathcal{A}$  is a set of linear hyperplanes in  $\mathbb{R}^{\mathbb{I}}$ , and  $T$  is a Tits cone in  $\mathbb{R}^{\mathbb{I}}$ , satisfying some additional conditions. One may refer to [9] for details.

## 2.2 Reduced sequences and an equivalent form of Cartan graphs

One of our purposes in this paper is to show reflections of Nichols algebras over coquasi-Hopf algebras with bijective antipode will give rise to Cartan graphs, which means (CG3) and (CG4) hold automatically in our settings. However, it is difficult to directly verify these two axioms, and therefore equivalent conditions are required.

Let  $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$  be a semi-Cartan graph. Let  $X \in \mathcal{X}, l \geq 0$ , and  $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$ . For all  $1 \leq k \leq l$  let

$$\beta_k^{X, \kappa} = \text{id}_X s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}),$$

and let

$$\Lambda^X(\kappa) = \{\beta_k^{X, \kappa} \mid 1 \leq k \leq l\}.$$

**Definition 2.6.** We say that  $\kappa$  is *X-reduced* if for any  $1 \leq k < l$ ,

$$\alpha_{i_k} \notin \Lambda^{r_{i_k} \cdots r_{i_1}(X)}(i_{k+1}, \dots, i_l).$$

The integer  $l$  is called the length of  $\kappa$ .

**Remark 2.7.** (1) Let  $X \in \mathcal{X}, l \geq 2$  and  $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$ . If  $i_j = i_{j+1}$  for some  $1 \leq j < l$ , then

$$\alpha_{i_j} \in \Lambda^{r_{i_j} \cdots r_{i_1}(X)}(i_{j+1}, \dots, i_l),$$

and hence  $\kappa$  is not  $X$ -reduced.

(2) A sequence  $\kappa = (i_1, \dots, i_l)$  is  $X$ -reduced if and only if  $(i_2, \dots, i_l)$  is  $r_{i_1}(X)$ -reduced and  $\alpha_{i_1} \notin \Lambda^{r_{i_1}(X)}(i_2, \dots, i_l)$ .

(3)  $\kappa$  is  $X$ -reduced if and only if  $\beta_p^{X, \kappa} \neq -\beta_q^{X, \kappa}$  for any  $1 \leq p < q \leq l$ .

Now we consider a special case.

**Definition 2.8.** Let  $i, j \in \mathbb{I}$  with  $i \neq j$  and  $\kappa = (i_k)_{k \geq 1} = (i, j, i, \dots)$  with  $i_k = i$  if  $k$  is odd and  $i_k = j$  if  $k$  is even. Let  $X \in \mathcal{X}$ . We call any  $X$ -reduced sequence with entries in  $\{i, j\}$  and starting with  $i$  a beginning of  $\kappa$ . Let  $\kappa_{ij}^X$  be the longest  $X$ -reduced beginning of  $\kappa$ , if it exists, and  $\kappa$  otherwise. We write  $\bar{m}_{ij}^X$  for the length of  $\kappa_{ij}^X$ .

With the help of the notation above, we are in the position to introduce axioms characterizing Cartan graphs.

(CG3') For any  $X \in \mathcal{X}$  and any  $X$ -reduced sequence  $\gamma$ ,  $\Lambda^X(\gamma) \subseteq \mathbb{N}_0^{\mathbb{I}}$ .

(CG4') For any  $X \in \mathcal{X}$  and any  $i, j \in \mathbb{I}$  with  $i \neq j$  and  $\bar{m}_{ij}^X < \infty$ , we have

$$(r_i r_j)^{\bar{m}_{ij}^X}(X) = X, \quad \text{id}_X(s_i s_j)^{\bar{m}_{ij}^X}(\alpha_k) = \alpha_k$$

for all  $k \in \mathbb{I}$ .

The following lemma is important for our purposes.

**Lemma 2.9.** [15, Corollary 9.2.20] For any semi-Cartan graph  $\mathcal{G}$ , the following are equivalent.

- (1)  $\mathcal{G}$  is a Cartan graph.
- (2)  $\mathcal{G}$  satisfies (CG3') and (CG4').

We collect some lemmas needed later, which can be found in [15, Section 9.2].

**Lemma 2.10.** Assume that  $|\mathbb{I}| \geq 2$ . Let  $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$  be a semi-Cartan graph,  $X \in \mathcal{X}, i, j \in \mathbb{I}$  with  $i \neq j$

(1) Assume  $\kappa = (i, j, i, j, \dots)$  is a  $X$ -reduced sequence beginning with  $\kappa_{ij}^X$  with length  $m \geq 2$ . If  $\bar{m}_{ji}^{r_j(X)} = m$ , then  $\beta_m^{X, \kappa} = \alpha_j$ .

(2) Assume  $\mathcal{G}$  satisfies (CG3'). Then  $\bar{m}_{ij}^{r_k(X)} = \bar{m}_{ji}^{r_k(X)} = \bar{m}_{ij}^X$  for  $k \in \{i, j\}$ .

(3) Assume  $\kappa = (i, j, i, j, \dots)$ ,  $\bar{m}_{ij} < \infty$  and  $\mathcal{G}$  satisfies (CG3'), then  $\beta_1^{X, \kappa} = \alpha_i$ ,  $\beta_{\bar{m}_{ij}}^{X, \kappa} = \alpha_j$ ,  $\text{id}_X(s_i s_j)^{\bar{m}_{ij}^X}(\alpha_i) = \alpha_i$  and  $\text{id}_X(s_i s_j)^{\bar{m}_{ij}^X}(\alpha_j) = \alpha_j$ .

The following lemma can be found in [15, Section 9.3], which is used to give a new criterion to determine the finite-dimensionality of a Nichols algebra.

**Lemma 2.11.** (1) Let  $\mathcal{G}$  be a Cartan graph, and  $\omega = \text{id}_X s_{i_1} \cdots s_{i_l} \in \text{Hom}(\mathcal{W}(\mathcal{G}), X)$  be a reduced decomposition of  $\omega$ . Then  $\kappa = (i_1, \dots, i_l)$  is  $X$ -reduced.

(2) Let  $\mathcal{G}$  be a finite Cartan graph and  $(i_1, \dots, i_n)$  be a reduced decomposition of longest element  $\omega_0 \in \text{Hom}(\mathcal{W}(\mathcal{G}), X)$ , then

$$\Delta_+^{X, \text{re}} = \Lambda^X(\kappa).$$

### 2.3 Coquasi-Hopf algebras

We follow the standard definition of a coquasi-Hopf algebra as introduced in [10].

**Definition 2.12.** A coquasi-Hopf algebra  $H$  is a coalgebra  $(H, \Delta, \varepsilon)$  equipped with a compatible quasi-algebra structure and an antipode  $(\mathcal{S}, \alpha, \beta)$ . Namely, there exist:

- Two coalgebra homomorphisms:

$$m : H \otimes H \rightarrow H, \quad a \otimes b \mapsto ab,$$

$$\mu : \mathbb{k} \rightarrow H, \quad \lambda \mapsto \lambda 1_H,$$

- A convolution-invertible map  $\Phi : H^{\otimes 3} \rightarrow \mathbb{k}$  called an associator,
- A coalgebra antimorphism  $\mathcal{S} : H \rightarrow H$ ,
- Two linear functions  $\alpha, \beta : H \rightarrow \mathbb{k}$

such that for all  $a, b, c, d \in H$  the following equalities hold:

$$a_1(b_1 c_1) \Phi(a_2, b_2, c_2) = \Phi(a_1, b_1, c_1)(a_2 b_2) c_2, \quad (2.1)$$

$$1_H a = a = a 1_H, \quad (2.2)$$

$$\Phi(a_1, b_1, c_1 d_1) \Phi(a_2 b_2, c_2, d_2) = \Phi(b_1, c_1, d_1) \Phi(a_1, b_2 c_2, d_2) \Phi(a_2, b_3, c_3), \quad (2.3)$$

$$\Phi(a, 1_H, b) = \varepsilon(a) \varepsilon(b), \quad (2.4)$$

$$\mathcal{S}(a_1) \alpha(a_2) a_3 = \alpha(a) 1_H, \quad a_1 \beta(a_2) \mathcal{S}(a_3) = \beta(a) 1_H, \quad (2.5)$$

$$\Phi(a_1, \mathcal{S}(a_3), a_5) \beta(a_2) \alpha(a_4) = \Phi^{-1}(\mathcal{S}(a_1), a_3, \mathcal{S}(a_5)) \alpha(a_2) \beta(a_4) = \varepsilon(a). \quad (2.6)$$

Throughout this paper, we use the Sweedler sigma notation  $\Delta(a) = a_1 \otimes a_2$  for the co-product and  $a_1 \otimes a_2 \otimes \cdots \otimes a_{n+1}$  for the result of the  $n$ -iterated application of  $\Delta$  on  $a$ . We say  $H$  has a bijective antipode if  $\mathcal{S}$  is bijective.

We now turn our attention to the Yetter-Drinfeld module category structure. The definition of the Yetter-Drinfeld module category  ${}^H_H \mathcal{YD}$  over an arbitrary coquasi Hopf-algebra  $H$  was already given in [8]. Since it plays a crucial role in our paper, we recall this definition.

**Definition 2.13.** [8, Definition 3.1] Let  $H$  be a coquasi-Hopf algebra with associator  $\Phi$ . A left-left Yetter-Drinfeld module over  $H$  is a triple  $(V, \delta_V, \triangleright)$  such that:

- $(V, \delta_V)$  is a left comodule of  $H$  and we denote  $\delta_V(v)$  by  $v_{-1} \otimes v_0$  as usual;
- $\triangleright : H \otimes V \rightarrow V$  is a  $\mathbb{k}$ -linear map satisfying for all  $h, l \in H$  and  $v \in V$ :

$$(hl) \triangleright v = \frac{\Phi(h_2, (l_2 \triangleright v_0)_{-1}, l_3)}{\Phi(h_1, l_1, v_{-1}) \Phi((h_3 \triangleright (l_2 \triangleright v_0)_0)_{-1}, h_4, l_4)} (h_3 \triangleright (l_2 \triangleright v_0)_0)_0, \quad (2.7)$$

$$1_H \triangleright v = v, \quad (2.8)$$

$$(h_1 \triangleright v)_{-1} h_2 \otimes (h_1 \triangleright v)_0 = h_1 v_{-1} \otimes (h_2 \triangleright v_0). \quad (2.9)$$

A morphism  $f : (V, \delta_V, \triangleright) \rightarrow (V', \delta_{V'}, \triangleright')$  is a colinear map  $f : (V, \delta_V) \rightarrow (V', \delta_{V'})$  such that  $f(h \triangleright v) = h \triangleright' f(v)$  for all  $h \in H$ .

The category  ${}^H_H\mathcal{YD}$  is a  $\mathbb{k}$ -linear braided monoidal abelian category over the field  $\mathbb{k}$ . The unit object of  ${}^H_H\mathcal{YD}$  is  $\mathbb{k}$ , which is regarded as an object in  ${}^H_H\mathcal{YD}$  via trivial structures. For  $V, W \in {}^H_H\mathcal{YD}$ , the tensor product of Yetter-Drinfeld modules is defined by:

$$(V, \delta_V, \triangleright) \otimes (W, \delta_W, \triangleright) = (V \otimes W, \delta_{V \otimes W}, \triangleright),$$

where  $\delta_{V \otimes W}$  is given by

$$\delta_{V \otimes W}(v \otimes w) = v_{-1}w_{-1} \otimes v_0 \otimes w_0,$$

and

$$h \triangleright (v \otimes w) = \frac{\Phi(h_1, v_{-1}, w_{-2})\Phi((h_2 \triangleright v_0)_{-1}, (h_4 \triangleright w_0)_{-1}, h_5)}{\Phi((h_2 \triangleright v_0)_{-2}, h_3, w_{-1})}(h_2 \triangleright v_0)_0 \otimes (h_4 \triangleright w_0)_0. \quad (2.10)$$

The braiding  $c_{V,W} : V \otimes W \rightarrow W \otimes V$  is given by:

$$c_{V,W}(v \otimes w) = (v_{-1} \triangleright w) \otimes v_0. \quad (2.11)$$

**Lemma 2.14.** [20, Lemma 2.5] *Let  $H$  be a coquasi-Hopf algebra with bijective antipode. The category  ${}^H_H\mathcal{YD}$  is braided monoidal isomorphic to the right center of the category of left  $H$ -comodules  $\mathcal{Z}_r({}^H\mathcal{M})$ .*

Bosonizations and coinvariants are fundamental constructions that build new Hopf algebras. These operations for coquasi-Hopf algebras first appeared in [8]. We omit these details.

**Lemma 2.15.** [20, Lemma 2.7] *Let  $H$  be a coquasi-Hopf algebra. Suppose  $R$  is a Hopf algebra in  ${}^H_H\mathcal{YD}$ . With appropriate operations,  $M := R \otimes H$  can be turned into a coquasi-Hopf algebra. We denote by  $R \# H$ .*

Now suppose  $N$  and  $H$  are both coquasi-Hopf algebras. Furthermore, assume there exist morphisms of coquasi-Hopf algebras

$$\pi : N \rightarrow H \text{ and } \sigma : H \rightarrow N$$

such that  $\pi\sigma = \text{id}_N$ .

**Lemma 2.16.** [20, Lemma 2.9] *Under above assumptions,  $L := N^{\text{co } H}$  is a Hopf algebra in  ${}^H_H\mathcal{YD}$ . We emphasize that the map satisfying Equation (2.8) hold by*

$$\text{ad} : H \otimes L \rightarrow L, h \otimes l \mapsto \text{ad}(h)(l).$$

**Remark 2.17.** For any monoidal category  $\mathcal{C}$  and Hopf algebra  $A \in \mathcal{C}$ , the category  ${}^A_A\mathcal{YD}(\mathcal{C})$  of Yetter-Drinfeld module over  $A$  in  $\mathcal{C}$  is well-defined. One may refer to [15, Section 3.4] for more details.

## 2.4 Monoidal equivalence between Yetter-Drinfeld module categories related by a dual pair

In this subsection, we recall an important braided monoidal equivalence

$$(\Omega, \beta) : {}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}} \rightarrow {}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}$$

constructed in [20, Section 3].

Let  $A$  and  $B$  be locally finite  $\mathbb{N}_0$ -graded Hopf algebras in  $\mathcal{C} = {}^H_H\mathcal{YD}$  with bijective antipodes related by a non-degenerate Hopf pairing  $\omega$  satisfying

$$\omega(A(n), B(m)) = 0, \text{ if } n \neq m.$$

We briefly review the relevant definitions.

**Remark 2.18.** (1) Let  $R = \bigoplus_{n \geq 0} R(n)$  be an  $\mathbb{N}_0$ -graded Hopf algebra in  $\mathcal{C}$ . A left module  $(X, \lambda_X)$  over  $R$  in  $\mathcal{C}$  is called rational if for any  $x \in X$ , there exists a natural number  $n_0$  such that  $\lambda(R(n) \otimes x) = 0$  for all  $n \geq n_0$ . We denote the category of left rational modules over  $R$  in  $\mathcal{C}$  by  ${}_R\mathcal{C}_{\text{rat}}$ . A left Yetter-Drinfeld module  $X$  in  ${}^R_R\mathcal{YD}(\mathcal{C})$  is called rational if  $X$  is rational as a module of  $R$  in  $\mathcal{C}$ . We denote the corresponding category by  ${}^R_R\mathcal{YD}(\mathcal{C})_{\text{rat}}$ . This category was proved to be braided monoidal in [20, Lemma 3.3].

(2) We have a braided monoidal equivalence

$$\Pi : {}^R_R\mathcal{YD}(\mathcal{C}) \cong {}^{R\#H}_{R\#H}\mathcal{YD},$$

where  $\mathcal{C} = {}^H_H\mathcal{YD}$  by [20, Remark 2.16]. Now restricting to rational Yetter-Drinfeld modules, we denote the image of  $\Pi$  by  ${}^{R\#H}_{R\#H}\mathcal{YD}_{\text{rat}}$ , which is a monoidal full subcategory of  ${}^{R\#H}_{R\#H}\mathcal{YD}$ .

The following lemma is necessary when constructing  $\Omega$ . We omit certain structural details here for brevity. We use the notation  $\bar{\mathcal{C}}$  to denote the reverse category of  $\mathcal{C}$ .

**Lemma 2.19.** (1) [20, Theorem 3.11] *The functor*

$$\Gamma : \overline{{}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}}} \longrightarrow {}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}, \quad V \mapsto V, \quad (2.12)$$

and the morphisms  $f$  are mapped onto  $f$ , is an equivalence of braided monoidal categories.

(2) [15, Theorem 3.4.15] *We have a braided monoidal equivalence:*

$$(F_{rl}, \alpha) : \mathcal{YD}(\mathcal{C})^A_{A, \text{rat}} \rightarrow {}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}, \quad V \mapsto V \quad (2.13)$$

$$\alpha_{X,Y} = c_{Y,X}^{\mathcal{YD}(\mathcal{C})^A_A} \circ \bar{c}_{X,Y}, \quad (2.14)$$

where morphisms  $f$  are mapped onto  $f$ .

(3) [15, Theorem 3.4.16] *We have a braided strict monoidal equivalence:*

$$F_{lr} : {}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}} \rightarrow \overline{\mathcal{YD}(\mathcal{C})^A_{A, \text{rat}}}, \quad V \mapsto V, \quad (2.15)$$

where morphisms  $f$  are mapped onto  $f$ .

The construction of the functor  $\Omega$  relies on the interplay between the equivalences established above, as illustrated in the following diagram:

$$\begin{array}{ccc} \overline{{}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}}} & \xrightarrow[\cong]{\Gamma} & {}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}} \\ \downarrow \Omega & & \downarrow F_{lr} \\ \overline{{}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}} & \xleftarrow{(F_{rl}, \alpha)} & \mathcal{YD}(\mathcal{C})^A_{A, \text{rat}} \end{array}$$

**Theorem 2.20.** [20, Theorem 3.16] *The following functor is a braided monoidal equivalence:*

$$(\Omega, \beta) : {}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}} \rightarrow {}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}, \quad \text{where } (V, \lambda_B, \delta_B) \mapsto (V, \lambda_A, \delta_A),$$

where morphisms  $f$  are mapped onto  $f$ . Here we omit the definition of  $\lambda_A, \delta_A$  for simplicity. The monoidal structure is given by

$$(\Omega, \beta) : {}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}} \rightarrow {}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}, \quad \beta_{X,Y} = c_{Y,X}^{{}^B_B\mathcal{YD}(\mathcal{C})} \circ \bar{c}_{X,Y}.$$

## 2.5 Nichols algebras

Let  $\mathcal{C}$  be an abelian braided monoidal category. We first recall the definition of Nichols algebras  $\mathcal{B}(V)$  in  $\mathcal{C}$ , where  $V \in \mathcal{C}$  is an arbitrary object.

We denote by

$$T(V) = \bigoplus_{n \geq 0} V^{\otimes n} = \bigoplus_{n \geq 0} (\cdots ((V \otimes V) \otimes V) \cdots) \otimes V$$

the tensor algebra in  $\mathcal{C}$  generated freely by  $V$ . The tensor algebra  $T(V)$  is naturally a graded Hopf algebra in  $\mathcal{C}$ .

**Definition 2.21.** Let  $V \in \mathcal{C}$ . The Nichols algebra of  $V$  is defined to be the quotient Hopf algebra

$$\mathcal{B}(V) := T(V)/I(V).$$

Here,  $I(V)$  is the largest Hopf ideal of  $T(V)$  contained in  $\bigoplus_{n \geq 2} T^n(V)$ .

To facilitate further analysis, we assume  $\mathcal{C}$  is a  $\mathbb{k}$ -linear braided monoidal abelian category. In this setting, the following equivalent definition of the Nichols algebra is more convenient.

**Definition 2.22.** [21, Definition 2.4] For a given object  $V \in \mathcal{C}$ , the Nichols algebra  $\mathcal{B}(V)$  is the unique Hopf algebra in  $\mathcal{C}$  that satisfies the following conditions:

- (1) The Hopf algebra  $\mathcal{B}(V)$  is graded by the non-negative integers.
- (2) The zeroth component of the grading satisfies  $\mathcal{B}(V)_0 = \mathbb{k}$ .
- (3) The first component of the grading satisfies  $\mathcal{B}(V)_1 = V$ , and  $\mathcal{B}(V)$  is generated by  $V$  as an algebra in  $\mathcal{C}$ .
- (4) The subobject of primitive elements of  $\mathcal{B}(V)$  is  $V$ .

Now we return to the case of a coquasi-Hopf algebra  $H$  with a bijective antipode. Let  $V$  be a finite-dimensional object in  ${}^H_H\mathcal{YD}$ . There is a well-defined dual pair

$$\omega : \mathcal{B}(V^*) \otimes \mathcal{B}(V) \rightarrow \mathbb{k}.$$

One may refer to [20, Section 4.1] for more details.

**Corollary 2.23.** [20, Corollary 4.3] For any finite-dimensional  $V \in {}^H_H\mathcal{YD}$ , there exists a braided monoidal equivalence:

$$(\Omega_V, \beta_V) : {}_{\mathcal{B}(V)}^{\mathcal{B}(V)}\mathcal{YD}(\mathcal{C})_{\text{rat}} \longrightarrow {}_{\mathcal{B}(V^*)}^{\mathcal{B}(V^*)}\mathcal{YD}(\mathcal{C})_{\text{rat}}. \quad (2.16)$$

## 2.6 Reflection theory over coquasi-Hopf algebras

In this subsection, we recall the reflection of simple Yetter-Drinfeld modules; this construction is crucial for the present paper.

**Definition 2.24.** Let  $\mathcal{F}_\theta$  denote the class of all  $\theta$ -tuples  $M = (M_1, \dots, M_\theta)$ , where  $M_1, \dots, M_\theta \in {}^H_H\mathcal{YD}$  are finite-dimensional simple Yetter-Drinfeld modules. If  $M \in \mathcal{F}_\theta$ , we define

$$\mathcal{B}(M) := \mathcal{B}(M_1 \oplus \cdots \oplus M_\theta).$$

Two tuples  $M, M' \in \mathcal{F}_\theta$  are called isomorphic, denoted  $M \cong M'$ , if  $M_j \cong M'_j$  in  ${}^H_H\mathcal{YD}$  for all  $j$ . The isomorphism class of  $M \in \mathcal{F}_\theta$  is denoted by  $[M]$ .

For  $1 \leq i \leq \theta$  and  $M \in \mathcal{F}_\theta$ , we say the tuple  $M$  admits the  $i$ -th reflection  $R_i(M)$  if for all  $j \neq i$  there is a natural number  $m_{ij}^M \geq 0$  such that  $\text{ad}(M_i)^{m_{ij}^M}(M_j)$  is a non-zero finite-dimensional subspace of  $\mathcal{B}(M)$ , and  $(\text{ad } M_i)^{m_{ij}^M+1}(M_j) = 0$ .

Assume  $M$  admits the  $i$ -th reflection. Then we set  $R_i(M) = (R_i(M_1)_1, \dots, R_i(M)_\theta)$ , where

$$R_i(M)_j = \begin{cases} M_i^*, & \text{if } j = i, \\ \text{ad}(M_i)^{m_{ij}^M}(M_j), & \text{if } j \neq i. \end{cases}$$

Having defined individual reflections, we now extend this notion to sequences of reflections, which will be important for our study of repeated reflections of tuples.

**Definition 2.25.** Let  $M \in \mathcal{F}_\theta$ , with each  $M_i$  being simple for  $i \in \mathbb{I}$ . For  $l \in \mathbb{N}_0$  and  $i_1, i_2, \dots, i_l \in \mathbb{I}$ .

(1) We say  $M$  admits the reflection sequence  $(i_1, i_2, \dots, i_l)$  if  $l = 0$  or  $M$  admits the  $i_1$ -th reflection and  $R_{i_1}(M)$  satisfies the reflection sequence  $(i_2, i_3, \dots, i_l)$ .

(2) We say  $M$  admits all reflection sequences if  $M$  admits all reflection sequences  $(i_1, i_2, \dots, i_l)$  for all  $l \in \mathbb{N}_0$  and  $i_1, i_2, \dots, i_l \in \mathbb{I}$ .

We also say that a tuple satisfying Definition 2.25(2) admits all reflections. For such a tuple, we write

$$\mathcal{F}_\theta(M) = \{R_{i_l}(\dots R_{i_1}(M) \dots) \mid l \in \mathbb{N}_0, i_1, \dots, i_l \in \mathbb{I}\}.$$

**Remark 2.26.** [20, Lemma 5.9]

(1) Suppose  $M \in \mathcal{F}_\theta$  and  $M$  admits the  $i$ -th reflection for each  $i \in \mathbb{I}$ . We define  $a_{ii}^M = 2$  for all  $1 \leq i \leq \theta$  and define  $a_{ij}^M = -m_{ij}^M$ . Then  $(a_{ij}^M)_{i,j \in \mathbb{I}}$  is a generalized Cartan matrix.

(2) Suppose  $M \cong N$  in  $\mathcal{F}_\theta$ . If  $M$  admits the  $i$ -th reflection for some  $i \in \mathbb{I}$ , then so does  $N$ . Furthermore,  $R_i(M) \cong R_i(N)$  and  $a_{ij}^M = a_{ij}^N$  for each  $j \in \mathbb{I}$ .

Let  $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$ , with each component  $M_i$  being simple for  $i \in \mathbb{I}$ . By [20, Corollary 4.3], we have such a braided tensor equivalence for each  $i \in \mathbb{I}$ .

$$\Omega_i, :_{\mathcal{B}(M_i)}^{\mathcal{B}(M_i)} \mathcal{YD}(\mathcal{C})_{\text{rat}} \longrightarrow_{\mathcal{B}(M_i)}^{\mathcal{B}(M_i^*)} \mathcal{YD}(\mathcal{C})_{\text{rat}}. \quad (2.17)$$

which is induced by the dual pair  $\omega_i : \mathcal{B}(M_i^*) \otimes \mathcal{B}(M_i) \rightarrow \mathbb{k}$ .

We also proved in [20, Lemma 5.10] that  $M$  admitting the  $i$ -th reflection for some  $i \in \mathbb{I}$  is equivalent to the condition that  $\mathcal{B}(M)^{\text{co} \mathcal{B}(M_i)}$  belongs to  $_{\mathcal{B}(M_i) \# H}^{\mathcal{B}(M_i) \# H} \mathcal{YD}_{\text{rat}}$ . The most important result in [19, 20] is as follows.

**Theorem 2.27.** [20, Theorem 5.12] With the above assumptions on  $M$ , if  $M$  admits the  $i$ -th reflection, there exists an isomorphism of Hopf algebras in  $^H_H \mathcal{YD}$ :

$$\Theta_i : \mathcal{B}(R_i(M)) \cong \Omega_i \left( \mathcal{B}(M)^{\text{co} \mathcal{B}(M_i)} \right) \# \mathcal{B}(M_i^*). \quad (2.18)$$

**Corollary 2.28.** [20, Corollary 5.15] Suppose  $M$  admits all reflections. We define the set

$$\mathcal{X} = \{[P] \mid P \in \mathcal{F}_\theta(M)\},$$

and the map

$$r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}, i \times [X] \mapsto [R_i(X)].$$

Then

$$\mathcal{G}(M) = (\mathbb{I}, \mathcal{X}, r, (A^{[X]})_{[X] \in \mathcal{X}}),$$

where  $A^{[X]} = (a_{ij}^{[X]})_{i,j \in \mathbb{I}}$  for all  $[X] \in \mathcal{X}$ , is a semi-Cartan graph.

### 3 One-sided coideal subalgebras of braided Hopf algebras in $^H_H \mathcal{YD}$

In this section, we study one-sided coideal subalgebras of braided Hopf algebras in  $^H_H \mathcal{YD}$ . We construct bijections between these subalgebras under reflections. These results provide the main algebraic tools to decompose Nichols algebras and verify the Cartan graph axioms in the next section.

### 3.1 Basic properties of one-sided coideal subalgebras

**Definition 3.1.** Let  $X$  be a bialgebra in  $\mathcal{C} = {}^H_H\mathcal{YD}$ . A left (respectively, right) coideal subalgebra  $Y$  is a subobject of  $X$ , and an algebra in  $\mathcal{C}$  such that the inclusion map  $Y \subseteq X$  is an algebra morphism, and  $\Delta(Y) \subseteq X \otimes Y$  (respectively,  $\Delta(Y) \subseteq Y \otimes X$ ).

Suppose  $A$  and  $P$  are Hopf algebras in  $\mathcal{C}$ . Let  $\gamma : A \rightarrow P$  be an injective Hopf algebra morphism in  $\mathcal{C}$ , and assume there exists a surjective Hopf algebra morphism  $\pi : P \rightarrow A$  such that  $\pi \circ \gamma = \text{id}_A$ . The triple  $(P, \pi, \gamma)$  is called a Hopf algebra triple.

By [20, Lemma 2.15], there exists a Hopf algebra  $R = P^{\text{co}A}$  in the braided monoidal category  ${}^A_A\mathcal{YD}(\mathcal{C})$  such that  $P \cong R \# A$ . Denoting the comultiplication of  $P$  by  $\Delta_P : P \rightarrow P \otimes P$ ,  $p \mapsto p_1 \otimes p_2$ , there is a well-defined map

$$\vartheta : P \rightarrow R, \quad p \mapsto p_1 \mathcal{S}_A(\pi(p_2)). \quad (3.1)$$

Moreover,  $R$  is a subalgebra of  $P$  in  $\mathcal{C}$  such that  $\Delta_P(R) \subseteq P \otimes R$ . The  $A$ -action on  $R$  is given by  $\text{ad} : A \otimes R \rightarrow R$ , satisfying

$$\text{ad}(a)(r) = \vartheta(ar), \quad (3.2)$$

$$\vartheta(pa) = \vartheta(p)\varepsilon(a) \quad (3.3)$$

for all  $a \in A, r \in R$ , and  $p \in P$ . The  $A$ -coaction  $\delta : R \rightarrow A \otimes R$  is defined by  $\delta(r) = \pi(r_1) \otimes r_2$ . The inverse of the Hopf algebra isomorphism  $P \rightarrow R \# A$ ,  $p \mapsto \vartheta(p_1) \otimes \pi(p_2)$ , is given by the multiplication map.

For our purposes, we introduce the following notation.

**Definition 3.2.** (1) Let

$$S_r^+(P) = \{E \mid E \subseteq P \text{ right coideal subalgebra in } \mathcal{C}, A \subseteq E\},$$

$$S_r(P, X) = \{E \mid E \subseteq P \text{ right coideal subalgebra in } \mathcal{C}, E \subseteq X\},$$

where  $X \subseteq P$  is a subobject in  ${}^H_H\mathcal{YD}$ , and

$$T_r(P) = \{F \mid F \subseteq R \text{ subalgebra in } \mathcal{C}, \Delta_R(F) \subseteq F \otimes R, F \subseteq R \text{ is left } A\text{-submodule}\}.$$

(2) Let

$$S_l^+(P) = \{E \mid E \subseteq P \text{ left coideal subalgebra in } \mathcal{C}, A \subseteq E\},$$

$$S_l(P, X) = \{E \mid E \subseteq P \text{ left coideal subalgebra in } \mathcal{C}, E \subseteq X\},$$

where  $X \subseteq P$  is a subobject in  ${}^H_H\mathcal{YD}$ , and

$$T_l(P) = \{F \mid F \subseteq R \text{ subalgebra in } \mathcal{C}, \Delta_R(F) \subseteq R \otimes F, F \subseteq R \text{ is left } A\text{-subcomodule}\}.$$

**Lemma 3.3.** (1) For all  $E \in S_r^+(P)$ , the multiplication map

$$(E \cap R) \otimes A \rightarrow E$$

is an isomorphism in  ${}^H_H\mathcal{YD}$ .

(2) The map  $S_r^+(P) \rightarrow T_r(P)$ ,  $E \mapsto E \cap R$ , is bijective with inverse given by  $T_r(P) \rightarrow S_r^+(P)$ ,  $F \mapsto FA$ .

(3)  $S_l(P, R) = T_l(P)$ .

*Proof.* (1) This multiplication map is well-defined since  $A \subseteq E$ . Now we consider the map

$$\psi : E \rightarrow (E \cap R) \otimes A, \quad x \mapsto \vartheta(x_1) \otimes \pi(x_2),$$

For  $x \in E \cap R, a \in A$ , we have

$$\psi(xa) = \vartheta(x_1 a_1) \otimes \pi(x_2 a_2) = \vartheta(x_1) \varepsilon(a_1) \otimes \varepsilon(x_2) a_2 = \vartheta(x) \otimes a = x \otimes a.$$

Thus,  $\psi$  is inverse to the multiplication map, making both maps bijective. The map

$$(E \cap R) \otimes A \rightarrow E$$

is a morphism in  ${}^H_H\mathcal{YD}$  because it is induced by the isomorphism  $R \otimes A \rightarrow P$ .

(2) We first show that both maps are well-defined. Let  $E \in S_r^+(P)$ . Then  $E \cap R \subseteq R$  is a subalgebra in  $\mathcal{C}$ . For all  $x \in E \cap R$ , we have  $\Delta_P(x) = x_1 \otimes x_2 \in (E \otimes P) \cap (P \otimes R)$ . Since  $E$  is a right coideal subalgebra of  $P$ , we have  $\Delta_P(e) = e_1 \otimes e_2 \in E \otimes P$  for any  $e \in E$ . By definition,  $\vartheta(e) = e_1 S_A(\pi(e_2))$ . Moreover,  $S_A(\pi(e_2)) \in A$ . Since  $E$  is an algebra in  ${}^H_H\mathcal{YD}$  and  $A \subseteq E$ , we have

$$e_1 S_A(\pi(e_2)) \in E \cdot A \subseteq E \cdot E \subseteq E.$$

Therefore  $\vartheta(E) \subseteq E$ . We can deduce that

$$\Delta_R(x) = \vartheta(x_1) \otimes x_2 \in (E \otimes P) \cap (R \otimes R) = (E \cap R) \otimes R.$$

Let  $c$  be the braiding isomorphism in  ${}^H_H\mathcal{YD}$ . Since  $c(A \otimes E) \subseteq E \otimes A$ , we have

$$\Delta_P(ax) = (a_1 \otimes a_2)(x_1 \otimes x_2) \in AE \otimes AP \subseteq E \otimes P$$

for all  $x \in E, a \in A$ . A direct computation shows that for  $x \in E \cap R$ ,

$$\text{ad}(a)(x) = \vartheta(ax) = (ax)_1 S_A \pi((ax)_2) \in EA \cap R = E \cap R$$

Thus,  $E \cap R$  is an  $A$ -submodule of  $R$ . This shows the map  $S_r^+(P) \rightarrow T_r(P), E \mapsto E \cap R$ , is well-defined.

Let  $F \in T_r(P)$  and  $y \in F$ . Since  $F \subseteq R$  and

$$\Delta_P(y) = \vartheta(y_1) \pi(y_2) \otimes y_3,$$

we obtain  $\Delta_P(F) \subseteq FA \otimes R$ . Thus, for all  $y \in F, a \in A, \Delta_P(ya) = \Delta_P(y) \Delta_P(a) \in FA \otimes P$ .

To show that  $FA \subseteq P$  is a subalgebra, it suffices to prove  $AF \subseteq FA$ . For all  $a \in A, y \in F$ ,

$$\Delta_P(ay) = \Delta_P(a) \Delta_P(y) \in A(FA) \otimes P.$$

Since  $F \subseteq R$  is a  $A$ -submodule, we have

$$ay = \vartheta((ay)_1) \pi((ay)_2) \in \vartheta(A(FA))A \stackrel{(3.3)}{\subseteq} \vartheta((AF)A)A \subseteq \vartheta(AF)A \stackrel{(3.2)}{=} \text{ad}(A)(F)A \subseteq FA.$$

Therefore  $FA$  is a subalgebra of  $P$  and the map  $T_r(P) \rightarrow S_r^+(P), F \mapsto FA$  is well-defined.

If  $E \in S_r^+(P)$ , the following composition is the identity map by (1):

$$S_r^+(P) \rightarrow T_r(P) \rightarrow S_r^+(P), \quad E \mapsto E \cap R \mapsto (E \cap R)A.$$

If  $F \in T_r(P)$ , we have the composition:

$$T_r(P) \rightarrow S_r^+(P) \rightarrow T_r(P), \quad F \mapsto FA \mapsto (FA) \cap R.$$

Since  $F \subseteq R$ , the multiplication maps  $(FA \cap R) \otimes A \rightarrow (FA \cap R)A = FA$  and  $F \otimes A \rightarrow FA$  are bijective. This yields  $F = FA \cap R$ . It follows that the two maps are mutually inverse bijections.

(3) Let  $E \in S_l(P, R)$ , then  $E \subseteq R$  is a subalgebra in  $\mathcal{C}$  and  $\Delta_R(x) = \vartheta(x_1) \otimes x_2 \in R \otimes E$  for  $x \in E$ . Moreover,  $\delta(x) = \pi(x_1) \otimes x_2 \in A \otimes E$ , hence  $E$  is a  $A$ -subcomodule. This shows  $E \in T_l(P)$ .

Conversely, let  $F \in T_l(P)$ , then  $F \subseteq P$  is a subalgebra automatically. For  $x \in F, \Delta_P(x) = \vartheta(x_1) \pi(x_2) \otimes x_3 \in P \otimes F$ , because  $\Delta_R(x) \in R \otimes F$ . Hence,  $F \in S_l(P, R)$ .  $\square$

### 3.2 Comparing one-sided coideal subalgebras via the isomorphism $T$

Let  $K$  be a Hopf algebra in  ${}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}}$  with a bijective antipode. We obtain a Hopf algebra triple  $(P := K\#B, \pi, \gamma)$  in  $\mathcal{C}$ , where  $\pi : P \rightarrow B$  and  $\gamma : B \rightarrow P$ . Let  $\Omega(K)$  denote the corresponding Hopf algebra in  ${}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}$  obtained via the braided monoidal equivalence  $\Omega$ . This yields another Hopf algebra triple  $(Q := \Omega(K)\#A, \bar{\pi}, \bar{\gamma})$  in  $\mathcal{C}$ , where

$$\bar{\pi} : Q \rightarrow A, \bar{\gamma} : A \rightarrow Q.$$

We also obtain a Hopf algebra triple  $(Q^{\text{cop}}, \bar{\pi}, \bar{\gamma})$  in the reverse braided category  $\bar{\mathcal{C}}$ . The antipodes of  $\Omega(K)$  and  $Q$  are bijective because the antipode of  $K$  is bijective.

By definition,  $\Omega(K) = Q^{\text{co}A}$ , which consists of the right coinvariant elements with respect to the projection  $\bar{\pi}$ . Meanwhile, we define

$$L := (Q^{\text{cop}})^{\text{co}\bar{\pi}}.$$

Then  $L$  is an object in  ${}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}$ . The following proposition establishes the relation between  $L$  and  $K$ .

**Proposition 3.4.** *With the above notation, there is an isomorphism of objects in  $\mathcal{C}$*

$$T : L \longrightarrow K, x \mapsto \mathcal{S}_K^{-1}\mathcal{S}_Q(x). \quad (3.4)$$

*In fact, the morphism  $T$  is an isomorphism of Hopf algebras in  ${}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}$*

$$T : L \longrightarrow \Gamma(K^{\text{cop}}). \quad (3.5)$$

*Proof.* By Lemma 2.19 (2) and (3),  $\Omega(K)^{\text{cop}} \in \overline{\mathcal{YD}(\mathcal{C})}_A^A$ . Then we have

$$F_{lr} \circ F_{rl}^{-1}(\Omega(K)^{\text{cop}}) \in {}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}.$$

By [15, Theorem 3.10.6], there is an isomorphism of objects  $T : L \rightarrow K$  in  $\mathcal{C}$ . This map is an isomorphism of Hopf algebras in  ${}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}$ :

$$T : L \longrightarrow F_{lr} \circ F_{rl}^{-1}(\Omega(K)^{\text{cop}})$$

satisfying  $\bar{\gamma} \circ T^{-1} = \mathcal{S}_Q^{-1} \circ \bar{\gamma} \circ \mathcal{S}_{\Omega(K)}$ .

Since  $\Omega$  is a braided monoidal functor, we have  $\Omega(K)^{\text{cop}} \cong \Omega(K^{\text{cop}})$ . Hence

$$F_{lr} \circ F_{rl}^{-1}(\Omega(K)^{\text{cop}}) \cong F_{lr} \circ F_{rl}^{-1}(\Omega(K^{\text{cop}})) = \Gamma(K^{\text{cop}}).$$

Therefore,  $T : L \rightarrow \Gamma(K^{\text{cop}})$  is a Hopf algebra isomorphism. For any  $x \in K$ , we compute  $T^{-1}(x) = \mathcal{S}_Q^{-1} \circ \bar{\gamma} \circ \mathcal{S}_{\Omega(K)}(x) = \mathcal{S}_Q^{-1} \circ \bar{\gamma} \circ \mathcal{S}_K(x) = \mathcal{S}_Q^{-1} \circ \mathcal{S}_K(x)$  using the identification  $\mathcal{S}_{\Omega(K)} = \mathcal{S}_K$ . Thus,  $T(x) = \mathcal{S}_K^{-1}\mathcal{S}_Q(x)$  for all  $x \in L$ .  $\square$

Let  $(P := K\#B, \pi, \gamma)$ ,  $(Q := \Omega(K)\#A, \bar{\pi}, \bar{\gamma})$  and  $L$  be defined as above. Define

$$\bar{K} := {}^{\text{co}\bar{\pi}} Q.$$

to be the space of left coinvariant elements with respect to the projection  $\bar{\pi}$ . The isomorphism  $T$  induces bijection between the one-sided coideal subalgebras of  $P$  and  $Q$ .

**Proposition 3.5.** (1) *With the above notation, there is a bijection*

$$\mathcal{S}_l(Q^{\text{cop}}, L) \longrightarrow \mathcal{S}_r^+(P), E \mapsto T(E)B$$

with inverse given by  $E \mapsto T^{-1}(E \cap K)$ . Furthermore, the multiplication map

$$T(E) \otimes B \rightarrow T(E)B$$

is bijective for all  $E \in S_l(Q^{\text{cop}}, L)$ .

(2) There is a bijection

$$S_l(P, K) \longrightarrow S_l^+(Q), E \mapsto T^{-1}(E)A$$

with inverse given by  $E \mapsto T(E \cap L)$ . Furthermore, the multiplication map

$$T^{-1}(E) \otimes A \rightarrow T^{-1}(E)A$$

is bijective for all  $E \in S_l(P, K)$ .

*Proof.* (1) By definition, the object  $\bar{K} \cong L$  in  $\mathcal{C}$ , hence

$$S_r(Q, \bar{K}) = S_l(Q^{\text{cop}}, L).$$

By Lemma 3.3 (3),  $S_l(Q^{\text{cop}}, L) = T_l(Q^{\text{cop}})$ . We now establish the bijection between  $T_l(Q^{\text{cop}})$  and  $S_r^+(P)$ .

By definition,

$$T_l(Q^{\text{cop}}) = \{E \mid E \subseteq L \text{ subalgebra in } \bar{\mathcal{C}}, \Delta_L(E) \subseteq L \otimes E, E \text{ is } A^{\text{cop}}\text{-subcomodule}\}.$$

We apply the Hopf algebra isomorphism  $T$ , which is an  $A^{\text{cop}}$ -colinear map in  $\bar{\mathcal{C}}$ . For any  $E \in T_l(Q^{\text{cop}})$ ,  $T(E)$  is a  $A^{\text{cop}}$ -subcomodule of  $\Gamma(K^{\text{cop}})$  in  $\bar{\mathcal{C}}$  and is a subalgebra of  $K$  in  $\mathcal{C}$ . Since  $D_2 : {}^{A^{\text{cop}}}\bar{\mathcal{C}} \rightarrow {}_B\mathcal{C}_{\text{rat}}$  is a braided monoidal equivalence by [20, Proposition 3.10],  $T(E)$  is a  $B$ -submodule of  $K$ . Finally,

$$\bar{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})} \circ \Delta_K(T(E)) = \Delta_{\Gamma(K^{\text{cop}})}(T(E)) \subseteq K \otimes T(E).$$

Here  $\bar{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})}$  denotes the reverse braiding of  $c_{K,K}^{B\mathcal{YD}(\mathcal{C})}$ . Since  $K$  has a bijective antipode,  $\bar{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})}$  is an isomorphism by [15, Proposition 3.4.8]. Now

$$\bar{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})} \circ \Delta_K(T(E)) \subseteq K \otimes T(E) \Leftrightarrow \Delta_K(T(E)) \subseteq T(E) \otimes K.$$

Consequently,  $T(E) \in T_r(P)$ . The inverse mapping can be established similarly. Therefore, the isomorphism  $T$  establishes a bijection between  $T_l(Q^{\text{cop}})$  and  $T_r(P)$ . By Lemma 3.3(2), the map

$$T_r(P) \longrightarrow S_r^+(P), F \mapsto FB$$

is bijective. Therefore, the map  $S_l(Q^{\text{cop}}, L) \longrightarrow S_r^+(P)$ ,  $E \mapsto T(E)B$  is bijective with inverse  $E \mapsto T^{-1}(E \cap K)$ .

The bijectivity of the multiplication map  $T(E) \otimes B \rightarrow T(E)B$  for all  $E \in S_l(Q^{\text{cop}}, L)$  follows directly from Lemma 3.3 (1).

(2) By a similar argument, there is a bijection

$$T_r(Q^{\text{cop}}) \longrightarrow T_l(P), E \mapsto T(E).$$

According to Lemma 3.3 (2), (3), there is a bijective map  $T_r(Q^{\text{cop}}) \rightarrow S_r^+(Q^{\text{cop}})$  given by  $F \mapsto FA$ , and  $T_l(P) = S_l(P, K)$ . Since  $S_r^+(Q^{\text{cop}}) = S_l^+(Q)$ , the map  $S_l(P, K) \rightarrow S_l^+(Q)$  is bijective.

For any  $E \in S_l(P, K)$ , since  $T^{-1}(E) \in T_r(Q^{\text{cop}})$ , the multiplication map

$$T^{-1}(E) \otimes A \rightarrow T^{-1}(E)A$$

is bijective by Lemma 3.3(1). □

### 3.3 One-sided coideal subalgebras of Nichols algebras

In this subsection, we fix a tuple  $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$ , where each  $M_i$  is a finite-dimensional simple Yetter-Drinfeld module in  ${}^H_H\mathcal{YD}$ . We endow  $\mathcal{B}(M)$  with an  $\mathbb{N}_0^\theta$ -grading by setting  $\deg(M_i) = \alpha_i$ . Consider the following maps:

$$\pi_i : \mathcal{B}(M) \rightarrow \mathcal{B}(M_i), \quad \gamma_i : \mathcal{B}(M_i) \rightarrow \mathcal{B}(M).$$

such that  $\pi_i \circ \gamma_i = \text{id}_{\mathcal{B}(M_i)}$ . Let

$$K_i^M = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)}, \quad L_i^M = {}^{\text{co } \mathcal{B}(M_i)} \mathcal{B}(M)$$

be the right and left coinvariant spaces with respect to  $\pi_i$ . A direct computation shows that

$$K_i^M = \mathcal{S}_{\mathcal{B}(M)}(L_i^M). \quad (3.6)$$

For any  $x \in \mathcal{B}(M)$ ,

$$\begin{aligned} (\pi_i \otimes \text{id})(\Delta(\mathcal{S}_{\mathcal{B}(M)})(x)) &= (\pi_i \otimes \text{id}) \circ (\mathcal{S}_{\mathcal{B}(M)} \otimes \mathcal{S}_{\mathcal{B}(M)}) \circ c_{\mathcal{B}(M), \mathcal{B}(M)}(\Delta(x)) \\ &= c_{\mathcal{B}(M), \mathcal{B}(M_i)} \circ (\mathcal{S}_{\mathcal{B}(M)} \otimes \mathcal{S}_{\mathcal{B}(M_i)}) \circ (\text{id} \otimes \pi_i)(\Delta(x)) \end{aligned}$$

Hence, the condition  $x \in K_i^M$  is equivalent to  $\mathcal{S}_{\mathcal{B}(M)}(x) \in L_i^M$ .

**Lemma 3.6.** *For each  $i \in \mathbb{I}$ .*

(1) *Let  $E \subseteq \mathcal{B}(M)$  be an  $\mathbb{N}_0^\theta$ -graded right coideal subalgebra in  ${}^H_H\mathcal{YD}$ . Then*

$$E \subseteq L_i^M \iff M_i \not\subseteq E.$$

(2) *Let  $F \subseteq \mathcal{B}(M)$  be an  $\mathbb{N}_0^\theta$ -graded left coideal subalgebra in  ${}^H_H\mathcal{YD}$ . Then*

$$F \subseteq K_i^M \iff M_i \not\subseteq F.$$

*Proof.* We only prove (1), as (2) is dual. Assume  $E \subseteq L_i^M$  and  $M_i \subseteq E$ ; it follows that  $M_i \subseteq L_i^M$ , which forces  $M_i = 0$ , yielding a contradiction.

Now we assume  $M_i \not\subseteq E$ . Since  $M_i$  is simple,  $M_i \cap E$  must be zero. Since  $E$  is an  $\mathbb{N}_0^\theta$ -graded right coideal subalgebra and  $\pi_i$  is  $\mathbb{N}_0^\theta$ -graded morphism, we deduce that  $\pi_i(E)$  is a right coideal subalgebra of  $\mathcal{B}(M_i)$ . Assume  $\pi_i(E) \neq \mathbb{k}1$ . Because  $M_i$  is simple,  $\pi_i(E) \cap P(\mathcal{B}(M_i)) \neq 0$ . Furthermore, the simplicity of  $M_i$  and the fact that  $P(\mathcal{B}(M_i)) = M_i$  jointly force  $M_i \subseteq \pi_i(E)$ . This contradicts  $M_i \cap E = 0$ . Therefore  $\pi_i(E) = \mathbb{k}1$ . Now for any  $x \in E$ ,

$$(\pi_i \otimes \text{id})\Delta(x) = \pi_i(x_1) \otimes x_2 = 1 \otimes x.$$

Hence  $E \subseteq L_i^M$ . □

**Definition 3.7.** *For each  $i \in \mathbb{I}$ .*

(1) *We define*

$$\begin{aligned} \mathcal{K}(M) &= \{E \mid E \subseteq \mathcal{B}(M) \text{ is an } \mathbb{N}_0^\theta\text{-graded right coideal subalgebra in } {}^H_H\mathcal{YD}\}, \\ \mathcal{K}_i^+(M) &= \{E \mid E \in \mathcal{K}(M), M_i \subseteq E\}, \\ \mathcal{K}_i^-(M) &= \{E \mid E \in \mathcal{K}(M), M_i \not\subseteq E\}. \end{aligned}$$

(2) *Dually, we define*

$$\begin{aligned} \mathcal{L}(M) &= \{F \mid F \subseteq \mathcal{B}(M) \text{ is an } \mathbb{N}_0^\theta\text{-graded left coideal subalgebra in } {}^H_H\mathcal{YD}\}, \\ \mathcal{L}_i^+(M) &= \{F \mid F \in \mathcal{L}(M), M_i \subseteq F\}, \\ \mathcal{L}_i^-(M) &= \{F \mid F \in \mathcal{L}(M), M_i \not\subseteq F\}. \end{aligned}$$

From now on, we assume that  $M$  admits the  $i$ -th reflection. Recall that

$$R_i(M) = (R_i(M)_1, \dots, R_i(M)_\theta),$$

where

$$R_i(M)_j = \begin{cases} M_i^*, & \text{if } j = i, \\ \text{ad}(M_i)^{m_{ij}^M}(M_j), & \text{if } j \neq i. \end{cases}$$

For simplicity, we denote  $R_i(M_j) = \widetilde{M}_j$  for each  $j$ .

**Lemma 3.8.** (1)

$$\text{ad}(M_i^*)^n(\widetilde{M}_j) = \begin{cases} 0, & \text{if } n > m_{ij}^M, \\ \text{ad}(M_i)^{m_{ij}^M - n}(M_j), & \text{if } 0 \leq n \leq m_{ij}^M. \end{cases}$$

(2) The Nichols algebra  $\mathcal{B}(R_i(M)) \cong \Omega(K_i^M) \# \mathcal{B}(M_i^*)$  is an  $\mathbb{N}_0^\theta$ -graded Hopf algebra in  ${}^H_H\mathcal{YD}$  with

$$\deg(x \otimes y) = s_i^M(\deg^{\mathcal{B}(M)}(x) + \deg(y))$$

for all homogeneous elements  $x \in K_i^M$  and  $y \in \mathcal{B}(M_i^*)$ , where  $\mathcal{B}(M_i^*)$  is a  $\mathbb{Z}^\theta$ -graded algebra with  $\deg(M_i^*) = -\alpha_i$ , and  $\deg^{\mathcal{B}(M)}$  denotes the degree in  $\mathcal{B}(M)$ .

*Proof.* (1) The first statement follows from [20, Lemma 5.11].

(2) According to [20, Lemmas 4.6 and 4.8],  $\mathcal{B}(M_i^*)$  is a  $\mathbb{Z}^\theta$ -graded Hopf algebra with  $\deg(M_i^*) = -\alpha_i$  and  $\Omega(K_i^M)$  is an  $\mathbb{Z}^\theta$ -graded Hopf algebra with  $\Omega(K_i^M)(n) = K_i^M(-n)$  for  $n \in \mathbb{Z}^\theta$ . Under this grading,  $\Omega_i(K_i^M) \# \mathcal{B}(M_i^*)$  is an  $\mathbb{Z}^\theta$ -graded Hopf algebra in  ${}^H_H\mathcal{YD}$ . We equip  $\mathcal{B}(R_i(M))$  with a new grading by shifting the degree via  $s_i^M$ . Explicitly, for  $x \in K_i^M$  and  $y \in \mathcal{B}(M_i^*)$ , we define

$$\deg(x \# y) = s_i^M(\deg^{\mathcal{B}(M)}(x) + \deg(y)).$$

Hence  $\deg(\widetilde{M}_i) = s_i^M(-\alpha_i) = \alpha_i$ , and for all  $j \neq i$ ,

$$\deg(\widetilde{M}_j) = s_i^M(\alpha_j - a_{ij}^M \alpha_i) = \alpha_j.$$

Hence  $\mathcal{B}(R_i(M))$  is an  $\mathbb{N}_0^\theta$ -graded Hopf algebra with  $\deg(\widetilde{M}_j) = \alpha_j$  for all  $j \in \mathbb{I}$  under this new grading. □

We consider the action of the isomorphism  $T$  mentioned in Proposition 3.4. The Hopf algebra isomorphism  $T$  restricts to an algebra isomorphism in  ${}^H_H\mathcal{YD}$ :

$$T_i^M : L_i^{R_i(M)} = {}^{\text{co} \mathcal{B}(M_i^*)}(\Omega_i(K_i^M) \# \mathcal{B}(M_i^*)) \xrightarrow{\sim} K_i^M. \quad (3.7)$$

**Lemma 3.9.** (1) With the above notation, for all  $j \neq i \in \mathbb{I}$ ,  $0 \leq n \leq m_{ij}^M$ , and  $x \in \text{ad}(M_i^*)^n(\widetilde{M}_j)$ ,

$$\begin{aligned} T_i^M(\mathcal{S}_{\mathcal{B}(R_i(M))}^{-1}(x)) &= -x, \\ T_i^M(\mathcal{S}_{\mathcal{B}(R_i(M))}^{-1}(\text{ad}(M_i^*)^n(\widetilde{M}_j))) &= \text{ad}(M_i)^{m_{ij}^M - n}(M_j). \end{aligned}$$

(2)  $T_i^M$  is an isomorphism of  $\mathbb{N}_0^\theta$ -graded objects. In particular

$$(T_i^M(L_i^{R_i(M)}(\alpha))) = K_i^M(s_i^{R_i(M)}(\alpha)),$$

where  $\alpha \in \mathbb{N}_0^\theta$ .

*Proof.* (1) We have

$$\mathrm{ad}(M_i^*)^n(\widetilde{M}_j) = \begin{cases} 0, & \text{if } n > m_{ij}^M, \\ \mathrm{ad}(M_i)^{m_{ij}^M-n}(M_j) & \text{if } 0 \leq n \leq m_{ij}^M. \end{cases}$$

Since  $x \in \mathrm{ad}(M_i^*)^n(\widetilde{M}_j) = \mathrm{ad}(M_i)^{m_{ij}^M-n}(M_j) \subseteq \mathrm{ad}(\mathcal{B}(M_i))(M_j)$ , we deduce that  $x$  is primitive in  $K_i^M$  by [20, Theorem 5.5].

Set  $y = \mathcal{S}_{\mathcal{B}(R_i(M))}^{-1}(x)$ . Then  $y \in L_i^{R_i(M)}$  by (3.6). By the definition of  $T$ ,

$$T_i^M(y) = \mathcal{S}_{K_i^M}^{-1} \mathcal{S}_{\mathcal{B}(R_i(M))}(y) = \mathcal{S}_{K_i^M}^{-1}(x) = -x.$$

The second equation holds because  $T_i^M$  is an isomorphism.

(2) (2) Let  $x \in \mathrm{ad}(M_i^*)^n(\widetilde{M}_j)$ . Under the  $\mathbb{N}_0^\theta$ -grading of  $\mathcal{B}(R_i(M))$ , we have  $\deg(x) = n\alpha_i + \alpha_j$ . Since  $\mathcal{S}_{\mathcal{B}(R_i(M))}$  preserves this grading,  $\deg(y) = n\alpha_i + \alpha_j$ . By (1),

$$\deg(T_i^M(y)) = (m_{ij}^M - n)\alpha_i + \alpha_j = s_i^M(\deg(y)) \in \mathbb{N}_0^\theta.$$

Since  $M$  admits the  $i$ -th reflection,  $R_i(M)$  does as well, and  $s_i^M = s_i^{R_i(M)}$  by [20, Corollary 5.15]. Hence, for every  $\alpha \in \mathbb{N}_0^\theta$ ,

$$\deg(T_i^M(L_i^{R_i(M)}(\alpha))) = s_i^{R_i(M)}(\alpha).$$

Therefore,

$$T_i^M(L_i^{R_i(M)}(\alpha)) = K_i^M(s_i^{R_i(M)}(\alpha)).$$

□

**Proposition 3.10.** Assume  $M$  admits the  $i$ -th reflection for some  $i \in \mathbb{I}$ .

(1) The map

$$t_i^M : \mathcal{K}_i^-(R_i(M)) \longrightarrow \mathcal{K}_i^+(M), \quad E \longmapsto T_i^M(E)\mathcal{B}(M_i),$$

is bijective with inverse given by  $E \longmapsto (T_i^M)^{-1}(E \cap K_i^M)$ . Furthermore, the following multiplication map is bijective for all  $E \in \mathcal{K}_i^-(R_i(M))$ .

$$T_i^M(E) \otimes \mathcal{B}(M_i) \longrightarrow T_i^M(E)\mathcal{B}(M_i).$$

(2) The map

$$\mathfrak{t}_i^M : \mathcal{L}_i^+(R_i(M)) \longrightarrow \mathcal{L}_i^-(M), \quad F \longmapsto T_i^M(F \cap L_i^{R_i(M)}),$$

is bijective with inverse given by  $F \longmapsto (T_i^M)^{-1}(F)\mathcal{B}(M_i^*)$ . Furthermore, the following multiplication map is bijective for all  $F \in \mathcal{L}_i^-(M)$ ,

$$(T_i^M)^{-1}(F) \otimes \mathcal{B}(M_i^*) \longrightarrow (T_i^M)^{-1}(F)\mathcal{B}(M_i^*).$$

*Proof.* (1) We apply Proposition 3.5(1) by setting  $P = \mathcal{B}(M)$ ,  $Q = \mathcal{B}(R_i(M))$ ,  $\overline{K} = L_i^{R_i(M)}$ ,  $K = K_i^M$  and  $B = \mathcal{B}(M_i)$ . We obtain a bijection:

$$S_r(\mathcal{B}(R_i(M)), L_i^{R_i(M)}) \longrightarrow S_r^+(\mathcal{B}(M)), \quad E \mapsto T_i^M(E)\mathcal{B}(M_i)$$

with inverse given by  $E \mapsto (T_i^M)^{-1}(E \cap K_i^M)$ . Since  $T_i^M$  preserves  $\mathbb{N}_0^\theta$ -grading, the bijection still holds when restricting to  $\mathbb{N}_0^\theta$ -graded right coideal subalgebras. By Lemma 3.6,  $E \subseteq L_i^{R_i(M)} \iff M_i^* \not\subseteq E$ . Thus, the bijection restricts to the desired bijection

$$t_i^M : \mathcal{K}_i^-(R_i(M)) \longrightarrow \mathcal{K}_i^+(M).$$

The bijectivity of the multiplication map

$$T_i^M(E) \otimes \mathcal{B}(M_i) \longrightarrow T_i^M(E)\mathcal{B}(M_i).$$

for all  $E \in \mathcal{K}_i^-(R_i(M))$  follows from Proposition 3.5(1).

(2) It follows from Proposition 3.5(2) and similar argument to (1). □

## 4 The associated Cartan graph

In this section, we prove that the associated semi-Cartan graph  $\mathcal{G}(M)$  is a Cartan graph by verifying the equivalent axioms (CG3') and (CG4').

### 4.1 Proof of (CG3')

Let  $l \in \mathbb{N}_0$  and  $i_1, \dots, i_l \in \mathbb{I}$ , and assume that  $M$  admits the reflection sequence  $(i_1, \dots, i_l)$ . We construct the simple Yetter–Drinfeld modules  $M_{\beta_k}$  associated with the corresponding roots and the right coideal subalgebra  $E^M(i_1, \dots, i_l)$ .

For simplicity, we write

$$R_{(i_1, \dots, i_k)}(M) = R_{i_k}(\dots R_{i_1}(M))$$

inductively for  $1 \leq k \leq l$ . For clarity, we denote  $R_{()} = M$ . Now we consider the following two maps:

$$\begin{aligned} T_{i_1}^M : L_{i_1}^{R_{(i_1)}(M)} &\xrightarrow{\sim} K_{i_1}^M, \\ T_{i_2}^{R_{(i_1)}(M)} : L_{i_2}^{R_{(i_1, i_2)}(M)} &\xrightarrow{\sim} K_{i_2}^{R_{(i_1)}(M)}. \end{aligned}$$

To compose these maps, the domain must be restricted to

$$(T_{i_2}^{R_{(i_1)}(M)})^{-1}(K_{i_2}^{R_{(i_1)}(M)} \cap L_{i_1}^{R_{(i_1)}(M)}).$$

Therefore  $T_{i_1}^M \circ T_{i_2}^{R_{(i_1)}(M)}$  is well-defined. We apply similar restrictions when considering the composition of  $t_i^M$  and  $\mathfrak{t}_i^M$ .

**Remark 4.1.** With above assumptions, for  $1 \leq k \leq l$ , we define inductively

$$\begin{aligned} L_{(i_1, \dots, i_k)}^M &= (T_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)})^{-1} \left( K_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} \cap L_{(i_1, \dots, i_{k-1})}^M \right), \\ T_{(i_1, \dots, i_k)}^M &= T_{i_1}^M \circ T_{i_2}^{R_{(i_1)}(M)} \circ \dots \circ T_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : L_{(i_1, \dots, i_k)}^M \longrightarrow K_{i_1}^M, \\ \mathcal{K}_{(i_1, \dots, i_k)}^-(R_{(i_1, \dots, i_k)}(M)) &= (t_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)})^{-1} \left( \mathcal{K}_{i_k}^+(R_{(i_1, \dots, i_{k-1})}(M)) \cap \mathcal{K}_{(i_1, \dots, i_{k-1})}^-(R_{(i_1, \dots, i_{k-1})}(M)) \right), \\ t_{(i_1, \dots, i_k)}^M &= t_{i_1}^M \circ \dots \circ t_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : \mathcal{K}_{(i_1, \dots, i_k)}^-(R_{(i_1, \dots, i_k)}(M)) \longrightarrow \mathcal{K}_{i_1}^+(M), \\ \mathcal{L}_{(i_1, \dots, i_k)}^+(R_{(i_1, \dots, i_k)}(M)) &= (\mathfrak{t}_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)})^{-1} \left( \mathcal{L}_{i_k}^-(R_{(i_1, \dots, i_{k-1})}(M)) \cap \mathcal{L}_{(i_1, \dots, i_{k-1})}^+(R_{(i_1, \dots, i_{k-1})}(M)) \right), \\ \mathfrak{t}_{(i_1, \dots, i_k)}^M &= \mathfrak{t}_{i_1}^M \circ \dots \circ \mathfrak{t}_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : \mathcal{L}_{(i_1, \dots, i_k)}^+(R_{(i_1, \dots, i_k)}(M)) \longrightarrow \mathcal{L}_{i_1}^-(M). \end{aligned}$$

By convention, we set

$$L_{()}^M = \mathcal{B}(M), \mathcal{K}_{()}^-(M) = \mathcal{K}(M), T_{()}^M = \text{id}, t_{()}^M = \text{id}, \mathcal{L}_{()}^+ = \mathcal{L}(M), \mathfrak{t}_{()}^M = \text{id}.$$

From now on, we assume  $M$  admits all reflections. Let the reflection sequence  $(i_1, \dots, i_l)$  be  $[M]$ -reduced in the semi-Cartan graph  $\mathcal{G}(M)$ . Note that in this case, the reflection sequence  $(i_k, \dots, i_l)$  is  $[R_{(i_1, \dots, i_{k-1})}(M)]$ -reduced in  $\mathcal{G}(R_{(i_1, \dots, i_{k-1})}(M))$  for all  $1 \leq k \leq l$  by Remark 2.7 (2). For all  $1 \leq k \leq l$ , let

$$\beta_k = \text{id}_{[M]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}),$$

The following observation is immediate and serves as the base case for our induction. Let  $l = 1$ , then  $\beta_1 = \alpha_{i_1}$ .

- $\mathbb{k}1 \in \mathcal{K}_{i_1}^-(R_{i_1}(M))$ , it follows from the fact that  $M_{i_1}^* \not\subseteq \mathbb{k}1$ . Then we have  $E^M(i_1) = t_{i_1}^M(\mathbb{k}1) = \mathcal{B}(M_{\beta_1})$ .
- $M_{\beta_1} = M_{i_1}$  is a finite-dimensional simple object of degree  $\beta_1 = \alpha_{i_1}$ .

For  $l > 1$ , the preceding argument corresponds to the first reflection step from  $R_{(i_1, \dots, i_{l-1})}(M)$  to  $R_{(i_1, \dots, i_l)}(M)$ . By induction, we assume the following:

- (i) For any  $2 \leq k \leq l$ , if we define  $\gamma_k = \text{id}_{[R_{i_1}(M)]^{s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k})}}$ , then  $\gamma_2, \dots, \gamma_l$  are pairwise distinct non-zero elements of  $\mathbb{N}_0^\theta$  (note that if  $l = 1$  this assumption is trivial).
- (ii) For any  $2 \leq k \leq l$ ,  $R_{(i_1, \dots, i_{k-1})}(M)_{i_k} \subseteq L_{(i_2, \dots, i_{k-1})}^{R_{i_1}(M)}$  and  $\mathbb{k}1 \in \mathcal{K}_{(i_2, \dots, i_l)}^-(R_{(i_1, \dots, i_l)}(M))$ . Therefore the following two objects are well-defined.

$$\begin{aligned}\tilde{M}_{\gamma_k} &:= T_{(i_2, \dots, i_{k-1})}^{R_{i_1}(M)}(R_{(i_1, \dots, i_{k-1})}(M)_{i_k}) \\ E^{R_{i_1}(M)}(i_2, \dots, i_l) &:= t_{(i_2, \dots, i_l)}^{R_{i_1}(M)}(\mathbb{k}1)\end{aligned}$$

for all  $2 \leq k \leq l$ . (Note that for  $k = 1$ , it reduces to  $M_{i_1} \subseteq \mathcal{B}(M)$ , which is trivial.)

- (iii) For any  $2 \leq k \leq l$ ,  $\tilde{M}_{\gamma_k} \subseteq E^{R_{i_1}(M)}(i_2, \dots, i_l)$  is a finite-dimensional simple subobject in  ${}^H_H\mathcal{YD}$  of degree  $\gamma_k$ .
- (iv) The multiplication map

$$(\cdots (\mathcal{B}(\tilde{M}_{\gamma_l}) \otimes \mathcal{B}(\tilde{M}_{\gamma_{l-1}})) \cdots) \otimes \mathcal{B}(\tilde{M}_{\gamma_2}) \longrightarrow E^{R_{i_1}(M)}(i_2, \dots, i_l)$$

is an isomorphism of  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ .

With the above notation and hypothesis, we establish the following proposition.

**Proposition 4.2.** *Let  $M \in \mathcal{F}_\theta$  be a tuple of simple objects admitting all reflections. Let*

$$\kappa = (i_1, \dots, i_l)$$

*be an  $[M]$ -reduced sequence, and put*

$$\beta_k = \text{id}_{[M]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}), \quad 1 \leq k \leq l.$$

*Then the following statements hold.*

- (1) *For any  $1 \leq k \leq l$ ,  $R_{(i_1, \dots, i_{k-1})}(M)_{i_k} \subseteq L_{(i_1, \dots, i_{k-1})}^M$  and  $\mathbb{k}1 \in \mathcal{K}_{(i_1, \dots, i_l)}^-(R_{(i_1, \dots, i_l)}(M))$ . Therefore the following two objects are well-defined.*

$$\begin{aligned}M_{\beta_k} &:= T_{(i_1, \dots, i_{k-1})}^M(R_{(i_1, \dots, i_{k-1})}(M)_{i_k}), \\ E^M(i_1, \dots, i_l) &:= t_{(i_1, \dots, i_l)}^M(\mathbb{k}1)\end{aligned}$$

*for all  $1 \leq k \leq l$ .*

- (2) *The elements  $\beta_1, \dots, \beta_l$  are pairwise distinct non-zero elements of  $\mathbb{N}_0^\theta$ .*
- (3) *The multiplication map*

$$(\cdots (\mathcal{B}(M_{\beta_l}) \otimes \mathcal{B}(M_{\beta_{l-1}})) \cdots) \otimes \mathcal{B}(M_{\beta_1}) \longrightarrow E^M(i_1, \dots, i_l)$$

*is an isomorphism of  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ .*

- (4) *For any  $1 \leq k \leq l$ ,  $M_{\beta_k} \subseteq E^M(i_1, \dots, i_l)$  is a finite-dimensional simple subobject in  ${}^H_H\mathcal{YD}$  of degree  $\beta_k$ .*

*Proof.* Since the sequence  $(i_1, \dots, i_l)$  is  $[M]$ -reduced, it follows from the definition that

$$\alpha_{i_1} \notin \Lambda^{R_{i_1}(M)}(i_2, \dots, i_l) = \{\gamma_2, \dots, \gamma_l\}.$$

For  $2 \leq k \leq l$ , assumption (ii) and (iii) say  $\tilde{M}_{\gamma_k}$  has degree  $\gamma_k$ . By assumption (iv), every degree occurring in  $E^{R_{i_1}(M)}(i_2, \dots, i_l)$  is a sum of  $\gamma_2, \dots, \gamma_l$ . Since  $\alpha_{i_1} \neq \gamma_k$  for  $2 \leq k \leq l$ , we deduce that

$$R_{i_1}(M)_{i_1} \not\subseteq E^{R_{i_1}(M)}(i_2, \dots, i_l).$$

Therefore by definition of  $\mathcal{K}_{i_1}^-(R_{i_1}(M))$ , we have

$$E^{R_{i_1}(M)}(i_2, \dots, i_l) \in \mathcal{K}_{i_1}^-(R_{i_1}(M)). \quad (4.1)$$

That is,  $t_{(i_2, \dots, i_l)}^{R_{i_1}(M)}(\mathbb{k}1) \in \mathcal{K}_{i_1}^-(R_{i_1}(M))$ , hence  $\mathbb{k}1 \in \mathcal{K}_{(i_1, \dots, i_l)}^-(R_{(i_1, \dots, i_l)}(M))$  and  $E^M(i_1, \dots, i_l) := t_{(i_1, \dots, i_l)}^M(\mathbb{k}1)$  is well-defined. By Lemma 3.6,

$$\tilde{M}_{i_1} = R_{i_1}(M)_{i_1} \not\subseteq E^{R_{i_1}(M)}(i_2, \dots, i_l) \Leftrightarrow E^{R_{i_1}(M)}(i_2, \dots, i_l) \subseteq L_{i_1}^{R_{i_1}(M)}.$$

By assumption (iii),  $\tilde{M}_{\gamma_k} \subseteq L_{i_1}^{R_{i_1}(M)}$  for any  $2 \leq k \leq l$ , hence

$$T_{i_1}^M(\tilde{M}_{\gamma_k}) = T_{i_1}^M(T_{(i_2, \dots, i_{k-1})}^{R_{i_1}(M)}(R_{(i_1, \dots, i_{k-1})}(M)_{i_k})) \subseteq K_{i_1}^M,$$

which implies  $R_{(i_1, \dots, i_{k-1})}(M)_{i_k} \subseteq L_{(i_1, \dots, i_{k-1})}^M$  and  $M_{\beta_k} := T_{(i_1, \dots, i_{k-1})}^M(R_{(i_1, \dots, i_{k-1})}(M)_{i_k})$  is well-defined. So we have proved (1).

By Lemma 3.9(2),

$$\deg(T_{i_1}^M(L_{i_1}^{R_{i_1}(M)}(\alpha))) = s_{i_1}^{R_{i_1}(M)}(\alpha).$$

Hence the elements

$$\beta_k = s_{i_1}^{R_{i_1}(M)}(\gamma_k) \in \mathbb{N}_0^\theta, \quad 2 \leq k \leq l,$$

are pairwise distinct by assumption (i). If  $\beta_1 = \beta_i$  for some  $2 \leq i \leq l$ , then  $-\beta_1 = \gamma_i \in \mathbb{N}_0^\theta$ , which yields a contradiction. This establishes (2).

Recall (4.1),  $E^{R_{i_1}(M)}(i_2, \dots, i_l) \in \mathcal{K}_{i_1}^-(R_{i_1}(M))$ , hence by Proposition 3.10, we have a bijective map:

$$T_{i_1}^M(E^{R_{i_1}(M)}(i_2, \dots, i_l)) \otimes \mathcal{B}(M_{i_1}) \rightarrow t_{i_1}^M(E^{R_{i_1}(M)}(i_2, \dots, i_l)) = E^M(i_1, \dots, i_l).$$

The map  $T$  itself is an algebra isomorphism. By assumption (iv), we have a bijective map

$$(\dots (T_{i_1}^M(\mathcal{B}(\tilde{M}_{\gamma_l})) \otimes T_{i_1}^M(\mathcal{B}(\tilde{M}_{\gamma_{l-1}}))) \dots) \otimes \mathcal{B}(M_{i_1}) \longrightarrow E^M(i_1, \dots, i_l).$$

By the above construction, for  $2 \leq k \leq l$ , we have  $M_{\beta_k} = T_{i_1}^M(\tilde{M}_{\gamma_k})$ . Then

$$(\dots (\mathcal{B}(M_{\beta_l}) \otimes \mathcal{B}(M_{\beta_{l-1}})) \dots) \otimes \mathcal{B}(M_{\beta_1}) \longrightarrow E^M(i_1, \dots, i_l)$$

is an isomorphism of  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ , which shows (3).

Note that  $M_{\beta_1} = M_{i_1} \subseteq E^M(i_1, \dots, i_l)$  is simple. By the above construction, for  $2 \leq k \leq l$ , we have

$$M_{\beta_k} = T_{i_1}^M(\tilde{M}_{\gamma_k}) \in E^M(i_1, \dots, i_l).$$

Since  $T$  is an isomorphism and  $\tilde{M}_{\gamma_k}$  is finite-dimensional and simple in  ${}^H_H\mathcal{YD}$  by assumption (iii), we deduce that for any  $1 \leq k \leq l$ ,  $M_{\beta_k} \subseteq E^M(i_1, \dots, i_l)$  is a finite-dimensional simple object in  ${}^H_H\mathcal{YD}$  of degree  $\beta_k$ . This implies (4).  $\square$

We now proceed to verify (CG3').

**Theorem 4.3.** *Let  $M \in \mathcal{F}_\theta$ , with each  $M_i$  simple, and assume that  $M$  admits all reflections. Then, for every  $X \in \mathcal{F}_\theta(M)$  and every  $[X]$ -reduced sequence  $\kappa$ ,*

$$\Lambda^{[X]}(\kappa) \subseteq \mathbb{N}_0^\theta.$$

*Proof.* For any  $X \in \mathcal{F}_\theta(M)$ , there is a sequence  $(i_1, \dots, i_l)$  such that  $[X] = [R_{(i_1, \dots, i_l)}(M)]$ . Clearly, the tuple  $X$  admits all reflections and gives rise to a semi-Cartan graph  $\mathcal{G}(X)$ . For any  $[X]$ -reduced sequence  $\kappa$ , we have  $\Lambda^{[X]}(\kappa) \subseteq \mathbb{N}_0^\theta$  by Proposition 4.2 (2). This proves (CG3').  $\square$

## 4.2 Decomposition of Nichols algebras

Let  $M \in \mathcal{F}_\theta$ . Assume that  $M$  admits all reflections. Let  $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$  be an  $[M]$ -reduced sequence. Recall that in Remark 4.1, we defined

$$\mathfrak{t}_{(i_1, \dots, i_k)}^M = \mathfrak{t}_{i_1}^M \circ \dots \circ \mathfrak{t}_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : \mathcal{L}_{(i_1, \dots, i_k)}^+(R_{(i_1, \dots, i_k)}(M)) \longrightarrow \mathcal{L}_{i_1}^-(M).$$

We use  $\mathfrak{t}_\kappa^M$  and  $E^M(\kappa)$  to obtain a decomposition of  $\mathcal{B}(M)$ .

**Lemma 4.4.** *With above assumption, we have  $\mathcal{B}(R_\kappa(M)) \in \mathcal{L}_\kappa^+(R_\kappa(M))$ . Furthermore, the multiplication map*

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes E^M(\kappa) \rightarrow \mathcal{B}(M)$$

*is bijective in  ${}^H_H\mathcal{YD}$ .*

*Proof.* We proceed by induction on  $l$ . The assertion is immediate for  $l=0$ . For  $l = 1$ ,  $M_{i_1}^* \subseteq \mathcal{B}(R_{i_1}(M))$  holds automatically and

$$\mathfrak{t}_{i_1}^M(\mathcal{B}(R_{i_1}(M))) = T_{i_1}^M(\mathcal{B}(R_{i_1}(M))) \cap \mathcal{L}_{i_1}^{R_{i_1}(M)} = K_{i_1}^M.$$

Recall  $E^M(i_1) = \mathcal{B}(M_{i_1})$  in this case, hence the multiplication map  $\mathfrak{t}_{i_1}^M(\mathcal{B}(R_{i_1}(M))) \otimes E^M(i_1) \rightarrow \mathcal{B}(M)$  is bijective.

Now for  $l \geq 2$ , we denote  $M' = R_{i_1}(M)$  and  $\kappa' = (i_2, \dots, i_l)$ . We assume that  $\mathcal{B}(R_\kappa(M)) \in \mathcal{L}_{\kappa'}^+(R_{\kappa'}(M'))$  and the multiplication map  $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \otimes E^{M'}(\kappa') \rightarrow \mathcal{B}(M')$  is bijective. From our assumptions,  $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \in \mathcal{L}_{i_2}^-(M')$ . Note that  $M_{i_1}' \not\subseteq E^{M'}(\kappa')$ . Because the multiplication map  $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \otimes E^{M'}(\kappa') \rightarrow \mathcal{B}(M')$  is bijective, it follows that  $M_{i_1}' \subseteq \mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M)))$ . Hence  $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \in \mathcal{L}_{i_1}^+(M')$ ,  $\mathfrak{t}_{i_1}^M(\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M)))) \in \mathcal{L}_{i_1}^-(M)$  and  $\mathcal{B}(R_\kappa(M)) \in \mathcal{L}_\kappa^+(R_\kappa(M))$ . By Lemma 3.6 (2), the condition  $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \in \mathcal{L}_{i_1}^-(M)$  is equivalent to

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \subseteq K_{i_1}^M.$$

By Proposition 4.2 (3) and 3.10 (1)

$$E^M(\kappa) \cong T_{i_1}^M(E^{M'}(\kappa')) \otimes \mathcal{B}(M_{i_1}).$$

To show  $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes E^M(\kappa) \rightarrow \mathcal{B}(M)$  is bijective, it suffices to show  $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes T_{i_1}^M(E^{M'}(\kappa'))$  is isomorphic to  $K_{i_1}^M$  in  ${}^H_H\mathcal{YD}$ . Since  $T_{i_1}^M$  is an algebra isomorphism, it suffices to show

$$(T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa') \rightarrow L_{i_1}^{M'} \quad (4.2)$$

is an isomorphism. The multiplication map  $(T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes \mathcal{B}(M_{i_1}') \rightarrow \mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M)))$  is bijective. Hence

$$((T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes \mathcal{B}(M_{i_1}')) \otimes E^{M'}(\kappa') \rightarrow \mathcal{B}(M')$$

is bijective. By Proposition 4.2(2), and the definition of  $T_{i_1}^M$ ,

$$(T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa') \subseteq L_{i_1}^{M'}.$$

Hence (4.2) is an injective map in  ${}^H_H\mathcal{YD}$ .

Now fix any  $\alpha \in \mathbb{N}_0^\theta$ ,

$$\dim \mathcal{B}(M')(\alpha) = \sum_{k=0}^{\infty} \dim((T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa'))(\alpha - k\alpha_{i_1}) \dim \mathcal{B}(M')_{i_1}(k\alpha_{i_1}).$$

On the other hand, for any  $\alpha \in \mathbb{N}_0^\theta$ , we have

$$\begin{aligned} \dim \mathcal{B}(M')(\alpha) &= \sum_{k=0}^{\infty} \dim K_{i_1}^{M'}(\alpha - k\alpha_{i_1}) \dim \mathcal{B}(M')_{i_1}(k\alpha_{i_1}) \\ &= \sum_{k=0}^{\infty} \dim L_{i_1}^{M'}(\alpha - k\alpha_{i_1}) \dim \mathcal{B}(M')_{i_1}(k\alpha_{i_1}), \end{aligned}$$

where the second equation follows by (3.6). By comparing dimensions, we deduce that for any  $\alpha \in \mathbb{N}_0^\theta$ ,  $\dim L_{i_1}^{M'}(\alpha) = \dim((T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa'))(\alpha)$ . Therefore (4.2) is bijective and

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes E^M(\kappa) \rightarrow \mathcal{B}(M)$$

is an isomorphism of  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ .  $\square$

The next proposition characterizes  $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))$  and  $E^M(\kappa)$  for a special reduced sequence  $\kappa = \kappa_{ij} = (i, j, i, j, \dots)$ , where  $i \neq j \in \mathbb{I}$ .

**Proposition 4.5.** *Let  $i, j \in \mathbb{I}$  with  $i \neq j$ . Assume that  $M$  admits all reflections and that  $m := \overline{m}_{ij}^{[M]} < \infty$ . Put  $\kappa = \kappa_{ij}^{[M]}$ . Then*

$$E^M(\kappa) = \mathcal{B}(M_i + M_j), \quad \mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}.$$

*Proof.* We know that  $\mathcal{G}(M)$  is a semi-Cartan graph satisfying (CG3') by Theorem 4.3. Since  $\kappa$  only involves  $i$  and  $j$ ,  $\beta_k^{[M], \kappa}$  must be linear combination of  $\alpha_i$  and  $\alpha_j$  for all  $1 \leq k \leq \overline{m}_{ij}^{[M]}$ . According to Proposition 4.2(4), we have

$$E^M(\kappa) \subseteq \bigoplus_{k_1, k_2 \geq 0} \mathcal{B}(M)(k_1\alpha_i + k_2\alpha_j) = \mathcal{B}(M_i + M_j).$$

Conversely, by Lemma 2.10 (3),  $\beta_1^{[M], \kappa} = \alpha_i$  and  $\beta_{\overline{m}_{ij}^{[M]}}^{[M], \kappa} = \alpha_j$ . According to Proposition 4.2(4), we have  $M_i + M_j \subseteq E^M(\kappa)$ , and then

$$\mathcal{B}(M_i + M_j) \subseteq E^M(\kappa),$$

since  $E^M(\kappa)$  is a subalgebra. Hence  $E^M(\kappa) = \mathcal{B}(M_i + M_j)$ .

For the second statement, we first claim that  $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \subseteq \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}$ . Without loss of generality, assume  $i < j$  and set  $M' = (M_1, \dots, M_i + M_j, \dots, M_{j-1}, M_{j+1}, \dots, M_\theta)$ . Since the multiplication map

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes \mathcal{B}(M_i + M_j) \rightarrow \mathcal{B}(M)$$

is bijective, we have  $\pi_i^{M'}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \cap (M_i + M_j) = 0$ . Note that  $\pi_i^{M'}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))))$  is a nonzero left coideal subalgebra of  $\mathcal{B}(M_i + M_j)$  and  $\mathcal{B}(M_i + M_j)$  is a strictly graded coalgebra. Hence,  $\pi_i^{M'}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) = \mathbb{k}1$ . For any  $x \in \mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))$ , this implies  $\pi_i^{M'}(x) = \epsilon(\pi_i^{M'}(x)) = \epsilon(x)$ . Therefore  $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \subseteq \mathcal{B}(M)^{\text{co } \pi_i^{M'}} = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}$ . We have

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes \mathcal{B}(M_i + M_j) \cong \mathcal{B}(M) \cong \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)} \otimes \mathcal{B}(M_i + M_j).$$

Since both isomorphisms are multiplication maps, we obtain

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}.$$

□

### 4.3 Proof of (CG4')

The main result of this section is the following theorem.

**Theorem 4.6.** *Let  $M \in \mathcal{F}_\theta$ , with each  $M_i$  simple, and assume that  $M$  admits all reflections. Then  $\mathcal{G}(M)$  satisfies (CG4') and is therefore a Cartan graph.*

For  $i \neq j \in \mathbb{I}$ , Let  $\kappa_{ij}^{[M]}$  be the  $[M]$ -reduced sequence beginning of  $(i, j, i, j, \dots)$ . Assuming  $m := \bar{m}_{ij}^{[M]} < \infty$ , we rewrite  $\kappa_{ij}^{[M]} = (i_1, i_2, \dots, i_m)$ . Now we denote  $\kappa_{ji}^{[M]} = (i_2, \dots, i_m, i_{m+1})$ , where  $i_{m+1} = i_{m-1}$ . By Lemma 2.10 (2),  $\kappa_{ji}^{[M]}$  is  $[M]$ -reduced. To consider  $(r_i r_j)^m$ , it is natural to consider the composition of these two reflection sequence:  $\kappa_{ij}^{[M]} \circ (\kappa_{ji}^{[M]})^{-1}$ .

**Lemma 4.7.** *Under the above notation, we consider two following reflection sequences:*

$$\begin{aligned} \kappa' &= (i_1, \dots, i_m, k), \\ \kappa'' &= (i_2, \dots, i_{m+1}, k) \end{aligned}$$

where  $k \neq i, j$ . Then  $\kappa'$  and  $\kappa''$  are  $[M]$ -reduced sequences.

*Proof.* Since  $\kappa_{ij}^{[M]}$  is  $[M]$ -reduced,  $\beta_p^{[M], \kappa_{ij}^{[M]}} \neq -\beta_q^{[M], \kappa_{ij}^{[M]}}$  for all  $1 \leq p < q \leq m$ . It is straightforward to see that for any  $1 \leq n \leq m$ ,  $\beta_n^{[M], \kappa'} \in \mathbb{Z}\alpha_i + \mathbb{Z}\alpha_j$  and  $\beta_{m+1}^{[M], \kappa'} = \text{id}_{[M]} s_{i_1} \cdots s_{i_m}(\alpha_k) \in \alpha_k + \mathbb{Z}\alpha_i + \mathbb{Z}\alpha_j$ . Hence,

$$\beta_p^{[M], \kappa'} \neq -\beta_q^{[M], \kappa'}$$

for all  $1 \leq p < q \leq m+1$ . Therefore, by Remark 2.7 (3),  $\kappa'$  is  $[M]$ -reduced. The proof that  $\kappa''$  is  $[M]$ -reduced is analogous. □

Now we define  $M' = R_{\kappa_{ij}^{[M]}}(M)$ , and  $M'' = R_{\kappa_{ji}^{[M]}}(M)$ . By Proposition 4.2(1),  $M'_k := R_{(i_1, \dots, i_m)}(M)_k \subseteq L_{(i_1, \dots, i_m)}^M$ , and  $M''_k := R_{(i_2, \dots, i_{m+1})}(M)_k \subseteq L_{(i_2, \dots, i_{m+1})}^M$ . Furthermore, recall that  $\mathfrak{t}_i^M : \mathcal{L}_i^+(R_i(M)) \longrightarrow \mathcal{L}_i^-(M)$ ,  $F \longmapsto T_i^M(F \cap L_i^{R_i(M)})$ . Therefore, by Proposition 4.5, we have

$$\begin{aligned} T_{(i_1, \dots, i_m)}^M(M'_k) &\subseteq T_{(i_1, \dots, i_m)}^M(\mathcal{B}(M') \cap L_{(i_1, \dots, i_m)}^M) = \mathfrak{t}_{(i_1, \dots, i_m)}^M(\mathcal{B}(M')) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}, \\ T_{(i_2, \dots, i_{m+1})}^M(M''_k) &\subseteq T_{(i_2, \dots, i_{m+1})}^M(\mathcal{B}(M'') \cap L_{(i_2, \dots, i_{m+1})}^M) = \mathfrak{t}_{(i_2, \dots, i_{m+1})}^M(\mathcal{B}(M'')) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}. \end{aligned}$$

With above preparations, we can now prove our main result.

*Proof of Theorem 4.6.* Regardless of whether  $i_m = i$  or  $i_m = j$ , the inclusions

$$T_{(i_1, \dots, i_m)}^M(M'_k), \quad T_{(i_2, \dots, i_{m+1})}^M(M''_k) \subseteq \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}$$

show that the relevant compositions are well-defined. Since  $T$  preserves the  $\mathbb{N}_0^\theta$ -grading and simplicity, we obtain

$$\begin{aligned} (T_{(i_2, \dots, i_{m+1})}^M)^{-1} \circ T_{(i_1, \dots, i_m)}^M(M'_k) &\subseteq \mathcal{B}(M''), \\ (T_{(i_1, \dots, i_m)}^M)^{-1} \circ T_{(i_2, \dots, i_{m+1})}^M(M''_k) &\subseteq \mathcal{B}(M') \end{aligned}$$

as  $\mathbb{N}_0^\theta$ -graded simple subobjects. For all  $l \in \mathbb{I}$ , we have

$$\begin{aligned} s_{i_{m+1}} s_{i_m} \cdots s_{i_2} s_{i_1} \cdots s_{i_m}(\alpha_l) &= (s_{i_{m+1}} s_{i_m})^m(\alpha_l), \\ s_{i_m} s_{i_{m-1}} \cdots s_{i_1} s_{i_2} \cdots s_{i_{m+1}}(\alpha_l) &= (s_{i_m} s_{i_{m+1}})^m(\alpha_l). \end{aligned}$$

By Lemma 2.10 (3), we have  $(s_{i_{m+1}} s_{i_m})^m(\alpha_i) = \alpha_i$  and  $(s_{i_{m+1}} s_{i_m})^m(\alpha_j) = \alpha_j$ . On the other hand, since  $T$  preserves  $\mathbb{N}_0^\theta$ -grading, we have

$$(s_{i_{m+1}} s_{i_m})^m(\alpha_k), (s_{i_m} s_{i_{m+1}})^m(\alpha_k) \in \alpha_k + \mathbb{N}_0 \alpha_i + \mathbb{N}_0 \alpha_j.$$

We assume  $(s_{i_{m+1}} s_{i_m})^m(\alpha_k) = \alpha_k + a\alpha_i + b\alpha_j$  for  $a, b \in \mathbb{N}_0$ . Then

$$(s_{i_m} s_{i_{m+1}})^m \circ (s_{i_{m+1}} s_{i_m})^m(\alpha_k) = \alpha_k$$

implies

$$(s_{i_m} s_{i_{m+1}})^m(\alpha_k) = \alpha_k - a\alpha_i - b\alpha_j.$$

By the above observations, the degree condition forces  $a = b = 0$ . Hence

$$(s_{i_{m+1}} s_{i_m})^m(\alpha_k) = (s_{i_m} s_{i_{m+1}})^m(\alpha_k) = \alpha_k,$$

which implies:

$$\begin{aligned} (T_{(i_1, \dots, i_m)}^M)^{-1} \circ T_{(i_2, \dots, i_{m+1})}^M(M''_k) &\subseteq M'_k, \\ (T_{(i_2, \dots, i_{m+1})}^M)^{-1} \circ T_{(i_1, \dots, i_m)}^M(M'_k) &\subseteq M''_k. \end{aligned}$$

Therefore  $M'_k \cong M''_k$  for  $k \neq i, j$ .

We now proceed to show:  $M'_i \cong M''_i$  and  $M'_j \cong M''_j$ . Note that  $M'_{i_m} = R_{(i_1, \dots, i_m)}(M)_{i_m} = R_{(i_1, \dots, i_{m-1})}(M)_{i_m}^*$ . According to Lemma 2.10 (3), we have  $\beta_m^{[M], \kappa} = \alpha_j$ . Hence  $M'_{i_m} \cong M''_{i_m}$ . On the other hand,  $M'_{i_{m+1}} = R_{(i_1, \dots, i_m)}(M)_{i_{m+1}} = R_{(i_2, \dots, i_m)}(R_{i_1}(M))_{i_{m+1}}$ . Then we have  $M'_{i_{m+1}} \cong M''_{i_{m+1}}$ . Similarly,

$$\begin{aligned} M''_{i_{m+1}} &= R_{(i_2, \dots, i_{m+1})}(M)_{i_{m+1}} = R_{(i_2, \dots, i_m)}(M)_{i_{m+1}}^*, \\ M''_{i_m} &= R_{(i_2, \dots, i_{m+1})}(M)_{i_m} = R_{(i_3, \dots, i_{m+1})}(R_{i_2}(M))_{i_m}. \end{aligned}$$

Therefore  $M''_{i_{m+1}} \cong M''_{i_m}$  and  $M''_{i_m} \cong M''_j$ . As a result  $[M'] = [M'']$ . Finally, we have

$$(r_i r_j)^m([M]) = [M], \quad \text{id}_{[M]}(s_i s_j)^m(\alpha_k) = \alpha_k$$

for all  $k \in \mathbb{I} \setminus \{i, j\}$ .

Now for any  $[X] \in \mathcal{X}$ , since  $M$  admits all reflections, there is a reflection sequence  $(j_1, \dots, j_l)$  such that  $[X] = [R_{(j_1, \dots, j_l)}(M)]$ . It is obvious that  $X$  admits all reflections for any representative  $X \in [X]$ . Therefore (CG4') holds for any  $[X] \in \mathcal{X}$ . Combined with Theorem 4.3, we deduce that  $\mathcal{G}(M)$  is a Cartan graph.  $\square$

**Corollary 4.8.** *Under the assumptions of Theorem 4.6, the pair*

$$(\mathcal{G}(M), (\Delta^{[X], \text{re}})_{[X] \in \mathcal{X}})$$

*is a root system over  $\mathcal{G}(M)$ .*

*Proof.* This corollary follows from the definition of the Cartan graph and  $\Delta^{[X], \text{re}}$ .  $\square$

## 5 Nichols algebras with finite Cartan graphs

In this section, we study Nichols algebras with finite Cartan graphs. We introduce tensor decomposable Nichols algebras over coquasi-Hopf algebras and use this concept to give a criterion for the finite-dimensionality of  $\mathcal{B}(M)$ .

### 5.1 Tensor decompositions

**Definition 5.1.** Let  $V$  be an  $\mathbb{N}_0^\theta$ -graded object in  ${}^H_H\mathcal{YD}$ . We say  $V$  is tensor decomposable if there exist an integer  $n \geq 0$ , finite-dimensional and simple Yetter-Drinfeld modules  $Q_1, \dots, Q_n \in {}^H_H\mathcal{YD}$  such that

$$V \cong \bigotimes_{l=1}^n \mathcal{B}(Q_l) := (\cdots (\mathcal{B}(Q_1) \otimes \mathcal{B}(Q_2)) \otimes \cdots) \otimes \mathcal{B}(Q_n)$$

as  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ . Moreover, the degree of  $Q_l$  satisfies  $\deg(Q_l) = \beta_l$ , such that  $\beta_1, \dots, \beta_n$  are pairwise distinct elements in  $\mathbb{N}_0^\theta \setminus \{0\}$ .

For  $M \in \mathcal{F}_\theta$ , the Nichols algebra  $\mathcal{B}(M)$  is called tensor decomposable if  $\mathcal{B}(M)$  is tensor decomposable as an  $\mathbb{N}_0^\theta$ -graded object in  ${}^H_H\mathcal{YD}$  with the standard grading  $\deg(M_i) = \alpha_i$ .

To show that this definition is well-defined, we consider the  $\mathbb{N}_0$ -grading on  $V = \bigoplus_{\alpha \in \mathbb{N}_0^\theta} V(\alpha)$ . For any  $\alpha \in \mathbb{N}_0^\theta$ ,  $\alpha = \sum_{i=1}^\theta n_i \alpha_i$  with  $n_i \geq 0$ , let  $|\alpha| = \sum_{i=1}^\theta n_i$ . We define

$$V(n) = \bigoplus_{|\alpha|=n} V(\alpha).$$

Then  $V = \bigoplus_{n \geq 0} V(n)$  is a  $\mathbb{N}_0$ -grading.

**Lemma 5.2.** Let  $U, V$  and  $W$  be  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$  with finite-dimensional homogeneous components. Assume that  $W(0) \cong \mathbb{k}$  in  ${}^H_H\mathcal{YD}$ . If

$$U \otimes W \cong V \otimes W \quad \text{or} \quad W \otimes U \cong W \otimes V$$

as  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ , then  $U \cong V$  as  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ .

*Proof.* The proof is the same as in [15, Lemma 14.4.4] and is omitted here.  $\square$

**Lemma 5.3.** Let  $V, W$  be tensor decomposable  $\mathbb{N}_0$ -graded objects in  ${}^H_H\mathcal{YD}$  with finite-dimensional homogeneous components. Assume

$$V \cong \bigotimes_{l=1}^n \mathcal{B}(Q_l) \cong \bigotimes_{k=1}^m \mathcal{B}(P_k).$$

Then  $n = m$ , and there is a permutation  $\sigma \in S_n$  such that  $P_l \cong Q_{\sigma(l)}$  as  $\mathbb{N}_0^\theta$ -graded object in  ${}^H_H\mathcal{YD}$ .

*Proof.* Let  $\mathbb{I}_n = \{1, 2, \dots, n\}$  and  $\mathbb{I}_m = \{1, 2, \dots, m\}$ ,  $r = \min\{|\deg(Q_l)| \mid 1 \leq l \leq n\}$ . Let  $L$  be the set of all  $l$  such that  $1 \leq l \leq n$  and  $|\deg(Q_l)| = r$ . Then  $\bigoplus_{l \in L} Q_l$  is the  $\mathbb{N}_0$ -homogeneous component of  $\bigotimes_{l=1}^n \mathcal{B}(Q_l)$  of minimal positive degree. According to  $\bigotimes_{l=1}^n \mathcal{B}(Q_l) \cong \bigotimes_{k=1}^m \mathcal{B}(P_k)$ , we have  $r = \min\{|\deg(P_k)| \mid 1 \leq k \leq m\}$  and  $\bigoplus_{l \in L} Q_l \cong \bigoplus_{k \in K} P_k$ , where  $K$  is the set of all  $k$  such that  $1 \leq k \leq m$  and  $|\deg(P_k)| = r$ . Since  $Q_l$  and  $P_k$  are finite-dimensional simple Yetter-Drinfeld modules, their endomorphism rings are local. Thus, the Krull-Schmidt theorem applies, yielding  $|L| = |K|$  and there is a permutation  $\sigma_r \in S_{|L|}$  such that  $Q_l \cong P_{\sigma_r(k)}$ .

Now we have two such decompositions since  ${}^H_H\mathcal{YD}$  is braided

$$\bigotimes_{l=1}^n \mathcal{B}(Q_l) \cong \left( \bigotimes_{l' \in \mathbb{I}_n \setminus L} \mathcal{B}(Q_{l'}) \right) \otimes \left( \bigotimes_{l'' \in L} \mathcal{B}(Q_{l''}) \right),$$

$$\bigotimes_{k=1}^m \mathcal{B}(P_k) \cong \left( \bigotimes_{k' \in \mathbb{I}_m \setminus K} \mathcal{B}(P_{k'}) \right) \otimes \left( \bigotimes_{k'' \in K} \mathcal{B}(P_{k''}) \right).$$

Note that  $\bigotimes_{l'' \in L} \mathcal{B}(Q_{l''}) \cong \bigotimes_{k'' \in K} \mathcal{B}(P_{k''})$ . Since  $\bigotimes_{l'' \in L} \mathcal{B}(Q_{l''})(0)$  is the trivial object  $\mathbb{k}$  and each homogeneous component of  $V$  is finite-dimensional, we have

$$\left( \bigotimes_{l' \in \mathbb{I}_n \setminus L} \mathcal{B}(Q_{l'}) \right) \cong \left( \bigotimes_{k' \in \mathbb{I}_m \setminus K} \mathcal{B}(P_{k'}) \right)$$

by Lemma 5.2. Proceeding similarly, we deduce that  $n = m$  and there exists a permutation  $\sigma \in S_n$  such that  $P_l \cong Q_{\sigma(l)}$  as  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ .  $\square$

Lemma 5.3 shows that the following sets are independent of the chosen tensor decomposition.

**Definition 5.4.** Let  $s \in \text{Aut}(\mathbb{Z}^\theta)$ ,  $\alpha \in \mathbb{Z}^\theta$  and  $Q \in {}^H_H\mathcal{YD}$ . Let  $V$  be a tensor decomposable  $\mathbb{N}_0$ -graded object in  ${}^H_H\mathcal{YD}$ , we define:

$$\begin{aligned} s([Q], \alpha) &= ([Q], s(\alpha)), \\ \Phi_+^V &= \{([Q_l], \beta_l) \mid 1 \leq l \leq n\}, \\ \Phi_-^V &= \{([Q_l^*], -\beta_l) \mid 1 \leq l \leq n\}, \\ \Phi^V &= \Phi_+^V \cup \Phi_-^V. \end{aligned}$$

Here  $Q_l$  and  $\beta_l$ ,  $1 \leq l \leq n$ , are the simple objects and their degrees, respectively, in the tensor decomposition of  $V$ .

## 5.2 Tensor decomposable Nichols algebras

For  $i \in \mathbb{I}$ , recall that  $K_i^M = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)}$ . The canonical coinvariant decomposition gives a graded isomorphism

$$K_i^M \otimes \mathcal{B}(M_i) \xrightarrow{\sim} \mathcal{B}(M). \quad (5.1)$$

**Lemma 5.5.** If  $\mathcal{B}(M)$  is tensor decomposable, then  $K_i^M$  is tensor decomposable for every  $i \in \mathbb{I}$ . More precisely,

$$\Phi_+^{\mathcal{B}(M)} = \Phi_+^{K_i^M} \cup \{([M_i], \alpha_i)\}. \quad (5.2)$$

*Proof.* Choose a tensor decomposition  $\mathcal{B}(M) \simeq \bigotimes_{l=1}^n \mathcal{B}(Q_l)$ ,  $\deg Q_l = \beta_l$ . Since  $\mathcal{B}(M)(\alpha_i) = M_i$ , the degree- $\alpha_i$  component of the right-hand side is nonzero. As every  $\beta_l$  is a nonzero element of  $\mathbb{N}_0^\theta$ , the only way to obtain the degree  $\alpha_i$  is from a single factor whose generating degree is  $\alpha_i$ . The generating degrees are pairwise distinct; hence there is a unique index  $h$  such that  $\beta_h = \alpha_i$ , and comparison of the degree  $\alpha_i$  components gives  $Q_h \simeq M_i$ .

By the braiding, we may move this factor to the right and write

$$\mathcal{B}(M) \simeq U_i \otimes \mathcal{B}(M_i), \quad U_i = \bigotimes_{l \neq h} \mathcal{B}(Q_l).$$

Comparing this isomorphism with (5.1) and applying Lemma 5.2, we obtain  $K_i^M \simeq U_i$ . This proves both assertions.  $\square$

The next proposition is the main technical step. In particular, it removes the assumption that reflections are already known to exist.

**Proposition 5.6.** Let  $M = (M_1, \dots, M_\theta)$  be a tuple of finite-dimensional simple objects. If  $\mathcal{B}(M)$  is tensor decomposable, then, for every  $i \in \mathbb{I}$ , the following statements hold.

- (1) The tuple  $M$  admits the  $i$ -th reflection.
- (2) The Nichols algebra  $\mathcal{B}(R_i(M))$  is tensor decomposable.
- (3) One has

$$\Phi^{\mathcal{B}(R_i(M))} = s_i^M(\Phi^{\mathcal{B}(M)}).$$

*Proof.* (1) Fix  $i \in \mathbb{I}$ . By Lemma 5.5, choose a tensor decomposition

$$K_i^M \simeq \bigotimes_{l=1}^n \mathcal{B}(Q_l), \quad \deg Q_l = \beta_l. \quad (5.3)$$

We first prove that the  $i$ -th reflection exists. Define a grading on  $K_i^M$  by

$$\text{ht}_i \left( \sum_{r=1}^{\theta} n_r \alpha_r \right) = \sum_{r \neq i} n_r, \quad K_i^M[t] = \bigoplus_{\text{ht}_i(\gamma)=t} K_i^M(\gamma).$$

Since  $\mathcal{B}(M)(n\alpha_i) = \mathcal{B}(M_i)(n)$  and the right coinvariants of  $\mathcal{B}(M_i)$  under its coproduct are just  $\mathbb{k}$ , one has  $K_i^M(n\alpha_i) = 0$  for  $n > 0$ . Therefore every  $\beta_l$  in (5.3) has  $\text{ht}_i(\beta_l) \geq 1$ . It follows from that decomposition that  $K_i^M[1] \simeq \bigoplus_{\text{ht}_i(\beta_l)=1} Q_l$ . In particular,  $K_i^M[1]$  is finite-dimensional.

On the other hand, apply [20, Theorem 5.5] with the grading in which  $M_i$  has degree zero and every  $M_j, j \neq i$ , has degree one. It gives

$$K_i^M[1] = \bigoplus_{j \neq i} \text{ad}(\mathcal{B}(M_i))(M_j) = \bigoplus_{j \neq i} \bigoplus_{r \geq 0} (\text{ad } M_i)^r(M_j). \quad (5.4)$$

The last sum is direct because its summands have the distinct  $\mathbb{N}_0^\theta$ -degrees  $r\alpha_i + \alpha_j$ . Since the left-hand side of (5.4) is finite-dimensional, for each  $j \neq i$  only finitely many of the spaces  $(\text{ad } M_i)^r(M_j)$  are nonzero. Let

$$m_{ij}^M = \max\{r \geq 0 \mid (\text{ad } M_i)^r(M_j) \neq 0\}.$$

Then  $(\text{ad } M_i)^{m_{ij}^M}(M_j)$  is nonzero and finite-dimensional, whereas  $(\text{ad } M_i)^{m_{ij}^M+1}(M_j) = 0$ . Thus  $M$  admits the  $i$ -th reflection.

(2) We next transport the tensor decomposition through this reflection. By Lemma 3.9, the map  $T_i^M : L_i^{R_i(M)} \xrightarrow{\sim} K_i^M$  satisfies  $T_i^M(L_i^{R_i(M)}(\gamma)) = K_i^M(s_i^M(\gamma))$ . Here we use  $s_i^{R_i(M)} = s_i^M$ , which follows from [20, Corollary 5.15]. Consequently, regrading (5.3) by  $s_i^M$  gives

$$L_i^{R_i(M)} \simeq \bigotimes_{l=1}^n \mathcal{B}(Q_l), \quad \deg_{R_i(M)} Q_l = s_i^M(\beta_l). \quad (5.5)$$

The grading rule for  $T_i^M$  gives  $s_i^M(\text{supp } K_i^M) = \text{supp } L_i^{R_i(M)} \subseteq \mathbb{N}_0^\theta$ . Hence the degrees  $s_i^M(\beta_l)$  belong to  $\mathbb{N}_0^\theta \setminus \{0\}$ . They remain pairwise distinct because  $s_i^M$  is an automorphism of  $\mathbb{Z}^\theta$ . The antipode induces a degree-preserving isomorphism of objects  $L_i^{R_i(M)} \simeq K_i^{R_i(M)}$  by (3.6). Hence  $K_i^{R_i(M)}$  is tensor decomposable with the factors in (5.5). Finally, the canonical coinvariant factorization gives

$$\mathcal{B}(R_i(M)) \simeq K_i^{R_i(M)} \otimes \mathcal{B}(M_i^*). \quad (5.6)$$

Here  $M_i^*$  has degree  $\alpha_i$ . No degree  $s_i^M(\beta_l)$  is a positive multiple of  $\alpha_i$ , because  $K_i^{R_i(M)}(r\alpha_i) = 0$  for  $r > 0$ . Thus (5.6) is a tensor decomposition.

(3) It remains to verify  $\Phi^{\mathcal{B}(R_i(M))} = s_i^M(\Phi^{\mathcal{B}(M)})$ . Put  $\Psi_i = \Phi_+^{K_i^M}$ . By (5.2),  $\Phi_+^{\mathcal{B}(M)} = \Psi_i \cup \{([M_i], \alpha_i)\}$ . Equations (5.5) and (5.6) give

$$\Phi_+^{\mathcal{B}(R_i(M))} = s_i^M(\Psi_i) \cup \{([M_i^*], \alpha_i)\}.$$

Since  $s_i^M(\alpha_i) = -\alpha_i$ , the two simple-root pairs are exchanged:

$$s_i^M([M_i], \alpha_i) = ([M_i], -\alpha_i), \quad s_i^M([M_i^*], -\alpha_i) = ([M_i^*], \alpha_i).$$

The same calculation for the dual negative pairs proves (3).  $\square$

**Corollary 5.7.** *If  $\mathcal{B}(M)$  is tensor decomposable, then the following statements hold.*

- (1) *The tuple  $M$  admits all reflections, and  $\mathcal{B}(P)$  is tensor decomposable for every  $P \in \mathcal{F}_\theta(M)$ .*
- (2) *For every morphism  $w : [P] \rightarrow [Q]$  in  $\mathcal{W}(\mathcal{G}(M))$  one has*

$$\Phi^{\mathcal{B}(Q)} = w(\Phi^{\mathcal{B}(P)}). \quad (5.7)$$

- (3) *For every  $P \in \mathcal{F}_\theta(M)$ ,*

$$\Delta^{[P], \text{re}} \subseteq \{\gamma \in \mathbb{Z}^\theta \mid ([V], \gamma) \in \Phi^{\mathcal{B}(P)} \text{ for some simple } V \in {}^H_H \mathcal{YD}\}. \quad (5.8)$$

*Proof.* For (1), Proposition 5.6, applied inductively, shows that every reflection sequence exists and that tensor decomposability is preserved at every step. The components of all reflected tuples remain simple by [20, Lemma 5.8]. By Proposition 5.6 (3), we can prove (2), namely (5.7).

Let  $\gamma \in \Delta^{[P], \text{re}}$ . Then  $\gamma = w(\alpha_i)$  for some  $i \in \mathbb{I}$  and some morphism  $w : [Q] \rightarrow [P]$ . Since  $([Q_i], \alpha_i) \in \Phi^{\mathcal{B}(Q)}$ , equation (5.7) gives  $([Q_i], \gamma) \in \Phi^{\mathcal{B}(P)}$ . This proves (3), that is, (5.8).  $\square$

We can now characterize tensor decomposability purely in terms of the associated Cartan graph.

**Theorem 5.8.** *Let  $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$ , with every  $M_i$  finite-dimensional and simple. The following conditions are equivalent.*

- (1)  *$M$  admits all reflections and  $\mathcal{G}(M)$  is a finite Cartan graph.*
- (2)  *$M$  admits all reflections and  $\mathcal{B}(P)$  is tensor decomposable for some  $P \in \mathcal{F}_\theta(M)$ .*
- (3)  *$\mathcal{B}(M)$  is tensor decomposable.*

*Proof.* Assume first that (1) holds. Let  $w_0 \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), [M])$  be a longest morphism and let  $\kappa = (i_1, \dots, i_l)$  be a reduced expression of  $w_0$ . Put  $\beta_k = \text{id}_{[M]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k})$  for  $1 \leq k \leq l$ . By Lemma 2.11, the sequence  $\kappa$  is  $[M]$ -reduced and

$$\Delta_+^{[M], \text{re}} = \{\beta_1, \dots, \beta_l\}. \quad (5.9)$$

Proposition 4.2 provides a graded right coideal subalgebra  $E^M(\kappa)$  and a multiplication isomorphism

$$\mathcal{B}(M_{\beta_l}) \otimes \cdots \otimes \mathcal{B}(M_{\beta_1}) \xrightarrow{\sim} E^M(\kappa). \quad (5.10)$$

Every simple root  $\alpha_i$  belongs to the set in (5.9). If  $\beta_k = \alpha_i$ , then  $M_{\beta_k}$  is a nonzero subobject of  $\mathcal{B}(M)(\alpha_i) = M_i$ ; hence  $M_{\beta_k} = M_i$  because  $M_i$  is simple. Thus  $E^M(\kappa)$  contains every  $M_i$ . Since it is a subalgebra of  $\mathcal{B}(M)$  and  $\mathcal{B}(M)$  is generated by  $\bigoplus_i M_i$ , we obtain  $E^M(\kappa) = \mathcal{B}(M)$ . Equation (5.10) therefore proves (3).

Clearly, (3) implies (2), with  $P = M$ , once we know that all reflections exist. This existence follows from Corollary 5.7.

Suppose that (2) holds. Choose a reflection sequence carrying  $M$  to  $P$ . By [20, Corollary 5.15], the reverse sequence carries  $P$  back to a tuple isomorphic to  $M$ . Repeated application of Proposition 5.6 therefore shows that  $\mathcal{B}(M)$  is tensor decomposable. Hence (2) implies (3).

Finally, assume (3). By Corollary 5.7,  $M$  admits all reflections, and  $\mathcal{B}(P)$  is tensor decomposable for every  $P \in \mathcal{F}_\theta(M)$ . By Theorem 4.6,  $\mathcal{G}(M)$  is a Cartan graph. The set  $\Phi^{\mathcal{B}(P)}$  is finite. Equation (5.8) therefore implies that  $\Delta^{[P],\text{re}}$  is finite for every object  $[P]$  of  $\mathcal{G}(M)$ . Thus  $\mathcal{G}(M)$  is finite, proving (1).  $\square$

The root factorization is now a consequence, rather than an additional input, of the preceding theorem.

**Corollary 5.9.** *Assume the equivalent conditions of Theorem 5.8, and let  $P \in \mathcal{F}_\theta(M)$ . Let  $w_0 \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), [P])$  be a longest morphism, and let  $\kappa = (i_1, \dots, i_l)$  be a reduced expression of  $w_0$ . For  $\beta_k = \text{id}_{[P]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k})$ , one has*

$$\Delta_+^{[P],\text{re}} = \{\beta_1, \dots, \beta_l\},$$

and multiplication induces an isomorphism of  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ ,

$$\mathcal{B}(P_{\beta_l}) \otimes (\cdots (\mathcal{B}(P_{\beta_2}) \otimes \mathcal{B}(P_{\beta_1}))) \xrightarrow{\sim} \mathcal{B}(P).$$

*Proof.* Apply Lemma 2.11 and Proposition 4.2 to  $P$  and  $\kappa$ . As in the proof of Theorem 5.8, the resulting coideal subalgebra  $E^P(\kappa)$  contains every  $P_i$ , and hence equals  $\mathcal{B}(P)$ . The factorization in Proposition 4.2 gives the claimed isomorphism.  $\square$

### 5.3 A criterion for finite-dimensionality

In this subsection, we first assume that  $M$  admits all reflections and that  $\mathcal{G}(M)$  is finite. Fix  $P \in \mathcal{F}_\theta(M)$ , and let  $w_0 \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), [P])$  be a longest morphism. Choose a reduced expression  $\kappa = (i_1, \dots, i_m)$  of  $w_0$ , and set

$$\beta_k = \text{id}_{[P]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}), \quad 1 \leq k \leq m.$$

By Corollary 5.9, multiplication induces an isomorphism

$$\mathcal{B}(P_{\beta_m}) \otimes \cdots \otimes \mathcal{B}(P_{\beta_1}) \xrightarrow{\sim} \mathcal{B}(P).$$

**Lemma 5.10.** *Let  $Q \in \mathcal{F}_\theta(M)$ ,  $1 \leq k \leq m$ ,  $i \in \mathbb{I}$ , and let  $w : [Q] \rightarrow [P]$  be a morphism in  $\mathcal{W}(\mathcal{G}(M))$ . Assume that  $\beta_k = w(\alpha_i)$ . Then  $P_{\beta_k} \cong Q_i$  in  ${}^H_H\mathcal{YD}$ .*

*Proof.* By Theorem 5.8,  $\mathcal{B}(M)$  is tensor decomposable. Hence Corollary 5.7 applies to every tuple in  $\mathcal{F}_\theta(M)$ . In particular,  $\Phi^{\mathcal{B}(P)} = w(\Phi^{\mathcal{B}(Q)})$ . Since  $\mathcal{B}(Q)$  is tensor decomposable, Lemma 5.5 gives  $([Q_i], \alpha_i) \in \Phi_+^{\mathcal{B}(Q)}$ . Therefore,

$$w([Q_i], \alpha_i) = ([Q_i], w(\alpha_i)) = ([Q_i], \beta_k) \in \Phi^{\mathcal{B}(P)}.$$

Since  $\beta_k \in \mathbb{N}_0^\theta \setminus \{0\}$ , this pair belongs to  $\Phi_+^{\mathcal{B}(P)}$ .

On the other hand, Corollary 5.9 and the definition of  $\Phi_+^{\mathcal{B}(P)}$  give

$$\Phi_+^{\mathcal{B}(P)} = \{([P_{\beta_h}], \beta_h) \mid 1 \leq h \leq m\}.$$

The roots  $\beta_1, \dots, \beta_m$  are pairwise distinct. Hence the unique element of  $\Phi_+^{\mathcal{B}(P)}$  whose second component is  $\beta_k$  is  $([P_{\beta_k}], \beta_k)$ . Consequently,  $[Q_i] = [P_{\beta_k}]$ , and therefore  $Q_i \cong P_{\beta_k}$  in  ${}^H_H\mathcal{YD}$ .  $\square$

The next proposition is useful for our characterization of finite-dimensional Nichols algebras.

**Proposition 5.11.** *Let  $i, j \in \mathbb{I}$ ,  $i \neq j$ , and  $0 \leq t \leq -a_{ij}^{[P]}$ . Then there exists  $1 \leq k \leq m$  such that  $\alpha_j + t\alpha_i = \beta_k$  and  $\text{ad}(P_i)^t(P_j) \cong P_{\beta_k}$  in  ${}^H_H\mathcal{YD}$ . In particular,  $\text{ad}(P_i)^t(P_j)$  is simple in  ${}^H_H\mathcal{YD}$ .*

*Proof.* By Corollary 5.9 and Lemma 5.5, there is a unique  $1 \leq h \leq m$  such that  $\beta_h = \alpha_i$  and  $P_{\beta_h} \cong P_i$ , and

$$K_i^P \cong \bigotimes_{\substack{1 \leq k \leq m \\ k \neq h}} \mathcal{B}(P_{\beta_k}) \quad (5.11)$$

as  $\mathbb{N}_0^\theta$ -graded objects in  ${}^H_H\mathcal{YD}$ .  
Set

$$\text{ht}_i \left( \sum_{r=1}^{\theta} n_r \alpha_r \right) = \sum_{r \neq i} n_r.$$

Since  $K_i^P(n\alpha_i) = 0$  for  $n > 0$ , we have  $\text{ht}_i(\beta_k) \geq 1$  for every  $k \neq h$ . By [20, Theorem 5.5],  $K_i^P$  is generated by  $\bigoplus_{\substack{r \in \mathbb{I} \\ r \neq i}} \bigoplus_{s=0}^{-a_{ir}^{[P]}} \text{ad}(P_i)^s(P_r)$ . Every summand has  $\text{ht}_i$ -degree one. Hence, for  $\gamma = \alpha_j + t\alpha_i$ ,  $K_i^P(\gamma) = \text{ad}(P_i)^t(P_j)$ .

A homogeneous contribution of (5.11) to degree  $\gamma$  is determined by integers  $n_k \geq 0$  satisfying  $\sum_{k \neq h} n_k \beta_k = \gamma$ . Applying  $\text{ht}_i$  gives  $1 = \sum_{k \neq h} n_k \text{ht}_i(\beta_k)$ . Thus exactly one  $n_k$  equals 1, all the others vanish, and  $\beta_k = \gamma$ . Therefore

$$\text{ad}(P_i)^t(P_j) = K_i^P(\gamma) \cong P_{\beta_k}.$$

The last object is simple by Proposition 4.2(4).  $\square$

**Theorem 5.12.** *Let  $M \in \mathcal{F}_\theta$  such that  $M_j$  is simple in  ${}^H_H\mathcal{YD}$  for all  $j \in \mathbb{I}$ . The following are equivalent.*

- (1)  $\mathcal{B}(M)$  is finite-dimensional.
- (2)  $M$  admits all reflections,  $\mathcal{G}(M)$  is finite and  $\mathcal{B}(P_i)$  is finite-dimensional for all  $P \in \mathcal{F}_\theta(M)$  and  $i \in \mathbb{I}$ .

*Proof.* (1)  $\Rightarrow$  (2) For any  $i \in \mathbb{I}$ , the coinvariant object  $\mathcal{B}(M)^{\text{co}\mathcal{B}(M_i)}$  is finite-dimensional. Hence,  $\mathcal{B}(M)^{\text{co}\mathcal{B}(M_i)}$  is a rational Yetter-Drinfeld module and then  $M$  admits the  $i$ -th reflection. Inductively,  $M$  admits all reflections and  $\mathcal{B}(P_i)$  is finite-dimensional for every  $P \in \mathcal{F}_\theta(M)$  and  $i \in \mathbb{I}$ . Since reflections preserve dimension, one has

$$\dim \mathcal{B}(P) = \dim \mathcal{B}(M)$$

for every  $P \in \mathcal{F}_\theta(M)$ . Let  $\kappa = (i_1, \dots, i_l)$  be any  $[P]$ -reduced sequence. By Proposition 4.2, there is an isomorphism

$$\mathcal{B}(P_{\beta_{i_l}}) \otimes (\dots (\mathcal{B}(P_{\beta_{i_2}}) \otimes \mathcal{B}(P_{\beta_{i_1}}))) \xrightarrow{\sim} E^P(\kappa).$$

Every  $P_{\beta_k}$  is nonzero, and hence  $\dim \mathcal{B}(P_{\beta_k}) \geq 2$ . Therefore

$$2^l \leq \dim E^P(\kappa) \leq \dim \mathcal{B}(P).$$

Thus the lengths of all reduced sequences are uniformly bounded. Since  $\mathbb{I}$  is finite, only finitely many reduced sequences can occur. It follows that  $\Delta^{[P], \text{re}}$  is finite for every  $P \in \mathcal{F}_\theta(M)$ . Hence  $\mathcal{G}(M)$  is finite.

(2)  $\Rightarrow$  (1). Since  $M$  admits all reflections, Theorem 4.6 shows that  $\mathcal{G}(M)$  is a Cartan graph. Hence it is a finite Cartan graph by assumption. Let

$$w_0 \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), [M])$$

be a longest morphism, and choose a reduced expression  $\kappa = (i_1, \dots, i_l)$  of  $w_0$ . Set

$$\beta_k = \text{id}_{[M]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}), \quad 1 \leq k \leq l.$$

By Corollary 5.9, multiplication induces an isomorphism  $\mathcal{B}(M) \cong (\cdots (\mathcal{B}(M_{\beta_l}) \otimes \mathcal{B}(M_{\beta_{l-1}})) \otimes \cdots) \otimes \mathcal{B}(M_{\beta_1})$ .

For each  $1 \leq k \leq l$ , the root  $\beta_k$  is real. Thus there exist  $Q^{(k)} \in \mathcal{F}_\theta(M)$ ,  $j_k \in \mathbb{I}$ , and a morphism  $w_k : [Q^{(k)}] \rightarrow [M]$  in  $\mathcal{W}(\mathcal{G}(M))$  such that  $\beta_k = w_k(\alpha_{j_k})$ . By Lemma 5.10,  $M_{\beta_k} \cong Q_{j_k}^{(k)}$ . Assumption (2) therefore implies that  $\mathcal{B}(M_{\beta_k})$  is finite-dimensional for every  $1 \leq k \leq l$ . The above root factorization now shows that  $\mathcal{B}(M)$  is finite-dimensional.  $\square$

## 6 Cartan graphs under braided monoidal equivalences

In this section, we prove that braided monoidal equivalences preserve the existence of reflections and induce isomorphisms of the associated Cartan graphs.

Let  $H$  and  $H'$  be coquasi-Hopf algebras with bijective antipodes, and let

$$(F, \xi) : {}^H_H \mathcal{YD} \rightarrow {}^{H'}_{H'} \mathcal{YD}$$

be a braided monoidal equivalence, with tensor structure

$$\xi_{V,W} : F(V) \otimes F(W) \xrightarrow{\sim} F(V \otimes W).$$

Let  $X = (X_1, \dots, X_\theta) \in \mathcal{F}_\theta$ , where each  $X_i$  is finite-dimensional and simple. Then  $F(X) = (F(X_1), \dots, F(X_\theta))$  is a tuple of finite-dimensional simple objects in  ${}^{H'}_{H'} \mathcal{YD}$ . According to Theorem 4.6, if  $X$  ( $F(X)$ , respectively) admits all reflections, then  $\mathcal{G}(X)$  ( $\mathcal{G}(F(X))$ , respectively) is a Cartan graph. Now we recall the definition of isomorphism of Cartan graphs

**Definition 6.1.** Let  $\mathcal{G} = \mathcal{G}(I, \mathcal{X}, r, A)$  and  $\mathcal{G}' = \mathcal{G}(J, \mathcal{Y}, t, B)$  be semi-Cartan graphs. A morphism  $(\beta, \gamma) : \mathcal{G} \rightarrow \mathcal{G}'$  of semi-Cartan graphs is a pair  $(\beta, \gamma)$ , where  $\beta : I \rightarrow J$ ,  $\gamma : \mathcal{X} \rightarrow \mathcal{Y}$  are maps such that for all  $i, j \in I$  and  $X \in \mathcal{X}$ ,

$$\gamma(r_i(X)) = t_{\beta(i)}(\gamma(X)), \quad a_{ij}^X = b_{\beta(i)\beta(j)}^{\gamma(X)}.$$

A morphism  $(\beta, \gamma)$  is an isomorphism if and only if both maps  $\beta$  and  $\gamma$  are bijective. Furthermore, if both  $\mathcal{G}$  and  $\mathcal{G}'$  are Cartan graphs, we say  $\mathcal{G}$  is isomorphic to  $\mathcal{G}'$  as Cartan graphs.

Our main goal is as follows:

**Theorem 6.2.** Under the above assumptions, the tuple  $X$  admits all reflections if and only if  $F(X)$  does. Under this condition,  $\mathcal{G}(X)$  is isomorphic to  $\mathcal{G}(F(X))$  as Cartan graphs.

Now fix two simple objects  $V, W \in {}^H_H \mathcal{YD}$ . By [20, Lemma 5.2],

$$\text{ad}_H(\mathcal{B}(V))(W) = \bigoplus_{n \geq 0} \text{ad}_H(V)^n(W)$$

is an object in  ${}^{A(V)}_{A(V)} \mathcal{YD}$ , where  $A(V) = \mathcal{B}(V) \# H$ . Furthermore,  $\text{ad}_H(V)^n(W)$  is an object in  ${}^H_H \mathcal{YD}$  for every  $n \geq 0$ .

**Proposition 6.3.** For each  $n \geq 0$ , there is an isomorphism of objects in  ${}^{H'}_{H'} \mathcal{YD}$

$$\text{ad}_{H'}(F(V))^n(F(W)) \cong F(\text{ad}_H(V)^n(W)).$$

*Proof.* Set  $B = \mathcal{B}(V \oplus W)$ ,  $B' = \mathcal{B}(F(V) \oplus F(W))$ , and denote their multiplications by  $\mu$  and  $\mu'$ , respectively. Let  $c$  and  $d$  denote the braidings of  ${}^H_H\mathcal{YD}$  and  ${}^{H'}_{H'}\mathcal{YD}$ , respectively.

Since  $F$  is an additive equivalence, we identify  $F(V \oplus W) \cong F(V) \oplus F(W)$ . Applying [19, Lemma 2.16] to  $V \oplus W$ , we obtain an isomorphism of Hopf algebras

$$\Psi_B : B' \xrightarrow{\sim} F(B).$$

Let  $\iota_V : V \longrightarrow B$ ,  $\iota_W : W \longrightarrow B$  and  $\iota_{F(V)} : F(V) \longrightarrow B'$ ,  $\iota_{F(W)} : F(W) \longrightarrow B'$  be the canonical inclusions. Under the above identifications,  $\Psi_B \circ \iota_{F(V)} = F(\iota_V)$ ,  $\Psi_B \circ \iota_{F(W)} = F(\iota_W)$ .

Let  $D_0 = W$  and, for  $n \geq 1$ , define  $D_n = V \otimes D_{n-1}$ . We define morphisms  $a_n : D_n \longrightarrow B$  inductively by  $a_0 = \iota_W$  and

$$a_n = \mu \circ (\iota_V \otimes a_{n-1}) - \mu \circ (a_{n-1} \otimes \iota_V) \circ c_{V, D_{n-1}}$$

for  $n \geq 1$ . By the definition of the iterated braided adjoint action,  $\text{Im}(a_n) = \text{ad}_H(V)^n(W)$ . Similarly, let  $E_0 = F(W)$  and  $E_n = F(V) \otimes E_{n-1}$  for  $n \geq 1$ . Define morphisms  $b_n : E_n \longrightarrow B'$  by  $b_0 = \iota_{F(W)}$  and

$$b_n = \mu' \circ (\iota_{F(V)} \otimes b_{n-1}) - \mu' \circ (b_{n-1} \otimes \iota_{F(V)}) \circ d_{F(V), E_{n-1}}.$$

Then  $\text{Im}(b_n) = \text{ad}_{H'}(F(V))^n(F(W))$ .

We next compare  $a_n$  and  $b_n$ . Define isomorphisms  $\Xi_n : E_n \xrightarrow{\sim} F(D_n)$  inductively by  $\Xi_0 = \text{id}_{F(W)}$  and

$$\Xi_n = \zeta_{V, D_{n-1}} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1})$$

for  $n \geq 1$ . We claim that

$$\Psi_B \circ b_n = F(a_n) \circ \Xi_n \tag{6.1}$$

for every  $n \geq 0$ .

For  $n = 0$ , this follows from  $\Psi_B \circ b_0 = \Psi_B \circ \iota_{F(W)} = F(\iota_W) = F(a_0) \circ \Xi_0$ . Suppose that (6.1) holds for  $n - 1$ . For convenience, write  $u_n = \mu \circ (\iota_V \otimes a_{n-1})$ ,  $v_n = \mu \circ (a_{n-1} \otimes \iota_V) \circ c_{V, D_{n-1}}$ , so that  $a_n = u_n - v_n$ . Meanwhile, we write  $u'_n = \mu' \circ (\iota_{F(V)} \otimes b_{n-1})$ ,  $v'_n = \mu' \circ (b_{n-1} \otimes \iota_{F(V)}) \circ d_{F(V), E_{n-1}}$ , so that  $b_n = u'_n - v'_n$ . Since  $\Psi_B$  is an algebra morphism relative to the tensor structure of  $F$ , we have

$$\Psi_B \circ \mu' = F(\mu) \circ \zeta_{B, B} \circ (\Psi_B \otimes \Psi_B).$$

Using this equality, the induction hypothesis, and the naturality of  $\zeta$ , we obtain

$$\begin{aligned} \Psi_B \circ u'_n &= F(\mu) \circ \zeta_{B, B} \circ (F(\iota_V) \otimes F(a_{n-1}) \circ \Xi_{n-1}) \\ &= F(\mu) \circ F(\iota_V \otimes a_{n-1}) \circ \zeta_{V, D_{n-1}} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}) \\ &= F(u_n) \circ \Xi_n. \end{aligned}$$

For the second term, the naturality of the braiding gives

$$(\Xi_{n-1} \otimes \text{id}_{F(V)}) \circ d_{F(V), E_{n-1}} = d_{F(V), F(D_{n-1})} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}).$$

Moreover, since  $F$  is braided monoidal,  $\zeta_{D_{n-1}, V} \circ d_{F(V), F(D_{n-1})} = F(c_{V, D_{n-1}}) \circ \zeta_{V, D_{n-1}}$ . Therefore,

$$\begin{aligned} \Psi_B \circ v'_n &= F(\mu) \circ \zeta_{B, B} \circ (F(a_{n-1}) \circ \Xi_{n-1} \otimes F(\iota_V)) \circ d_{F(V), E_{n-1}} \\ &= F(\mu) \circ F(a_{n-1} \otimes \iota_V) \circ \zeta_{D_{n-1}, V} \circ d_{F(V), F(D_{n-1})} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}) \\ &= F(\mu) \circ F(a_{n-1} \otimes \iota_V) \circ F(c_{V, D_{n-1}}) \circ \zeta_{V, D_{n-1}} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}) \\ &= F(v_n) \circ \Xi_n. \end{aligned}$$

Consequently,

$$\Psi_B \circ b_n = \Psi_B \circ (u'_n - v'_n) = F(u_n - v_n) \circ \Xi_n = F(a_n) \circ \Xi_n.$$

This completes the induction.

Finally,  $F$  is an exact equivalence of abelian categories and hence preserves images. Since  $\Psi_B$  and  $\Xi_n$  are isomorphisms, equation (6.1) gives

$$\mathrm{ad}_{H'}(F(V))^n(F(W)) = \mathrm{Im}(b_n) \cong F(\mathrm{Im}(a_n)) = F(\mathrm{ad}_H(V)^n(W)).$$

This proves the proposition.  $\square$

Now we can prove Theorem 6.2.

*Proof of Theorem 6.2.* If  $X$  admits the  $i$ -th reflection for some  $i \in \mathbb{I}$ , then for each  $j \neq i$ , we have  $\mathrm{ad}_H(X_i)^{-a_{ij}^X}(X_j) \neq 0$  but  $\mathrm{ad}_H(X_i)^{-a_{ij}^X+1}(X_j) = 0$  for some  $a_{ij}^X \leq 0$ . By Proposition 6.3, we have

$$\mathrm{ad}_{H'}(F(X_i))^{-a_{ij}^X}(F(X_j)) \neq 0, \mathrm{ad}_{H'}(F(X_i))^{-a_{ij}^X+1}(F(X_j)) = 0.$$

Obviously, the converse is true, hence  $X$  admits the  $i$ -th reflection if and only if  $F(X)$  does. Since  $\mathrm{ad}_H(X_i)^{-a_{ij}^X}(X_j)$  is a simple object in  ${}^H_H\mathcal{YD}$  by [20, Lemma 5.8], it follows that

$$\mathrm{ad}_{H'}(F(X_i))^{-a_{ij}^X}(F(X_j)) \cong F(\mathrm{ad}_H(X_i)^{-a_{ij}^X}(X_j))$$

is a simple object in  ${}^{H'}_{H'}\mathcal{YD}$  as well. Meanwhile we have  $F(X_i^*) \cong F(X_i)^*$ . Hence

$$F(R_i(X)) \cong R_i(F(X)).$$

By applying Proposition 6.3 again, if  $R_i(X)$  admits the  $l$ -th reflection, then for all  $j \in \mathbb{I}$ ,

$$a_{lj}^{R_i(X)} = a_{lj}^{F(R_i(X))}.$$

By induction,  $X$  admits all reflections if and only if  $F(X)$  does, and for every  $Y \in \mathcal{F}_\theta(X)$  and  $i, j \in \mathbb{I}$ ,  $a_{ij}^Y = a_{ij}^{F(Y)}$ .

Suppose now that  $X$  admits all reflections. Then  $\mathcal{G}(X)$  is given by

$$\mathcal{G}(X) = (\mathbb{I}, \mathcal{X}, r, (A^{[Y]})_{[Y] \in \mathcal{X}}),$$

where  $\mathcal{X} = \{[Y] \mid Y \in \mathcal{F}_\theta(X)\}$ , the map  $r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}$ ,  $i \times [Y] \mapsto [R_i(Y)]$  and  $A^{[Y]} = (a_{ij}^Y)_{i,j \in \mathbb{I}}$  for all  $[Y] \in \mathcal{X}$ . Then  $\mathcal{G}(F(X))$  is well-defined and can be expressed as follows,

$$\mathcal{G}(F(X)) = (\mathbb{I}, \mathcal{Y}, t, (A^{[Y]})_{[Y] \in \mathcal{Y}}),$$

where  $\mathcal{Y} = \{[Y] \mid Y \in \mathcal{F}_\theta(F(X))\}$ , the map  $t : \mathbb{I} \times \mathcal{Y} \rightarrow \mathcal{Y}$ ,  $i \times [Y] \mapsto [R_i(Y)]$  and  $A^{[Y]} = (a_{ij}^Y)_{i,j \in \mathbb{I}}$  for all  $[Y] \in \mathcal{Y}$ . Let us define  $\beta = \mathrm{id}_{\mathbb{I}}$  and

$$\gamma : \mathcal{X} \rightarrow \mathcal{Y}, [Y] \mapsto [F(Y)].$$

It is straightforward to see that  $\gamma$  is well-defined. For any  $[Z] \in \mathcal{X}$ , choose a representative  $Z$ . Then

$$\gamma(r_i([Z])) = [F(R_i(Z))] = [R_i(F(Z))] = t_i(\gamma([Z])), \quad a_{ij}^{[Z]} = a_{ij}^{\gamma([Z])}.$$

The map  $\gamma$  is injective because  $F$  is an equivalence of categories,

$$F(Y_1) \cong F(Y_2) \implies Y_1 \cong Y_2.$$

To prove surjectivity, let  $[W] \in \mathcal{Y}$ . Then  $[W] = [R_{i_k} \cdots R_{i_1}(F(X))]$  for some  $(i_1, \dots, i_k) \in \mathbb{I}^k$ . Set

$$Y = R_{i_k} \cdots R_{i_1}(X) \in \mathcal{F}_\theta(X).$$

The isomorphism  $F(R_i(Z)) \cong R_i(F(Z))$ , applied inductively, gives  $F(Y) \cong W$ . Hence  $\gamma([Y]) = [W]$ , so  $\gamma$  is surjective. Hence  $\mathcal{G}(X)$  is isomorphic to  $\mathcal{G}(F(X))$  as Cartan graphs.  $\square$

**Corollary 6.4.** *Assume that  $X$  admits all reflections, and let*

$$\gamma : \mathcal{X} \longrightarrow \mathcal{Y}, \quad \gamma([Z]) = [F(Z)],$$

*be the object map induced by  $F$ . Then  $\mathcal{W}(\mathcal{G}(X))$  and  $\mathcal{W}(\mathcal{G}(F(X)))$  are isomorphic as Weyl groupoids, and*

$$\Delta^{[Z], \text{re}} = \Delta^{\gamma([Z]), \text{re}}$$

*for every  $[Z] \in \mathcal{X}$ .*

*Proof.* It follows from the definition immediately.  $\square$

A special and important case of interest to us is when the braided monoidal equivalence originates from a twist. Let  $H$  be a coquasi-Hopf algebra with bijective antipode. A convolution-invertible linear map

$$J : H \otimes H \rightarrow \mathbb{k}$$

is called a twist on  $H$  if  $J(h, 1) = \varepsilon(h) = J(1, h)$  for all  $h \in H$ . Given a twist, we can construct a new coquasi-Hopf algebra as follows:  $H^J = H$  as a coalgebra and the multiplication  $\circ$  on  $H^J$  is given by

$$a \circ b := J(a_1, b_1) a_2 b_2 J^{-1}(a_3, b_3)$$

for all  $a, b \in H$ . The associator  $\Phi^J$  and the quasi-antipode  $(S^J, \alpha^J, \beta^J)$  are given as

$$\Phi^J(a, b, c) = J(b_1, c_1) J(a_1, b_2 c_2) \Phi(a_2, b_3, c_3) J^{-1}(a_3 b_4, c_4) J^{-1}(a_4, b_5),$$

$$S^J = S, \quad \alpha^J(a) = J^{-1}(S(a_1), a_3) \alpha(a_2), \quad \beta^J(a) = J(a_1, S(a_3)) \beta(a_2)$$

for all  $a, b, c \in H$ .

It is standard that

$$\text{Comod}(H) \cong \text{Comod}(H^J)$$

as tensor categories and

$$(\text{id}, \mathcal{J}) : {}^H_H \mathcal{YD} \xrightarrow{\sim} {}^{H^J}_{H^J} \mathcal{YD}, \quad V \mapsto V^J$$

is an equivalence of braided monoidal categories, where  $V^J = V$  as vector spaces and  $\mathcal{J}_{V,W} : V^J \otimes W^J \rightarrow V \otimes W, v \otimes w \mapsto J^{-1}(v_{-1}, w_{-1}) v_0 \otimes w_0$ .

**Corollary 6.5.** *Let  $H$  be a coquasi-Hopf algebra with bijective antipode, let  $J$  be a twist on  $H$ , and let  $X \in \mathcal{F}_\theta$ . Then  $X$  admits all reflections if and only if  $X^J$  does. If these equivalent conditions hold, then*

$$\mathcal{G}(X) \cong \mathcal{G}(X^J)$$

*as Cartan graphs.*

*Proof.* It follows from Theorem 6.2 immediately.  $\square$

## 7 Applications

### 7.1 Reflections of Nichols algebras of diagonal type

Let  $G$  be a finite abelian group and let  $\Phi$  be a normalized 3-cocycle on  $G$ . Nichols algebras of diagonal type in  ${}^G_G \mathcal{YD}^\Phi$  were studied in [16]. In this subsection, we apply the reflection theory developed above to such Nichols algebras. We first recall the relevant Yetter–Drinfeld module structure and then compute the iterated braided adjoint actions, reflections, and Cartan matrix for tuples of diagonal type.

**Example 7.1.** The left  $\mathbb{k}G$ -comodule  $(V, \delta)$  is a left-left Yetter–Drinfeld module over the coquasi-Hopf algebra  $H = (\mathbb{k}G, \Phi)$  if there is a linear map  $\triangleright : G \otimes V \rightarrow V$  such that for all  $e, f \in G$  and  $v \in {}^sV$ :

$$e \triangleright (f \triangleright v) = \frac{\Phi(e, f, g)\Phi(g, e, f)}{\Phi(e, g, f)}(ef) \triangleright v, \quad (7.1)$$

$$1_G \triangleright v = v, \quad (7.2)$$

$$e \triangleright v \in {}^sV. \quad (7.3)$$

The category of all left-left Yetter–Drinfeld modules over  $(\mathbb{k}G, \Phi)$  is denoted by  ${}^c_G\mathcal{YD}^\Phi$ . We always denote

$$\tilde{\Phi}_g(e, f) := \frac{\Phi(e, f, g)\Phi(g, e, f)}{\Phi(e, g, f)}.$$

A Direct computation shows  $\tilde{\Phi}_g$  is a 2-cocycle on  $G$ . Furthermore, a special but important class of 3-cocycles is called abelian 3-cocycle. If  $\Phi$  is abelian, then, up to cohomology, it can be chosen in the standard form given in [18, Equation (2.13)]. For this representative, one has

$$\Phi(x, y, z) = \Phi(x, z, y)$$

for all  $x, y, z \in G$ .

Now let  $(H, \Phi)$  be a coquasi-Hopf algebra with bijective antipode and  $V$  be a one dimensional vector space. Let  $\triangleright : H \otimes V \rightarrow V$  and  $\delta : V \rightarrow H \otimes V$  be maps. Now we choose  $v \in V$  and  $g \in H$  and  $\chi \in \text{Hom}(H, \mathbb{k})$  such that

$$\delta(v) = g \otimes v, \quad h \triangleright v = \chi(h)v \text{ for all } h \in H.$$

It is straightforward to see that  $V$  is a  $H$ -comodule if and only if  $\Delta(g) = g \otimes g$  and  $\varepsilon(g) = 1$ . On the other hand, one may note that (2.9) holds if and only if  $gh_2 \otimes \chi(h_1)v = h_1g \otimes \chi(h_2)v$ . This is equivalent to

$$gh_2\chi(h_1) = h_1g\chi(h_2). \quad (7.4)$$

Now for any  $h, l \in H$ ,  $V$  satisfies (2.7) if and only if

$$\chi(hl) = \frac{\Phi(h_2, g, l_3)}{\Phi(h_1, l_1, g)\Phi(g, h_4, l_4)}\chi(h_3)\chi(l_2). \quad (7.5)$$

This implies  $\chi$  is far from being an algebra homomorphism. Nevertheless, it exhibits several useful properties. Since  $H$  has bijective antipode, if  $h \in G(H)$ , then  $h^{-1} \in G(H)$ . Moreover,

$$1 = \chi(1) = \frac{\Phi(h, g, h^{-1})}{\Phi(h, h^{-1}, g)\Phi(g, h, h^{-1})}\chi(h)\chi(h^{-1}). \quad (7.6)$$

Since elements in  $G(H)$  are invertible, one deduces that  $\chi(h) \neq 0$  for any  $h \in G(H)$ . According to (7.4), we have  $gh = hg$  for all  $h \in G(H)$ . Now for any one-dimensional Yetter-Drinfeld module  $(W = \text{span}\{w\}, \triangleright, \delta)$ , given by  $\delta(w) = l \otimes w$ ,  $h \triangleright w = \zeta(h)w$ , where  $\zeta \in \text{Hom}(H, \mathbb{k})$  satisfying (7.4). If  $h$  is group-like, the equation reduces to

$$\zeta(gh)\chi(h) = \zeta(hg)\chi(h).$$

Since  $hg = gh$  and  $\zeta(g) \neq 0$ , we can deduce that  $\zeta\chi(h) = \chi\zeta(h)$  for all  $h \in G(H)$ .

**Lemma 7.2.** Let  $V \in {}^H_H\mathcal{YD}$  be a Yetter-Drinfeld module of diagonal type. Denote

$$\text{Supp } V := \{g \in G(H) \mid {}^sV \neq 0\}.$$

Here  ${}^sV = \{v \in V \mid \delta(v) = g \otimes v\}$ . Then  $\text{Supp } V$  generates an abelian group  $G$ , and  $\Phi|_G$  is an abelian 3-cocycle on  $G$ .

*Proof.* The preceding analysis shows that  $\text{Supp } V$  generates an abelian group  $G$  and that  $\Phi|_G$  is a 3-cocycle on  $G$ . By [18, Lemma 2.10],  $V$  is of diagonal type if and only if  $\Phi|_G$  is an abelian 3-cocycle.  $\square$

Let  $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$  be a tuple of Yetter-Drinfeld modules of diagonal type. For all  $j \in \mathbb{I}$ , let  $x_j$  be a basis of  $M_j$ , and let  $g_j \in H, \chi_j \in \text{Hom}(H, \mathbb{k})$  such that

$$\delta_{M_j}(x_j) = g_j \otimes x_j, \quad h \cdot x_j = \chi_j(h)x_j$$

for all  $h \in H$ . Then  $g_j$  is an invertible group-like element for all  $j \in \mathbb{I}$ . For all  $j, k \in \mathbb{I}$  let  $q_{jk} \in \mathbb{k}^\times$  such that

$$c_{M_j, M_k}(x_j \otimes x_k) = q_{jk} x_k \otimes x_j.$$

Therefore  $G := G_M = \langle g_1, \dots, g_\theta \rangle$  and  $\Phi$  is an abelian 3-cocycle on  $G$  and  $M \in {}^G_G \mathcal{YD}^\Phi$ .

For  $n \geq 3$ , we will use the notation

$$x_i^n := ((x_i \cdot x_i) \cdots) \cdot x_i.$$

to represent iterated multiplication.

**Lemma 7.3.** *For each  $i \in \mathbb{I}$ ,  $x_i^n = 0$  in  $\mathcal{B}(M)$  if and only if  $(n)!_{q_{ii}} = 0$ .*

*Proof.* We first consider  $n = 3$  as a base case. By direct computation:

$$\begin{aligned} \Delta(x_i^3) &= (x_i^2 \otimes 1 + 1 \otimes x_i^2 + (1 + q_{ii})x_i \otimes x_i)(x_i \otimes 1 + 1 \otimes x_i) \\ &= x_i^3 \otimes 1 + 1 \otimes x_i^3 + (1 + q_{ii} + q_{ii}^2)x_i^2 \otimes x_i + \frac{1}{\Phi(g_i, g_i, g_i)}(1 + q_{ii} + q_{ii}^2)x_i \otimes x_i^2. \end{aligned}$$

It is easy to see that  $x_i^3 = 0$  if and only if  $(3)!_{q_{ii}} = 0$ , since  $\Phi(g_i, g_i, g_i) \neq 0$ .

We claim that

$$\Delta(x_i^n) = \sum_{j=0}^n \binom{n}{n-j}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} x_i^{n-j} \otimes x_i^j. \quad (7.7)$$

Here, the associators are trivial when  $j = 0$  and  $j = 1$ . We prove this claim by induction.

$$\begin{aligned} \Delta(x_i^{n+1}) &= \Delta(x_i^n) \Delta(x_i) \\ &= \left( \sum_{j=0}^n \binom{n}{n-j}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} x_i^{n-j} \otimes x_i^j \right) (1 \otimes x_i + x_i \otimes 1). \end{aligned}$$

Let us focus on coefficient of the term  $x_i^{n-j+1} \otimes x_i^j$ , which is given by

$$\begin{aligned} &\binom{n}{n-j+1}_{q_{ii}} \prod_{k=1}^{j-2} \frac{1}{\Phi(g_i^{n-j+1}, g_i^k, g_i)} \frac{1}{\Phi(g_i^{n-j+1}, g_i^{j-1}, g_i)} \\ &+ \binom{n}{n-j}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} \frac{\Phi(g_i^{n-j}, g_i, g_i^j)}{\Phi(g_i^{n-j}, g_i^j, g_i)} \prod_{k=1}^{j-1} \frac{1}{\tilde{\Phi}_{g_i}(g_i^k, g_i)} q_{ii}^j. \end{aligned}$$

A Direct computation shows

$$\Phi(g_i^{n-j}, g_i^k, g_i) = \Phi(g_i^{n-j}, g_i, g_i^k) = \frac{\Phi(g_i^{n-j+1}, g_i^k, g_i) \Phi(g_i^{n-j}, g_i, g_i^{k+1})}{\Phi(g_i^{n-j}, g_i^{k+1}, g_i) \Phi(g_i, g_i^k, g_i)}$$

Therefore

$$\prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} \frac{\Phi(g_i^{n-j}, g_i, g_i^j)}{\Phi(g_i^{n-j}, g_i^j, g_i)} \prod_{k=1}^{j-1} \frac{1}{\tilde{\Phi}_{g_i}(g_i^k, g_i)} = \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j+1}, g_i^k, g_i)}.$$

On the other hand,

$$\binom{n}{n-j+1}_{q_{ii}} + \binom{n}{n-j}_{q_{ii}} q_{ii}^j = \binom{n+1}{n-j+1}_{q_{ii}}.$$

The above calculation implies

$$\Delta(x_i^{n+1}) = \sum_{j=0}^{n+1} \binom{n+1}{n-j+1}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j+1}, g_i^k, g_i)} x_i^{n-j+1} \otimes x_i^j.$$

Since  $\Phi$  is invertible,  $\Delta(x^n) = 0$  if and only if  $\binom{n}{n-j}_{q_{ii}} = 0$  for all  $0 \leq j \leq n$ . This is equivalent to  $(n)!_{q_{ii}} = 0$ .  $\square$

**Lemma 7.4.** For any  $i, j \in \mathbb{I}$ , let  $n \in \mathbb{N}_0$ , then

$$\begin{aligned} \delta(\text{ad}(x_i)^n(x_j)) &= g_i^n g_j \otimes \text{ad}(x_i)^n(x_j), \\ g \triangleright \text{ad}(x_i)^n(x_j) &= \prod_{k=0}^{n-1} \tilde{\Phi}_g(g_i, g_i^k g_j) \chi_i^n(g) \chi_j(g) \text{ad}(x_i)^n(x_j). \end{aligned}$$

*Proof.* Both equations follow from induction on  $n$  and the fact that  $G$  is abelian.  $\square$

**Lemma 7.5.** For any  $i, j \in \mathbb{I}$ . The condition  $\text{ad}(x_i)^n(x_j) \neq 0$ , but  $\text{ad}(x_i)^{n+1}(x_j) = 0$  is equivalent to

$$(n+1)_{q_{ii}}(q_{ii}^n q_{ij} q_{ji} - 1) = 0, \quad (k)_{q_{ii}}(q_{ii}^{k-1} q_{ij} q_{ji} - 1) \neq 0, \text{ for all } 1 \leq k \leq n. \quad (7.8)$$

*Proof.* **We claim that**

$$\begin{aligned} \Delta(\text{ad}(x_i)^n(x_j)) &= \text{ad}(x_i)^n(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^n(x_j) \\ &+ \sum_{k=1}^n \binom{n}{k}_{q_{ii}} \prod_{l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j). \end{aligned} \quad (7.9)$$

The case  $n = 1$  is trivial, yielding

$$\Delta(\text{ad}(x_i)(x_j)) = \text{ad}(x_i)(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)(x_j) + (1 - q_{ij} q_{ji}) x_i \otimes x_j.$$

For  $n = 2$ , direct computation shows

$$\begin{aligned} \Delta(\text{ad}(x_i)^2(x_j)) &= \Delta(x_i) \Delta(\text{ad}(x_i)(x_j)) - \Delta(g_i \triangleright (\text{ad}(x_i)(x_j))) \Delta(x_i) \\ &= \text{ad}(x_i)^2(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^2(x_j) + (1 - q_{ii}^2 q_{ij} q_{ji} + q_{ii} - q_{ii} q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)(x_j) \\ &+ \Phi(g_i, g_i, g_j) (1 - q_{ij} q_{ji}) (1 - q_{ii} q_{ij} q_{ji}) x_i^2 \otimes x_j \\ &= \text{ad}(x_i)^2(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^2(x_j) + (1 + q_{ii}) (1 - q_{ii} q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)(x_j) \\ &+ \Phi(g_i, g_i, g_j) (1 - q_{ij} q_{ji}) (1 - q_{ii} q_{ij} q_{ji}) x_i^2 \otimes x_j. \end{aligned}$$

Now we assume (7.9) holds for  $\text{ad}(x_i)^n(x_j)$ . Then

$$\begin{aligned}
\Delta(\text{ad}(x_i)^{n+1}(x_j)) &= \Delta(x_i)\Delta(\text{ad}(x_i)^n(x_j)) - \Delta(g_i \triangleright (\text{ad}(x_i)^n(x_j)))\Delta(x_i) \\
&= \Delta(x_i)\Delta(\text{ad}(x_i)^n(x_j)) - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij}\Delta(\text{ad}(x_i)^n(x_j))\Delta(x_i) \\
&= \text{ad}(x_i)^{n+1}(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^{n+1}(x_j) \\
&\quad + x_i \otimes \text{ad}(x_i)^n(x_j) - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij}(g_i^n g_j \triangleright x_i) \otimes \text{ad}(x_i)^n(x_j) \\
&\quad + \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij} \text{ad}(x_i)^n(x_j) \otimes x_i - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij} \text{ad}(x_i)^n(x_j) \otimes x_i \\
&\quad + (1 \otimes x_i + x_i \otimes 1) \left( \sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) \\
&\quad - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij} \left( \sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) \right. \\
&\quad \left. x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) \Delta(x_i).
\end{aligned}$$

We focus on the coefficient of  $x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j)$ , where  $0 \leq k \leq n$ . For  $k = 0$ , this term equals:

$$\begin{aligned}
&(1 - q_{ii}^n q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)^n(x_j) + \frac{\Phi(g_i, g_i, g_i^{n-1}g_j)}{\Phi(g_i, g_i, g_i^{n-1}g_j)} \binom{n}{1}_{q_{ii}} (1 - q_{ii}^{n-1} q_{ij} q_{ji}) q_{ii} x_i \otimes x_i \text{ad}(x_i)^{n-1}(x_j) \\
&- \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij} \binom{n}{1}_{q_{ii}} (1 - q_{ii}^{n-1} q_{ij} q_{ji}) \frac{1}{\Phi(g_i, g_i^{n-1}g_j, g_i)} x_i \otimes \text{ad}(x_i)^{n-1}(x_j) x_i \\
&= (1 - q_{ii}^{2n} q_{ij} q_{ji} + \binom{n}{1}_{q_{ii}} (1 - q_{ii}^{n-1} q_{ij} q_{ji}) q_{ii}) x_i \otimes \text{ad}(x_i)^n(x_j) \\
&= \binom{n+1}{1}_{q_{ii}} (1 - q_{ii}^n q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)^n(x_j).
\end{aligned}$$

Now for  $1 \leq k \leq n-1$ , if we denote the coefficient  $\prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^n q_{ij}$  by  $\gamma$ , then the term  $x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j)$  is given by

$$\begin{aligned}
&(x_i \otimes 1) \left( \binom{n}{k}_{q_{ii}} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) \\
&+ (1 \otimes x_i) \left( \binom{n}{k+1}_{q_{ii}} \prod_{q_{ii} \mid l=n-k-1}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^{k+1} \otimes \text{ad}(x_i)^{n-k-1}(x_j) \right) \\
&- \gamma \left( \binom{n}{k}_{q_{ii}} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) (x_i \otimes 1) \\
&- \gamma \left( \binom{n}{k+1}_{q_{ii}} \prod_{q_{ii} \mid l=n-k-1}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^{k+1} \otimes \text{ad}(x_i)^{n-k-1}(x_j) \right) (1 \otimes x_i).
\end{aligned}$$

Let  $A_1, A_2, A_3, A_4$  denote these four terms, respectively. The associators  $\text{Asso}(A_i)$  are as fol-

lows:

$$\begin{aligned}
\text{Asso}(A_1) &= \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \Phi(g_i, g_i^k, g_i^{n-k} g_j) \prod_{r=1}^{k-1} \Phi(g_i, g_i^r, g_i) = \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j), \\
\text{Asso}(A_2) &= \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \frac{\Phi(g_i, g_i^{k+1}, g_i^{n-k-1} g_j)}{\Phi(g_i^{k+1}, g_i, g_i^{n-k-1} g_j)} \prod_{r=1}^k \tilde{\Phi}_{g_i}(g_i^r, g_i) = \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j), \\
\text{Asso}(A_3) &= \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p} g_j) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \prod_{r=1}^{n-k} \tilde{\Phi}_{g_i}^{-1}(g_i, g_i^{n-k-r} g_j) \\
&= \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j), \\
\text{Asso}(A_4) &= \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p} g_j) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \frac{1}{\Phi(g_i^{k+1}, g_i^{n-k-1} g_j, g_i)} \\
&= \prod_{r=1}^{n-k} \tilde{\Phi}_{g_i}^{-1}(g_i, g_i^{n-k-r} g_j) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j).
\end{aligned}$$

Combining these terms yields

$$\begin{aligned}
&\prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \left[ \binom{n}{k} \prod_{q_{ii} l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \right. \\
&\quad \left. + \binom{n}{k+1} \prod_{q_{ii} l=n-k-1}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \right] x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j) \\
&= \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \binom{n+1}{k+1} \prod_{q_{ii} l=n-k}^n (1 - q_{ii}^l q_{ij} q_{ji}) x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j).
\end{aligned}$$

For  $k = n$ , a similar computation shows the term is

$$\prod_{l=0}^n (1 - q_{ii}^l q_{ij} q_{ji}) \Phi(g_i^l, g_i, g_i^{n-l} g_j) x_i^{n+1} \otimes x_j.$$

This completes the proof of the claim (7.9).

Now according to (7.9), it is straightforward to see that the condition  $\text{ad}(x_i)^n(x_j) \neq 0$ ,  $\text{ad}(x_i)^{n+1}(x_j) = 0$  is equivalent to

$$(n+1)_{q_{ii}}(q_{ii}^n q_{ij} q_{ji} - 1) = 0, \quad (k)_{q_{ii}}(q_{ii}^{k-1} q_{ij} q_{ji} - 1) \neq 0, \text{ for all } 1 \leq k \leq n.$$

□

**Corollary 7.6.** *Let  $M = (M_1, \dots, M_\theta)$  be the  $\theta$ -tuple as above. Then  $M$  admits the  $i$ -th reflection if and only if for each  $j \neq i$ , there is  $n_{ij} \in \mathbb{N}_0$  such that*

$$(n_{ij} + 1)_{q_{ii}}(q_{ii}^{n_{ij}} q_{ij} q_{ji} - 1) = 0.$$

*Proof.* This follows from Lemma 7.5 immediately. □

**Lemma 7.7.** *Let  $i, j \in \mathbb{I}$ . Assume that  $M$  admits the  $i$ -th reflection.*

- (1) *If  $j \neq i$ ,  $a_{ij}^M = -\min\{m \in \mathbb{N}_0 \mid (m+1)_{q_{ii}}(q_{ii}^m q_{ij} q_{ji} - 1) = 0\}$ .*
- (2)

$$R_i(M)_j = \begin{cases} \mathbb{k}x_i^*, & \text{if } j = i, \\ \mathbb{k}(\text{ad } x_i)^{-a_{ij}^M}(x_j), & \text{if } j \neq i. \end{cases}$$

*Proof.* (1) and (2) follows from Lemma 7.5 and definition of reflection immediately.  $\square$

Now we choose basis for  $R_i(M)_j$ . We define

$$y_j := \begin{cases} x_i^*, & \text{if } j = i, \\ (\text{ad } x_i)^{-a_{ij}^M}(x_j), & \text{if } j \neq i. \end{cases}$$

Here we choose  $x_i^*$  such that  $x_i^*(x_i) = 1$ .

**Lemma 7.8.** *With the above notation and assume that  $M$  admits the  $i$ -th reflection. Then*

$$\delta(y_j) = g_j g_i^{-a_{ij}^M} \otimes y_j \quad (7.10)$$

for all  $j \in \mathbb{I}$ . Meanwhile, for any  $g \in G$ ,

$$g \triangleright y_j = \begin{cases} \chi_i(g)^{-1} \Phi(g, g_i^{-1}, g_i)^{-1} y_i, & \text{if } j = i, \\ \prod_{k=0}^{-a_{ij}^M-1} \tilde{\Phi}_g(g_i, g_i^k g_j) \chi_i^{-a_{ij}}(g) \chi_j(g) y_j, & \text{if } j \neq i. \end{cases} \quad (7.11)$$

*Proof.* For  $j \neq i$ , since  $G$  is abelian, the statement follows from Lemma 7.4.

For  $j = i$ , note that the evaluation map  $\text{ev} : \mathbb{k} y_i \otimes \mathbb{k} x_i \rightarrow \mathbb{k}$  is given by

$$\text{ev}(y_i \otimes x_i) = y_i(\alpha(g_i) x_i).$$

In  $(\mathbb{k}G, \Phi)$ ,  $\alpha(g) = 1$  for all  $g \in G$ . Hence  $\text{ev}(y_i \otimes x_i) = 1$ . Since  $\text{ev}$  is a morphism in  ${}^G_C \mathcal{YD}^\Phi$ , it follows that

$$g \triangleright \text{ev}(y_i \otimes x_i) = 1.$$

This implies  $\delta(y_i) = g_i^{-1} \otimes y_i$ . Meanwhile, we have

$$\tilde{\Phi}_g(g_i^{-1}, g_i) < g \triangleright y_i, g \triangleright x_i > = 1.$$

We denote  $g \triangleright y_i = \zeta(g) y_i$ , then  $\zeta(g) \chi(g) \tilde{\Phi}_g(g_i^{-1}, g_i) = 1$ . Hence

$$\zeta(g) = \chi_i(g)^{-1} \tilde{\Phi}_g^{-1}(g_i^{-1}, g_i) = \chi_i(g)^{-1} \Phi(g, g_i^{-1}, g_i)^{-1}.$$

For further statement, Equation (7.6) implies

$$\chi_i(g) \chi_i(g^{-1}) \tilde{\Phi}_{g_i}^{-1}(g, g^{-1}) = 1,$$

we obtain that

$$\zeta(g) = \tilde{\Phi}_{g_i}^{-1}(g, g^{-1}) \tilde{\Phi}_g^{-1}(g_i^{-1}, g_i) \chi_i(g^{-1}) = \Phi(g_i, g, g^{-1})^{-1} \Phi(g, g_i^{-1}, g_i)^{-1} \chi_i(g^{-1}).$$

$\square$

**Lemma 7.9.** *Let  $i \in \mathbb{I}$  and  $M$  admits the  $i$ -th reflection.*

(1) *Assume the braiding matrix of  $R_i(M)$  is  $(q'_{jk})_{j,k \in \mathbb{I}}$ , then for  $j, k \neq i$ ,*

$$\begin{aligned} q'_{ii} &= q_{ii}, \\ q'_{ji} &= \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^l, g_i) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i), \\ q'_{ij} &= \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}(g_i, g_i^l g_j) \chi_i^{-a_{ij}}(g_i^{-1}) \chi_j(g_i^{-1}), \\ q'_{jk} &= \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_k) \\ &\quad \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) \chi_i(g_i)^{-a_{ij}^M} \chi_i(g_j))^{-a_{ik}^M} \tilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) \chi_k(g_i)^{-a_{ij}^M} \chi_k(g_j). \end{aligned}$$

(2) The above elements  $(q'_{jk})_{j,k \in \mathbb{I}}$  satisfy the following relations.

$$\begin{aligned} q'_{jj} &= q_{jj} q_{ij}^{-a_{ij}^M} q_{ji}^{-a_{ij}^M} q_{ii}^{(a_{ij}^M)^2}, \\ q'_{ij} q'_{ji} &= q_{ij}^{-1} q_{ji}^{-1} q_{ii}^{2a_{ij}^M}, \\ q'_{kj} q'_{jk} &= (q_{jk} q_{kj}) (q_{ik} q_{ki})^{-a_{ij}^M} (q_{ij} q_{ji})^{-a_{ik}^M} q_{ii}^{2a_{ij}^M a_{ik}^M}. \end{aligned}$$

*Proof.* The proof proceeds by direct computation:

(1) We know that  $q'_{ii} = \zeta(g_i^{-1})$ , while

$$\zeta(g_i^{-1}) = \Phi(g_i, g_i^{-1}, g_i)^{-1} \Phi(g_i^{-1}, g_i^{-1}, g_i)^{-1} \chi_i(g_i) = q_{ii}.$$

For  $j, k \neq i$ ,

$$\begin{aligned} q'_{ji} &= \zeta(g_i^{-a_{ij}^M} g_j) \stackrel{(7.5)}{=} \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^l, g_i) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i) \\ q'_{ij} &\stackrel{(7.11)}{=} \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_j) \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \\ q'_{jk} &\stackrel{(7.11)}{=} \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M} g_j}^{-1}(g_i, g_i^l g_k) \chi_i^{-a_{ik}^M}(g_i^{-a_{ij}^M} g_j) \chi_k(g_i^{-a_{ij}^M} g_j) \\ &\stackrel{(7.5)}{=} \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M} g_j}^{-1}(g_i, g_i^l g_k) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^m, g_i) \chi_i(g_i)^{-a_{ij}^M} \chi_j(g_j))^{-a_{ik}^M} \\ &\quad \cdot \tilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) \chi_k(g_i)^{-a_{ij}^M} \chi_k(g_j) \end{aligned}$$

(2) Since  $a_{ii}^M = 2$ , it is straightforward to see  $q'_{ii} = q_{ii}^{1-2a_{ii}^M+(a_{ii}^M)^2}$ . For  $j \neq i$ ,

$$\begin{aligned} q'_{jj} &= \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M} g_j}^{-1}(g_i, g_i^l g_j) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^m, g_i) \chi_i(g_i)^{-a_{ij}^M} \chi_j(g_j))^{-a_{ij}^M} \tilde{\Phi}_{g_j^{-1}}^{-1}(g_j g_i^m, g_i) \chi_j(g_i)^{-a_{ij}^M} \chi_j(g_j) \\ &= q_{jj} q_{ij}^{-a_{ij}^M} q_{ji}^{-a_{ij}^M} q_{ii}^{(a_{ij}^M)^2}, \end{aligned}$$

and

$$\begin{aligned} q'_{ij} q'_{ji} &= \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_j) \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^l, g_i) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i) \\ &= \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i) \\ &= \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \chi_i(g_j)^{-1} \Phi(g_j, g_i^{-1}, g_i)^{-1} (\chi_i(g_i)^{-1} \Phi(g_i, g_i^{-1}, g_i)^{-1})^{-a_{ij}^M} \\ &= \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \chi_i(g_j)^{-1} \Phi(g_j, g_i^{-1}, g_i)^{-1} (\chi_i(g_i)^{-1} \chi_i(g_i)^{-1} \chi_i(g_i^{-1})^{-1})^{-a_{ij}^M} \\ &= \chi_j(g_i^{-1}) \chi_i(g_j)^{-1} \Phi(g_j, g_i^{-1}, g_i)^{-1} \chi_i(g_i)^{2a_{ij}^M} \\ &= \chi_j(g_i)^{-1} \chi_i(g_j)^{-1} \chi_i(g_i)^{2a_{ij}^M} = q_{ij}^{-1} q_{ji}^{-1} q_{ii}^{2a_{ij}^M}. \end{aligned}$$

Now for  $j, k \neq i$ ,

$$\begin{aligned}
q'_{jk}q'_{kj} &= \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M}g_j}^{-1}(g_i, g_i^l g_k) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) q_{ii}^{-a_{ij}^M} q_{ji})^{-a_{ik}^M} \tilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) q_{ik}^{-a_{ij}^M} q_{jk} \\
&\quad \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-a_{ik}^M}g_k}^{-1}(g_i, g_i^l g_j) \prod_{m=0}^{-a_{ik}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_k g_i^m, g_i) q_{ii}^{-a_{ik}^M} q_{ki})^{-a_{ij}^M} \tilde{\Phi}_{g_j}^{-1}(g_k g_i^m, g_i) q_{ij}^{-a_{ik}^M} q_{kj} \\
&= (q_{ii}^{-a_{ij}^M} q_{ji})^{-a_{ik}^M} q_{ik}^{-a_{ij}^M} q_{jk} (q_{ii}^{-a_{ik}^M} q_{ki})^{-a_{ij}^M} q_{ij}^{-a_{ik}^M} q_{kj} \\
&= (q_{jk}q_{kj})(q_{ik}q_{ki})^{-a_{ij}^M} (q_{ij}q_{ji})^{-a_{ik}^M} q_{ii}^{2a_{ij}^M a_{ik}^M}.
\end{aligned}$$

□

**Definition 7.10.** Let  $(V, c)$  be a finite-dimensional braided vector space of diagonal type and let  $q = (q_{ij})_{i,j \in \mathbb{I}}$  be a braiding matrix of  $V$ . We say that  $(V, c)$  is of Cartan type, if there exists a Cartan matrix  $A = (a_{ij})_{i,j \in \mathbb{I}}$  such that for all  $i, j \in \mathbb{I}$ ,

$$q_{ij}q_{ji} = q_{ii}^{a_{ij}}, \quad \text{where } 0 \leq -a_{ij} < \text{ord}(q_{ii}) \text{ if } i \neq j. \quad (7.12)$$

**Lemma 7.11.** Assume that  $M = (M_1, \dots, M_\theta) \in {}_{\mathbb{G}}\mathcal{YD}^\Phi$  is of Cartan type with Cartan matrix  $A$ . Then  $M$  admits the  $i$ -th reflection for all  $i \in \mathbb{I}$  and  $a_{ij}^M = a_{ij}$  for all  $i, j \in \mathbb{I}$ .

*Proof.* Let  $i \in \mathbb{I}$ . Recall that if  $j \neq i$ ,  $a_{ij}^M = -\min\{m \in \mathbb{N}_0 \mid (m+1)_{q_{ii}}(q_{ii}^m q_{ij}q_{ji} - 1) = 0\}$ . If  $a_{ij} = 0$ , then it follows from the definition that  $a_{ij}^M = 0$ . Now if  $-a_{ij} \geq 1$ , it is obvious that  $-a_{ij}^M \leq -a_{ij}$ . Suppose  $-a_{ij}^M < -a_{ij}$ . Since  $q_{ij}q_{ji} = q_{ii}^{a_{ij}}$  and  $0 \leq -a_{ij} < \text{ord}(q_{ii})$ , the condition  $-a_{ij}^M < -a_{ij}$  implies  $q_{ii}^{-a_{ij}^M} q_{ij}q_{ji} - 1 = q_{ii}^{-a_{ij}^M} q_{ii}^{a_{ij}} - 1 \neq 0$ . Therefore, to satisfy the minimality condition, we must have:

$$(-a_{ij}^M + 1)_{q_{ii}} = 0 \implies q_{ii}^{-a_{ij}^M+1} = 1.$$

Consequently, we obtain  $(q_{ij}q_{ji})^{-a_{ij}^M+1} = (q_{ii}^{a_{ij}})^{-a_{ij}^M+1} = 1$  for any  $j \in \mathbb{I} \setminus \{i\}$ . This forces  $-a_{ij}^M + 1$  to be a multiple of  $\text{ord}(q_{ii})$ . Since  $\text{ord}(q_{ii}) > -a_{ij}$ , it follows that:

$$-a_{ij}^M + 1 \geq \text{ord}(q_{ii}) > -a_{ij}.$$

This contradicts the assumption  $-a_{ij}^M < -a_{ij}$ . Thus,  $a_{ij}^M = a_{ij}$  for all  $i, j \in \mathbb{I}$ . □

**Lemma 7.12.** Assume that  $M = (M_1, \dots, M_\theta) \in {}_{\mathbb{G}}\mathcal{YD}^\Phi$  is of Cartan type with Cartan matrix  $A$ . Let  $i \in \mathbb{I}$ , then the following hold.

(1) The labels of the generalized Dynkin diagram of  $R_i(M)$  are

$$q'_{jj} = q_{jj}, \quad q'_{jk}q'_{kj} = q_{jk}q_{kj}$$

for all  $j, k \in \mathbb{I}$ .

(2) The tuple  $R_i(M)$  is of Cartan type with Cartan matrix  $A$ .

*Proof.* By Lemma 7.11,  $M$  admits the  $i$ -th reflection, so  $R_i(M)$  is well-defined. Since  $q_{ii}^{a_{ij}^M} = q_{ij}q_{ji}$ , the first statement follows from Lemma 7.9 (2), and the second follows from the definition of Cartan type. □

**Theorem 7.13.** Let  $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$  be a tuple of finite-dimensional simple objects in  ${}^G\mathcal{YD}^\Phi$ , and assume that  $M$  is of Cartan type. Then the following statements hold.

- (1) The tuple  $M$  admits all reflections.
- (2) Every  $M' \in \mathcal{F}_\theta(M)$  is of Cartan type.
- (3) The Cartan graph of  $M$  is standard; that is,  $A^{[X]} = A^{[Y]}$  for all  $[X], [Y] \in \mathcal{X}$ .
- (4) The Cartan graph of  $M$  is finite if and only if  $A^M$  is of finite type.
- (5) The Nichols algebra  $\mathcal{B}(M)$  is finite-dimensional if and only if  $A^M$  is of finite type and, for every  $i \in \mathbb{I}$ , there exists  $m \in \mathbb{N}_0$  such that  $(m+1)_{q_{ii}} = 0$ .

*Proof.* Statements (1), (2), and (3) follow from repeated applications of Lemma 7.11 and Lemma 7.12. Since  $\mathcal{G}(M)$  is standard, it follows from the definition of Weyl groupoid that

$$\text{Hom}(\mathcal{W}(\mathcal{G}(M)), X) = W(A^M)$$

for each  $X \in \mathcal{X}$ . Here  $W(A)$  denotes the Weyl group of the Cartan matrix  $A^M$ . Hence,  $\mathcal{G}(M)$  is finite if and only if  $W(A)$  is finite, which is equivalent to  $A^M$  being of finite type. This proves (4).

By Theorem 5.12,  $\mathcal{B}(M)$  is finite-dimensional if and only if  $M$  admits all reflections,  $\mathcal{G}(M)$  is finite, and  $\mathcal{B}(P_i)$  is finite-dimensional for every  $P \in \mathcal{F}_\theta(M)$  and  $i \in \mathbb{I}$ . Moreover, the last condition is equivalent to the existence, for every  $i \in \mathbb{I}$ , of an  $m \in \mathbb{N}_0$  such that  $(m+1)_{q_{ii}} = 0$ . This proves (5).  $\square$

## 7.2 Cartan graphs of diagonal tuples

In this subsection, we compare the Cartan graphs associated with diagonal tuples in  ${}^G\mathcal{YD}^\Phi$ , where  $G$  is finite abelian and  $\Phi$  is an abelian 3-cocycle, with those associated with diagonal tuples in  ${}^K\mathcal{YD}$ , where  $K$  is finite abelian. We first show that the first class properly contains the second. We then prove that every Cartan graph in the first class is a quotient of one in the second and compare their real-root sets.

**Proposition 7.14.** *There exists a rank-two diagonal tuple  $M$  over a finite abelian group equipped with an abelian 3-cocycle such that, for no finite abelian group  $K$  and no diagonal tuple  $P \in {}^K\mathcal{YD}$ , the Cartan graphs  $\mathcal{G}(M)$  and  $\mathcal{G}(P)$  are isomorphic.*

*Proof.* Let  $G = \mathbb{Z}_2 = \langle g \mid g^2 = e \rangle$  and define  $\Phi(g^a, g^b, g^c) = (-1)^{abc}$ ,  $a, b, c \in \{0, 1\}$ . This is a normalized abelian 3-cocycle on  $G$ . Let  $\zeta$  be a primitive fourth root of unity, and let  $S = \mathbb{k}x$  be the one-dimensional object in  ${}^G\mathcal{YD}^\Phi$  defined by

$$\delta(x) = g \otimes x, \quad g^a \triangleright x = \zeta^a x.$$

The tensor square of  $S$  is isomorphic to the unit object. In fact, its coaction is trivial, and Equation (2.10) gives

$$g^a \triangleright (x \otimes x) = \frac{\Phi(g^a, g, g)\Phi(g, g, g^a)}{\Phi(g, g^a, g)} \zeta^{2a} (x \otimes x) = x \otimes x.$$

Thus  $S \otimes S \cong \mathbb{k}, S^* \cong S$ .

Consider the rank-two tuple  $M = (S, S)$ . Its braiding matrix satisfies  $q_{rs} = \zeta, r, s \in \{1, 2\}$ . In particular,  $q_{rs}q_{sr} = \zeta^2 = -1$  for  $r \neq s$ . By Lemma 7.5, the integer  $-a_{rs}^M$  is the least  $m \in \mathbb{N}_0$  such that

$$(m+1)_\zeta(-\zeta^m - 1) = 0.$$

The expressions corresponding to  $m = 0$  and  $m = 1$  are nonzero, whereas for  $m = 2$  one has  $-\zeta^2 - 1 = 0$ . Hence

$$a_{12}^M = a_{21}^M = -2.$$

Moreover,  $\text{ad}(S)^2(S) \neq 0$ . The morphism defining this object has the one-dimensional source  $S^{\otimes 3}$ , and therefore its nonzero image is isomorphic to  $S^{\otimes 3}$ . Since  $S^{\otimes 2} \cong \mathbb{k}$ , it follows that  $\text{ad}(S)^2(S) \cong S$ . Together with  $S^* \cong S$ , this gives

$$R_1(M) \cong M, \quad R_2(M) \cong M.$$

Consequently,  $\mathcal{G}(M)$  has one object and its Cartan matrix is

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}.$$

Suppose now that  $\mathcal{G}(M)$  were isomorphic to  $\mathcal{G}(P)$  for some diagonal tuple  $P = (P_1, P_2) \in {}^K_K\mathcal{YD}$ , where  $K$  is a finite abelian group. Since  $\mathcal{G}(M)$  has one object, so does  $\mathcal{G}(P)$ . Hence  $[R_r(P)] = [P]$  for  $r = 1, 2$ . By the definition of isomorphism classes of tuples, comparison of the  $r$ -th components gives  $P_r^* \cong P_r$ .

Write  $P_r = \mathbb{k}y_r$ , where  $\delta(y_r) = h_r \otimes y_r$ ,  $h \triangleright y_r = \chi_r(h)y_r$  for some  $h_r \in K$  and  $\chi_r \in \widehat{K}$ . In the ordinary Yetter–Drinfeld category,  $P_r^*$  has degree  $h_r^{-1}$  and character  $\chi_r^{-1}$ . Therefore  $P_r^* \cong P_r$  implies  $h_r = h_r^{-1}$ ,  $\chi_r = \chi_r^{-1}$ . In particular, if  $q_{rr} = \chi_r(h_r)$ , then  $q_{rr}^2 = 1$ .

Let  $r \neq s$ . If  $q_{rr} = 1$ , then

$$(m+1)_{q_{rr}} = m+1 \neq 0$$

for every  $m \in \mathbb{N}_0$ . Since  $P$  admits the  $r$ -th reflection, the defining formula for  $a_{rs}^P$  forces  $q_{rs}q_{sr} = 1$ , and hence  $a_{rs}^P = 0$ . If  $q_{rr} = -1$ , then  $(2)_{-1} = 0$ , so the same formula gives  $-a_{rs}^P \leq 1$ . Thus, in either case,

$$a_{rs}^P \neq -2.$$

This contradicts the fact that the unique Cartan matrix of  $\mathcal{G}(P)$  must be

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}.$$

Therefore  $\mathcal{G}(M)$  cannot arise from any diagonal tuple in an ordinary Yetter–Drinfeld category over a finite abelian group.  $\square$

**Corollary 7.15.** *The isomorphism classes of Cartan graphs arising from diagonal tuples in ordinary Yetter–Drinfeld categories over finite abelian groups form a proper subset of those arising from diagonal tuples over finite abelian groups equipped with abelian 3-cocycles.*

*Proof.* The inclusion follows by taking the 3-cocycle to be trivial. It is proper by Proposition 7.14.  $\square$

**Remark 7.16.** The Cartan graph in the preceding proposition can also be obtained from a tuple with non-isomorphic components. Let

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle g, h \rangle, \quad \Phi(g^a h^b, g^c h^d, g^e h^f) = (-1)^{ace},$$

and let  $\zeta$  be a primitive fourth root of unity. Put  $u_1 = g$  and  $u_2 = gh$ , and define  $M_r = \mathbb{k}x_r$ ,  $r = 1, 2$ , by  $\delta(x_r) = u_r \otimes x_r$ ,  $(g^a h^b) \triangleright x_r = \zeta^a x_r$ . Their coaction degrees are different, and hence  $M_1 \not\cong M_2$ . A direct calculation gives

$$M_1^{\otimes 2} \cong M_2^{\otimes 2} \cong \mathbb{k}, \quad q_{11} = q_{12} = q_{21} = q_{22} = \zeta.$$

Thus both  $M_1$  and  $M_2$  are self-dual. As in the proof of Proposition 7.14, the Cartan entries are  $a_{12} = a_{21} = -2$ . Moreover,

$$\text{ad}(M_1)^2(M_2) \cong M_2, \quad \text{ad}(M_2)^2(M_1) \cong M_1.$$

Consequently,  $\mathcal{G}(M_1, M_2)$  is the same one-object Cartan graph as in the preceding proposition, although  $M_1 \not\cong M_2$ .

We now establish the general relationship between the two settings by constructing a covering of Cartan graphs.

Write

$$G = \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_n} = \langle g_1 \rangle \times \cdots \times \langle g_n \rangle, \quad m_i \mid m_{i+1}.$$

By [16, Proposition 3.8], every normalized 3-cocycle on  $G$  is cohomologous to one of the standard cocycles  $\Phi_{\underline{a}}$ . Since cohomologous cocycles give braided monoidally equivalent Yetter–Drinfeld categories, Corollary 6.5 allows us to assume that the cocycle is of this form.

Suppose that  $\Phi_{\underline{a}}$  is abelian. Set

$$\mathbb{G} = \mathbb{Z}_{m_1^2} \times \cdots \times \mathbb{Z}_{m_n^2} = \langle g_1 \rangle \times \cdots \times \langle g_n \rangle$$

and define

$$\pi : \mathbb{G} \longrightarrow G, \quad \pi(g_i) = g_i.$$

Choose the set-theoretic section

$$\iota(g_1^{r_1} \cdots g_n^{r_n}) = g_1^{r_1} \cdots g_n^{r_n}, \quad 0 \leq r_i < m_i.$$

By [16, Proposition 3.15], there is a normalized 2-cochain  $J_{\underline{a}}$  on  $\mathbb{G}$  such that

$$\pi^*(\Phi_{\underline{a}}) = \partial J_{\underline{a}}.$$

Let  $M = (M_1, \dots, M_\theta) \in {}^{\mathbb{G}}\mathcal{YD}^{\Phi_{\underline{a}}}$ . Following [16, Lemma 4.6 and Proposition 4.7], define  $\tilde{M} \in {}^{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$  by

$$\rho(x_i) = \tilde{d}_i \otimes x_i, \quad a \blacktriangleright x_i = \pi(a) \triangleright x_i, \quad (7.13)$$

The Nichols algebra  $\mathcal{B}(\tilde{M}) \in {}^{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$  is isomorphic to  $\mathcal{B}(M)$  in the sense of [16, Definition 4.3], while

$$M' = \tilde{M}^{J_{\underline{a}}^{-1}}$$

is a diagonal tuple in  ${}^{\mathbb{G}}\mathcal{YD}$ .

**Definition 7.17.** [15, Definition 10.1.1] Let  $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$ ,  $\mathcal{G}' = \mathcal{G}(\mathbb{I}, \mathcal{Y}, t, B)$  be semi-Cartan graphs, and let  $p : \mathcal{Y} \rightarrow \mathcal{X}$  be a map. We say that  $\mathcal{G}'$  is a covering of  $\mathcal{G}$ , and that  $p$  is a covering map, if  $p$  is surjective and

$$p(t_i(Y)) = r_i(p(Y)), \quad b_{jk}^Y = a_{jk}^{p(Y)}$$

for all  $i, j, k \in \mathbb{I}$  and  $Y \in \mathcal{Y}$ . In this case,  $\mathcal{G}$  is called a quotient semi-Cartan graph of  $\mathcal{G}'$ .

**Proposition 7.18.** The tuple  $M \in {}^{\mathbb{G}}\mathcal{YD}^{\Phi_{\underline{a}}}$  admits all reflections if and only if  $\tilde{M} \in {}^{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$  admits all reflections. If these equivalent conditions hold, then  $\mathcal{G}(\tilde{M})$  is a covering of  $\mathcal{G}(M)$ .

*Proof.* Write  $\iota(d_i) = \tilde{d}_i$ . For all  $i, j \in \mathbb{I}$ , we have

$$\rho(x_i) = \tilde{d}_i \otimes x_i, \quad \tilde{d}_j \blacktriangleright x_i = d_j \triangleright x_i = q_{ji}x_i.$$

Thus  $M$  and  $\tilde{M}$  have the same braiding matrix.

We claim that, along every reflection sequence, the corresponding tuples are simultaneously defined and their braiding matrices  $(q_{jk})$  and  $(\tilde{q}_{jk})$  satisfy

$$q_{jj} = \tilde{q}_{jj}, \quad q_{jk}q_{kj} = \tilde{q}_{jk}\tilde{q}_{kj} \quad (7.14)$$

for all  $j, k \in \mathbb{I}$ . We prove this by induction on the length of the reflection sequence. The assertion holds for  $M$  and  $\tilde{M}$ . Suppose that it holds for two corresponding tuples  $P$  and  $\tilde{P}$ . By Lemma 7.5,  $P$  admits the  $i$ -th reflection if and only if  $\tilde{P}$  does, and in this case

$$a_{jk}^P = a_{jk}^{\tilde{P}} \quad j, k \in \mathbb{I}.$$

If these reflections exist, the formulas in Lemma 7.9(2) show that the braiding matrices of  $R_i(P)$  and  $R_i(\tilde{P})$  again satisfy (7.14). This proves the claim.

Consequently,  $M$  admits all reflections if and only if  $\tilde{M}$  does. Moreover, corresponding tuples obtained by the same reflection sequence have the same Cartan matrix.

Assume that these equivalent conditions hold. Let  $Y = (Y_1, \dots, Y_\theta) \in \mathcal{F}_\theta(\tilde{M})$ . Each  $Y_j$  is one-dimensional. Choose  $0 \neq y_j \in Y_j$  and write

$$\rho(y_j) = h_j \otimes y_j, \quad a \blacktriangleright y_j = \chi_j^Y(a)y_j$$

for  $h_j, a \in \mathbb{G}$ . We claim that  $\chi_j^Y(a)$  depends only on  $\pi(a)$ . For  $Y = \tilde{M}$ , this follows from (7.13). The claim is preserved under reflections, since the components of a reflected tuple are obtained by taking duals and iterated braided adjoints, whose action formulas involve only the action scalars of the original components and the values of  $\pi^*(\Phi_a)$ . Since

$$\pi^*(\Phi_a)(a, b, c) = \Phi_a(\pi(a), \pi(b), \pi(c)),$$

the claim follows by induction on the length of a reflection sequence.

Define a one-dimensional object  $\bar{Y}_j \in {}^{\mathbb{G}}\mathcal{YD}^{\Phi_a}$ , with the same underlying vector space as  $Y_j$ , by

$$\delta_{\bar{Y}_j}(y_j) = \pi(h_j) \otimes y_j, \quad g \triangleright y_j = \chi_j^Y(a)y_j,$$

where  $a \in \mathbb{G}$  satisfies  $\pi(a) = g$ . The preceding claim shows that the action is well-defined. The Yetter–Drinfeld compatibility follows from that of  $Y_j$ , because the associator of  ${}^{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_a)}$  is the pullback of  $\Phi_a$  along  $\pi$ .

Set  $\bar{Y} = (\bar{Y}_1, \dots, \bar{Y}_\theta)$ . The definitions of duals and braided adjoint actions, together with  $\pi^*(\Phi_a) = \Phi_a \circ \pi^{\otimes 3}$ , give  $\bar{Y}_i^* \cong \bar{Y}_i^*$ ,  $\overline{\text{ad}(Y_i)^n(Y_j)} \cong \text{ad}(\bar{Y}_i)^n(\bar{Y}_j)$ . Hence

$$\overline{R_i(Y)} \cong R_i(\bar{Y}), \quad a_{jk}^Y = a_{jk}^{\bar{Y}}$$

for all  $i, j, k \in \mathbb{I}$ . In particular, if  $Y = R_{i_1} \cdots R_{i_l}(\tilde{M})$ , then  $\bar{Y} \cong R_{i_1} \cdots R_{i_l}(M)$ .

Let

$$\mathcal{X} = \{[X] \mid X \in \mathcal{F}_\theta(M)\}, \quad \mathcal{Y} = \{[Y] \mid Y \in \mathcal{F}_\theta(\tilde{M})\},$$

and define

$$p : \mathcal{Y} \longrightarrow \mathcal{X}, \quad p([Y]) = [\bar{Y}].$$

If  $Y \cong Y'$ , then  $\bar{Y} \cong \bar{Y}'$ , so  $p$  is well-defined. It is surjective because every reflection sequence starting from  $M$  is the image under  $p$  of the same reflection sequence starting from  $\tilde{M}$ .

Let  $r_i$  and  $t_i$  denote the reflection maps on  $\mathcal{X}$  and  $\mathcal{Y}$ , respectively. Then

$$p(t_i([Y])) = [\overline{R_i(Y)}] = [R_i(\bar{Y})] = r_i(p([Y]))$$

and

$$a_{jk}^{[Y]} = a_{jk}^{[\bar{Y}]} = a_{jk}^{p([Y])}.$$

Therefore  $(\text{id}_{\mathbb{I}}, p) : \mathcal{G}(\tilde{M}) \longrightarrow \mathcal{G}(M)$  is a covering of Cartan graphs.  $\square$

The covering in Proposition 7.18 also gives a precise relationship between the corresponding Weyl groupoids and real roots.

**Corollary 7.19.** *Let  $p : \mathcal{G}(\tilde{M}) \longrightarrow \mathcal{G}(M)$  be the covering map constructed in Proposition 7.18. Then  $p$  induces a covering functor*

$$F_p : \mathcal{W}(\mathcal{G}(\tilde{M})) \longrightarrow \mathcal{W}(\mathcal{G}(M))$$

*which is the identity on the linear parts of the morphisms. Moreover, for every  $[Y] \in \mathcal{Y}$ ,*

$$\Delta^{[Y], \text{re}} = \Delta^{p([Y]), \text{re}}$$

*as subsets of  $\mathbb{Z}^{\mathbb{I}}$ .*

*Proof.* The covering functor is determined on the generating reflections by

$$F_p(s_i^{[Y]}) = s_i^{p([Y])}.$$

This is well-defined because

$$A^{[Y]} = A^{p([Y])}, \quad p(t_i([Y])) = r_i(p([Y])).$$

In particular, a morphism and its image under  $F_p$  have the same linear part in  $\text{Aut}(\mathbb{Z}^I)$ .

Let  $\beta \in \Delta^{[Y],\text{re}}$ . Then  $\beta = w(\alpha_i)$  for some morphism  $w$  with target  $[Y]$ . Applying  $F_p$  gives a morphism with target  $p([Y])$  and the same linear part. Hence

$$\Delta^{[Y],\text{re}} \subseteq \Delta^{p([Y]),\text{re}}.$$

Conversely, let  $\beta = w(\alpha_i) \in \Delta^{p([Y]),\text{re}}$ , and choose an expression of  $w$  as a product of fundamental reflections. Starting from  $[Y]$  and lifting this reflection sequence backwards gives a morphism in  $\mathcal{W}(\mathcal{G}(\tilde{M}))$  with target  $[Y]$ . Since the Cartan matrices at corresponding objects agree, the lifted morphism has the same linear part  $w$ . Thus  $\beta \in \Delta^{[Y],\text{re}}$ , proving the reverse inclusion.  $\square$

**Theorem 7.20.** *Let  $M = (M_1, \dots, M_\theta) \in {}^{\mathbb{G}}\mathcal{YD}^{\Phi_a}$  be a tuple of Yetter–Drinfeld modules of diagonal type, and put*

$$M' = \tilde{M}^{J_a^{-1}} \in {}^{\mathbb{G}}\mathcal{YD}.$$

*Then  $M$  admits all reflections if and only if  $M'$  admits all reflections. If these equivalent conditions hold, then  $\mathcal{G}(M')$  is a covering of  $\mathcal{G}(M)$ .*

*Proof.* By Proposition 7.18,  $M$  admits all reflections if and only if  $\tilde{M}$  does. On the other hand, twisting by  $J_a^{-1}$  induces a braided monoidal equivalence. Hence Corollary 6.5 implies that  $\tilde{M}$  admits all reflections if and only if

$$M' = \tilde{M}^{J_a^{-1}}$$

does. Combining the two equivalences proves the first assertion.

If these conditions hold, Corollary 6.5 gives an isomorphism  $\mathcal{G}(M') \simeq \mathcal{G}(\tilde{M})$ . Composing this isomorphism with the covering  $\mathcal{G}(\tilde{M}) \rightarrow \mathcal{G}(M)$  from Proposition 7.18, we obtain the required covering

$$\mathcal{G}(M') \rightarrow \mathcal{G}(M).$$

$\square$

**Corollary 7.21.** *Every Cartan graph arising from a diagonal-type Nichols algebra over a finite abelian group with an abelian 3-cocycle is a quotient Cartan graph of one arising from a diagonal-type Nichols algebra over an finite abelian group. At corresponding objects, their real-root sets coincide.*

*Proof.* It follows from Theorem 7.20. The converse follows because a group algebra with trivial associator is the special case in which the associator is trivial.  $\square$

### 7.3 Root systems and GK-dimension for Nichols algebras of diagonal type

It is well known that a braided vector space of diagonal type has an associated generalized Dynkin diagram. We recall its definition in the present setting.

**Definition 7.22.** Let  $V$  be a  $\theta$ -dimensional braided vector space of diagonal type. Let  $(x_i)_{i \in \mathbb{I}}$  be a basis of  $V$ , and let  $(q_{ij})_{i,j \in \mathbb{I}}$  be its braiding matrix, so that

$$c_{V,V}(x_i \otimes x_j) = q_{ij} x_j \otimes x_i$$

for all  $i, j \in \mathbb{I}$ . The generalized Dynkin diagram of  $V$  is the labeled graph with vertices indexed by  $\mathbb{I}$ , where the  $i$ -th vertex is labeled by  $q_{ii}$ . For  $1 \leq i < j \leq \theta$ , the vertices  $i$  and  $j$  are joined by an edge if and only if  $q_{ij}q_{ji} \neq 1$ , in which case the edge is labeled by  $q_{ij}q_{ji}$ .

If  $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$  and  $\dim M_i = 1$  for all  $i \in \mathbb{I}$ , then the generalized Dynkin diagram of  $M$  is the generalized Dynkin diagram of  $M_1 \oplus \dots \oplus M_\theta$  with respect to a basis  $(x_i)_{i \in \mathbb{I}}$ , where  $0 \neq x_i \in M_i$ .

**Corollary 7.23.** Let  $M = (M_1, \dots, M_\theta)$  be of diagonal type in  ${}^{\mathbb{G}}\mathcal{YD}^{\Phi_{\mathbb{A}}}$  and admit all reflections. Then  $\mathcal{B}(M)$  has a finite real root system if and only if the generalized Dynkin diagram of  $M$  occurs in the classification of [12].

*Proof.* Let  $M' = \tilde{M}^{J_{\mathbb{A}}^{-1}}$  be the tuple in  ${}^{\mathbb{G}}\mathcal{YD}$  from Theorem 7.20. If  $(q'_{ij})$  is the braiding matrix of  $M'$ , then the construction of  $M'$  gives

$$q'_{ii} = q_{ii}, \quad q'_{ij}q'_{ji} = q_{ij}q_{ji}$$

for all  $i, j \in \mathbb{I}$ . Thus  $M$  and  $M'$  have the same generalized Dynkin diagram. Theorem 7.20 also gives

$$\Delta^{[M], \text{re}} = \Delta^{[M'], \text{re}}.$$

The conclusion now follows from the classification of finite arithmetic root systems in [12].  $\square$

For Nichols algebras of diagonal type in  ${}^K\mathcal{YD}$ , where  $K$  is a finite abelian group, Kharchenko's PBW theorem associates a set of positive roots with the multidegrees of the PBW generators. By [13, Corollary 6.12 and Theorem 6.11], together with [15, Theorem 10.4.7], this set is finite if and only if the tuple admits all reflections and its associated real-root system is finite. Hence [6] gives

$$\text{GKdim } \mathcal{B}(V) < \infty \iff V \text{ admits all reflections and } \mathcal{G}(V) \text{ is finite.}$$

**Corollary 7.24.** Let  $M = (M_1, \dots, M_\theta)$  be a tuple of Yetter–Drinfeld modules of diagonal type in  ${}^{\mathbb{G}}\mathcal{YD}^{\Phi_{\mathbb{A}}}$ . Then the following statements are equivalent.

- (1)  $\text{GKdim } \mathcal{B}(M) < \infty$ .
- (2)  $M$  admits all reflections and  $\mathcal{G}(M)$  is a finite Cartan graph.

*Proof.* Let  $M' = \tilde{M}^{J_{\mathbb{A}}^{-1}}$  be the tuple of diagonal type over an ordinary Hopf algebra constructed above. Notice that the construction of  $M'$  does not require  $M$  to admit all reflections.

The construction of  $M'$  preserves the  $\mathbb{N}_0^{\mathbb{I}}$ -graded dimensions of the corresponding Nichols algebras. Hence  $\mathcal{B}(M)$  and  $\mathcal{B}(M')$  have the same multigraded Hilbert series, and therefore

$$\text{GKdim } \mathcal{B}(M) = \text{GKdim } \mathcal{B}(M').$$

By [6] and the relation between PBW roots and real roots recalled above,

$$\text{GKdim } \mathcal{B}(M') < \infty \iff M' \text{ admits all reflections and } \mathcal{G}(M') \text{ is finite.}$$

By Theorem 7.20,  $M$  admits all reflections if and only if  $M'$  does. If these equivalent conditions hold, the covering map  $\pi : \mathcal{G}(M') \rightarrow \mathcal{G}(M)$  preserves the Cartan matrices and the reflection maps. Consequently, for every object  $[Y]$  of  $\mathcal{G}(M')$ ,

$$\Delta^{[Y], \text{re}} = \Delta^{\pi([Y]), \text{re}}.$$

Since  $\pi$  is surjective, it follows that

$$\mathcal{G}(M') \text{ is finite} \iff \mathcal{G}(M) \text{ is finite.}$$

Combining these equivalences proves the assertion.  $\square$

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